

S3 File Supporting Information: Species Narratives

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This document uses the Gulf of America to refer to the area formerly known as the Gulf of Mexico consistent with Executive Order (E.O.) 14172 (Restoring Names That Honor American Greatness).

Atlantic angel shark - *Squatina dumeril*Overall Vulnerability Rank = moderate ■ (99% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 15% of scores ≥ 2

<i>Squatina dumeril</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.8	1.5		Low Moderate High Very High
	Habitat Specificity	1.5	1.9		
	Mobility and Dispersal of Early Life Stages	1.4	1.2		
	Other Stressors	1.8	0.8		
	Population Growth Rate	2.5	1.6		
	Prey Specificity	1.4	2.7		
	Reproductive Cycle	1.2	1.7		
	Reproductive Strategy Sensitivity	1.7	0.9		
	Sensitivity to Ocean Acidification	1.2	1.5		
	Sensitivity to Temperature	1.8	2		
	Site Fidelity	2.2	1.4		
	Specificity in Early Life History Requirements	1	1.8		
	Stock Size Status	1.8	1.5		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	2.9	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.7	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Atlantic angel shark (*Squatina dumeril*)

Contributors: B. Curtis, W. Driggers, K. Hannan, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (99% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), Sea Surface Oxygen (3.7), and pH (4.0). Atlantic angel shark habitat associations differ regionally. In the western North Atlantic, they are found in cool, deep waters throughout the year, and occasionally in shallow nearshore waters in winter (Driggers et al. 2018). The majority of angel sharks throughout New England and mid-Atlantic waters (southern extent off North Carolina) were found in temperatures of 13.5 to 19.5°C and depths of 17 to 94 m. In the Gulf of America, the majority of angel sharks were found in temperatures of 15.7 to 19.4°C and depths of 92.3 to 171.9 m (Driggers et al. 2018).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored equal to or above 2.5: Population Growth Rate (2.5). Atlantic angel sharks mature between 8 to 13 years of age, but little else is known about their population growth parameters (Baremore et al. 2009; Baremore 2010).

Data Quality: 15% of the data quality scores were 2 or greater. Two sensitivity attributes received scores below 1: Other Stressors (0.8) and Reproductive Strategy Sensitivity (0.9). Little information is available regarding possible stressors of Atlantic angel sharks and their reproductive habits.

Climate Effects on Distribution and Abundance: There is limited data available for assessing the potential impacts of climate change on Atlantic angel sharks. Recent studies indicate discontinuous distribution with a primary range from southern New Jersey to Cape Lookout, North Carolina that may be temperature and depth driven (Driggers et al. 2018). Angel sharks are ectothermic, relying on their environment to regulate body temperature. They are found in a relatively narrow and cool water temperature range of 13.5 to 19.5°C (Driggers et al. 2018), which would indicate that this species is on the cusp of being impacted by temperature changes and may experience large fluctuations in temperature in shallow waters during winter months. However, they inhabit deeper, cooler water throughout the remainder of the year (Driggers et al. 2018; Castro 2011). The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). Angel shark distribution within and ability to move to deeper depths may afford some resilience to changes in water temperature caused by climate change. It is unclear how climate change will affect Atlantic angel shark abundance.

Life History Synopsis: The Atlantic angel shark is a benthic shark found primarily in water temperatures of 13.5 to 19.5°C and depths of 17 to 94 m along the Atlantic Coast of the United States, and 15.7 to 19.4°C and depths of 92.3 to 171.9 m in the northern Gulf of America from spring to fall (Driggers et al. 2018). While Atlantic angel sharks exhibit seasonal inshore-offshore migration patterns, adult members of the *Squatina* family generally have limited movement between larger geographic areas, resulting in localized stocks or discrete populations, and have displayed site fidelity to coastal waters in the Pacific (Ellis et al. 2021).

This species gives birth in the spring or early summer months, and the reproductive cycle for Gulf of America Atlantic angel sharks is considered to be at least biennial with a gestation period of 10 to 12

months (Baremore 2010). While average litter size is 7 (Baremore 2010), up to 16 pups have been observed in a single litter (Castro 1983). Angel sharks reach a total length of 90 to 105 cm (NMFS 2017), with females maturing at smaller sizes than males; median length at maturity for males has been reported at 92.9 cm and females at 85.8 cm (Baremore 2010).

Additionally, adult sharks have been shown to have a mid-range specialist-generalist diet that includes a breadth of prey species (e.g. molluscs, crustaceans, skates, and demersal teleost fishes, such as flounder) that narrows with maturity (Baremore et al 2010). Northeastern Florida stomach samples showed the highest frequency of prey fish to be jacks, then porgies, goatfishes, and butterfishes; prey selection for squids occurred in winter and spring, scorpionfishes, hake and croakers only in spring (Baremore 2008). Juveniles have a diet that is similar to adults, but with broader variety; demersal teleost fishes, squid and crustaceans are important for all size classes throughout all seasons (Baremore et al. 2010).

Population growth parameters and natural mortality rates are not well described for angel sharks. There is possible genetic isolation between the United States Atlantic Coast and Gulf of America populations (Driggers et al. 2018). There is incomplete knowledge about the status of genetic variation in Atlantic Coast populations that might result in impacts to reproductive success, substantial fluctuations in population size, or local extinctions. Landing of Atlantic angel sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Atlantic bigeye tuna - *Thunnus obesus*Overall Vulnerability Rank = low ■ (75% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 77% of scores ≥ 2

<i>Thunnus obesus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.8		Low Moderate High Very High
	Habitat Specificity	1.4	2.6		
	Mobility and Dispersal of Early Life Stages	1.9	2.1		
	Other Stressors	1.5	1.3		
	Population Growth Rate	2	2.8		
	Prey Specificity	1.4	2.6		
	Reproductive Cycle	1.5	2.8		
	Reproductive Strategy Sensitivity	1.9	1.9		
	Sensitivity to Ocean Acidification	1.7	1.6		
	Sensitivity to Temperature	1.3	2.6		
	Site Fidelity	1.2	2		
	Specificity in Early Life History Requirements	2.5	2.3		
	Stock Size Status	2.5	2.6		
	Sensitivity Score	low			
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.4	-		
	Sea Surface Salinity	2.3	-		
	Sea Surface Temperature	3.5	-		
	Exposure Score	high			
Overall Vulnerability Rank		low			

Atlantic bigeye tuna (*Thunnus obesus*)

Contributors: B. Galuardi, W. Goldsmith, J. Kneebone, T. Loughran, D. Richardson

Overall Climate Vulnerability Rank: **Low** (75% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.5), Sea Surface Oxygen (3.4), and pH (4.0). Bigeye tuna are an epi- and mesopelagic species, generally inhabiting pelagic surface waters at night and diving to depths of 500 m during dawn (Brill et al. 2005; Dagorn et al. 2000). Bigeye tuna thermal tolerance ranges from 17°C to 22°C, though they can tolerate temperatures down to 5°C while diving up to 1,200 m (Lam et al. 2014). Bigeye tuna can withstand lower concentrations of dissolved oxygen than any of the other Atlantic tuna species (lower tolerance limit of approximately 0.5 ml/l) (Brill 1994).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Specificity in Early Life History Requirements (2.52). Bigeye tuna larvae are found in the northern Gulf of America and tropical Atlantic, namely in the Gulf of Guinea, in the margins of warm ocean currents and eddies (Cayre et al. 1991; Cornic and Rooker 2021). Tuna larvae generally feed on zooplankton which are influenced by primary productivity. Temporal shifts in primary production may affect food availability for larval stages of bigeye tuna (Llopiz and Hobday 2015).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Several studies have described the impact of climate change to bigeye tuna. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal and temporal shifts of bigeye tuna catch throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Timing of the first 25% of bigeye tuna catch was estimated to have shifted 50 days earlier for 2019 relative to 2002, and bigeye tuna catch was estimated to have shifted poleward 14 ± 2 km for every 1°C increase (average of 3 ± 1 km/year).

These findings are supported by multiple studies modeling future distribution of bigeye tuna. Erauskin-Extramiana et al. (2019) used a species distribution model to project that bigeye tuna abundance in the Atlantic would decline in the tropics by the end of the century but increase in more northern latitudes, particularly in the eastern Atlantic. In contrast, Townhill et al. (2021) projected an increase in habitat around oceanic islands in the tropical Atlantic, where Erauskin-Extramiana projected a decrease in abundance. Braun et al. (2023) predicted a similar northeastern shift in core habitat, along with approximately 30% loss in habitat in the western North Atlantic. These contrasting results point to some of the uncertainty in projecting how bigeye tuna will respond to climate change. Further, expansion of the oxygen minimum zone may limit the vertical distribution of large pelagic fishes (Stramma et al. 2012). While bigeye tuna are uniquely adapted to enter hypoxic areas during deep foraging dives, expansion of the oxygen thermal zone coupled with rising sea surface temperatures may affect the distribution and abundance of prey (Duffy et al. 2017; Stramma et al. 2012).

Early life stages of bigeye tuna may also be impacted by changing conditions of the Loop Current in the Gulf of America and other oceanographic features important to larval survival (Cornic and Rooker 2021; Dell’Apa et al. 2018). Projected declines in Loop Current extent may affect transport and forage opportunities provided by warm- and cold-core eddies, possibly impacting survival of early life stages and overall population growth (Cornic and Rooker 2021; Dell’Apa et al. 2018; Dell’Apa et al. 2023). The average body size of bigeye tuna is also projected to decrease by 22.9% by 2050 (Erauskin-Extramiana et al. 2023).

Life History Synopsis: Bigeye tuna range from approximately 50°S to 55°N latitude in the Atlantic Ocean. Within the Atlantic Ocean, bigeye tuna are considered to exist as a single population, though alternate population configurations remain a possibility with further research needed on this topic.

In general, juvenile bigeye tuna tend to occur in equatorial areas, forming mixed schools with other species of tuna. North-south migrations of these smaller size classes in the eastern Atlantic are evident through conventional tagging data. Conventional tagging has also revealed migrations between Africa and the western Atlantic. Longline fisheries operating off North America catch the larger size classes of bigeye tuna, with one electronic tagging study in the western North Atlantic documenting a roundtrip migration from off the Atlantic Coast of the United States to the equator off the coast of South America, potentially for spawning (Lam et al. 2014). However, electronic tagging efforts for bigeye tuna have been limited, and as a result we lack the detailed knowledge of bigeye tuna migrations that is available for Atlantic bluefin tuna and some other species.

As with other *Thunnus* species, bigeye tuna spawn pelagic eggs approximately 1 mm in diameter in warm (>24 °C) surface waters. Eggs hatch within a few days and the larvae grow rapidly, feeding on zooplankton at the earliest stages before broadening their diet with the inclusion of small fishes. Egg and larval stages are dependent on currents and eddies for transport (Cornic and Rooker 2021). By approximately age 1, bigeye tuna reach 50 to 60 cm in length. At approximately 2 to 3 years of age and 100 cm in length, about 50% of bigeye tuna are mature.

Cayre et al. (1991) summarized larval and reproductive data on the spatial and temporal distribution of bigeye tuna spawning in the Atlantic. Their work estimated that spawning occurred year-round and was most prevalent between 15°S and 15°N. Within this broad area, there were a number of areas and times with more concentrated spawning activity. These presumed dominant spawning areas included: December through April in the Gulf of Guinea; January through June in an area east of Brazil at 5° to 10°S; and September through December in a large area centered on 10°N, 45°W (roughly northeast of French Guiana). Cornic and Rooker (2021) also documented the occurrence of larval bigeye tuna in the northern Gulf of America. Notably, there are substantial seasonal and spatial gaps in the distribution of both reproductive and larval sampling for highly migratory species. As with other *Thunnus* species, there are likely undocumented spawning grounds for bigeye tuna.

The physiology of bigeye tuna is well adapted to feeding at depth and in waters with low oxygen concentrations that exclude other tuna species. Coupled with their unique physiology is a behavior pattern that involves a shallow (<100 m) distribution at night and a deeper distribution during the day that is punctuated by periodic returns to warmer surface waters. This behavioral pattern allows bigeye

tuna to effectively target organisms of the deep scattering layer. Bigeye tuna maintain a generalist diet of mesopelagic teleosts, cephalopods, and euphausiacea (krill) (Duffy et al. 2017; Logan et al. 2013).

The maximum recorded age for a bigeye tuna is 17 years, with an asymptotic size of about 180 cm, though individuals up to 230 cm and 250 kg have been reported (Waterhouse et al. 2022). The intrinsic rate of increase and von Bertalanffy growth coefficient are 0.05 and 0.392/yr, respectively (Winker et al. 2019; Waterhouse et al. 2022). Natural mortality estimates are between 0.62 and 0.67 per year (Gaertner and Hallier 2003).

The last stock assessment on Atlantic bigeye tuna was conducted in 2021 and found that the stock was overfished and not subject to overfishing ($SSB_{2019}/SSB_{MSY} = 0.94$ (0.71-1.37); $F_{2019}/F_{MSY} = 1$ (0.63-1.35)) (SCRS 2023).

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Atlantic sharpnose shark - Atlantic stock (*Rhizoprionodon terraenovae*)Overall Vulnerability Rank = moderate  (100% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = very high  (41% certainty)

Directional Effect = neutral (98% certainty)

Data Quality = 69% of scores ≥ 2

<i>Rhizoprionodon terraenovae</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	3		Low Moderate High Very High
	Habitat Specificity	1.5	2.8		
	Mobility and Dispersal of Early Life Stages	1.4	1.9		
	Other Stressors	1.7	2		
	Population Growth Rate	2.1	2.9		
	Prey Specificity	1.1	3		
	Reproductive Cycle	1.4	2.4		
	Reproductive Strategy Sensitivity	1.3	1.7		
	Sensitivity to Ocean Acidification	1.3	1.5		
	Sensitivity to Temperature	1.5	2.9		
	Site Fidelity	1.6	2.7		
	Specificity in Early Life History Requirements	1.1	2.2		
	Stock Size Status	1.9	2.7		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3.2	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1.3	-		
	pH	4	-		
	Primary Production	1.3	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.7	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Atlantic sharpnose shark - Atlantic stock (*Rhizoprionodon terraenovae*)

Contributors: C. Bangley, J. Carlson, W. Driggers, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (100% certainty from bootstrap analysis)

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), Sea Surface Oxygen (3.7), and pH (4.0). Atlantic sharpnose sharks are found in shallow, temperate to tropical coastal waters in the southeastern United States. In the summer, adult males can now be found regularly in estuarine waters as far north as Delaware Bay, but the majority are found in coastal waters from North Carolina to southern Georgia (Castro 2011; C. McCandless pers comm). Male adults will remain in shallow waters until winter, during which they migrate south offshore to deeper waters, and adult females are primarily found offshore. Beginning in spring, adults move to shallower waters in the north. Atlantic sharpnose in nearshore and estuarine waters have been documented over sand, shell, and mud bottoms, with temperatures between 12.0 and 37.3°C, salinities between 11 and 39 psu, dissolved oxygen between 3.1 and 9.6 mg/l, depths from 0.5 to 21.9 m, and pH between 7.6 and 8.6 (C. McCandless pers comm; McCandless et al. 2007; NMFS SEFSC, unpublished data; Ulrich et al. 2007).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Few studies have examined the effect of climate change on the abundance and distribution of Atlantic sharpnose sharks. A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that Atlantic sharpnose sharks showed significant loss of suitable habitat compared to 24 other carcharhinid species that inhabit Mexican waters (Diaz-Carballido et al. 2022). The largest loss in distribution occurred in the northern Gulf of America under RCP2.6 and RCP8.5, while little change in their suitable habitat occurred north of the Bahamas (Diaz-Carballido et al. 2022). Additionally, sharpnose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: The Atlantic sharpnose shark is a small (max. length approx. 81.6 cm fork length (FL)) coastal species distributed within the United States Atlantic Coast of the western North Atlantic Ocean, ranging from low-salinity waters of estuaries to the continental shelf (SEDAR 2013). This species occurs from the Bay of Fundy to the Yucatan (Bigelow and Schroeder 1948; Compagno 1984; Cortés et al. 2009; Castro 2011). It is abundant from North Carolina to the Florida Keys off the Atlantic Coast of the United States (Coles 1915; Bigelow and Schroeder 1948; Castro 2011).

Along the Atlantic Coast of the United States, adult female Atlantic sharpnose sharks remain exclusively in high salinity waters once mature (Ulrich et al. 2007). Adult males and juveniles undergo seasonal inshore-offshore migrations with their winter habitat being deeper waters (Castro 1993; Ulrich et al. 2007).

Atlantic sharpnose sharks are viviparous with a gestation period of 10 to 12 months; parturition occurs from May to July and pups are about 30 cm total length at birth (Parsons 1983; Loefer and Sedberry 2003; Hoffmayer et al. 2013). They exhibit relatively rapid growth with von Bertalanffy growth coefficients of 0.48 for females and 0.50 for males; size at 50% maturity for females was 60.5 cm FL (2 years); maximum age of 19.8 years (tag-recapture data) and 18.5 years (direct age estimates) and natural mortality is relatively low at 0.209 to 0.256 (Frazier et al. 2015; Loefer and Sedberry 2003; SEDAR 2013).

Atlantic sharpnose sharks are opportunistic predators that exhibit an ontogenetic shift in their diet (Hoffmayer and Parsons 2003; Bethea et al. 2006). In the Atlantic, adults tend to feed primarily on teleosts, crustaceans, and molluscs (Delorenzo et al. 2015; Gelsleichter et al. 1999). Small juveniles collected off the coast of Georgia were found to feed primarily on teleost prey with some preference for crustaceans and molluscs (Gurshin 2007). Since the Atlantic sharpnose shark is considered an opportunistic predator (Gelsleichter et al. 1999; Hoffmayer and Parsons 2003; Bethea et al. 2006; Gurshin 2007) it feeds on the most abundantly available prey species in the area and may be less susceptible to changes in prey groups under climate change conditions (Viana-Morayta et al. 2020).

The last stock assessment for the Atlantic sharpnose sharks, Atlantic stock was conducted in 2013 and found that the stock is not overfished and overfishing is not occurring ($SSF_{2011}/SSF_{MSY} = 2.07$; $F_{2011}/F_{MSY} = 0.23$) (SEDAR 2013). Both SSF_{2011}/SSF_{MSY} and F_{2011}/F_{MSY} values suggested that the Atlantic stock had a more optimistic status compared to the Gulf of America stock for Atlantic sharpnose sharks.

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Atlantic sharpnose shark - Gulf of America stock (*Rhizoprionodon terraenovae*)Overall Vulnerability Rank = moderate ■ (100% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = very high ■ (63% certainty)

Directional Effect = neutral (89% certainty)

Data Quality = 69% of scores ≥ 2

<i>Rhizoprionodon terraenovae</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	3		Low Moderate High Very High
	Habitat Specificity	1.4	2.8		
	Mobility and Dispersal of Early Life Stages	1.4	1.8		
	Other Stressors	1.6	1.8		
	Population Growth Rate	2.1	2.7		
	Prey Specificity	1.2	3		
	Reproductive Cycle	1.4	2.6		
	Reproductive Strategy Sensitivity	1.3	1.8		
	Sensitivity to Ocean Acidification	1.3	1.3		
	Sensitivity to Temperature	1.5	2.7		
	Site Fidelity	1.6	2.5		
	Specificity in Early Life History Requirements	1.1	2.3		
	Stock Size Status	2	2.7		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	2.9	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score		very high		
Overall Vulnerability Rank		moderate			

Atlantic sharpnose shark - Gulf of America stock (*Rhizoprionodon terraenovae*)

Contributors: J. Carlson, M. Drymon, W. Driggers, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (69% certainty from bootstrap analysis)

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), Sea Surface Oxygen (3.9), and pH (4.0). Atlantic sharpnose sharks are found in shallow, temperate to tropical coastal waters in the Gulf of America (15 to 33.9°C), with adults segregating by sex in spring and autumn months with males inshore and offshore, and females exclusively offshore (J. Carlson pers comm; Hoffmayer 2003; McCandless et al. 2007; NMFS SEFSC, unpublished data; Parsons and Hoffmayer 2005). Juveniles prefer warm (18.2 to 33.7°C) nearshore waters throughout most of the year (J. Carlson pers comm; Parsons and Hoffmayer 2005, Ward-Paige et al. 2015, Bethea et al. 2015). Juveniles have been documented in the estuarine waters of the Gulf from Texas to the 10,000 Islands, Florida in depths of 0.6 to 8.2 m, temperatures down to 16°C, salinities from 10 to 38 psu, dissolved oxygen from 2.9 to 8.6 mg/l and over seagrass, sand, mud, and mud and shell bottoms (McCandless et al. 2007).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Several studies have described the distribution and abundance of Atlantic sharpnose sharks throughout the northern Gulf of America (Parsons and Hoffmayer 2005, 2007; Drymon et al. 2010, 2013; Peterson and Grubbs 2023); however, only Peterson and Grubbs (2023) addressed the potential effects of climate change suggesting that as inshore water temperatures rise, Atlantic sharpnose sharks may move into deeper offshore waters during summer. Atlantic sharpnose sharks have shown extensive movements within the Gulf (Carlson et al. 2008, Bethea and Grace 2013, Hendon et al. 2013, Kohler and Turner 2019), and the timing of their inshore-offshore movements may be shifted with increasing water temperatures. Additionally, sharpnose sharks in the Atlantic have also been found to shift reproductive timing due to an 3.0°C increase in spring sea surface temperature over a 29-year period (Hoffmayer et al. 2010). Sharpnose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that Atlantic sharpnose sharks showed significant loss of suitable habitat compared to 24 other carcharhinid species that inhabit Mexican waters (Diaz-Carballido et al. 2022). The largest loss in distribution occurred in the northern Gulf of America under RCP2.6 and RCP8.5, while the southernmost range changed minimally under the same climate projections (Diaz-Carballido et al. 2022).

Due to their nearshore distribution, Atlantic sharpnose sharks are potentially susceptible to hypoxic events and harmful algal blooms, and mortalities were documented in the northern Gulf of America

during a bloom of *Karenia brevis* (Flewelling et al. 2010). There is a possibility that negative effects to abundance could stem from the disruption of nearshore habitat.

Life History Synopsis: The Atlantic sharpnose shark is a small (max. length approx. 79.5 cm fork length (FL)) coastal species with a ubiquitous distribution ranging from low-salinity waters of estuaries, across the continental shelf, and to offshore depths up to 140 m in the Gulf of America (Carlson et al. 2021; Drymon et al. 2010; NMFS SEFSC, unpublished data; SEDAR 2013; Ward-Paige et al. 2015). Atlantic sharpnose sharks have been shown to make substantial (>100km) north-south movements in the western Gulf (Kohler and Turner 2019) and east-west movements in the northern Gulf (Carlson et al. 2008).

Adult females remain exclusively in offshore, cooler waters once mature (Parsons and Hoffmayer 2005; Drymon et al. 2010). Adult males are highly mobile and undergo a seasonal inshore-offshore migration with their winter habitat being deeper, offshore waters (Compagno 1984; Parsons and Hoffmayer 2005). During summer, adult males tend to move to inshore waters, although the extent of their vertical migrations is unknown (Parsons and Hoffmayer 2005; Hoffmayer et al. 2006; Drymon et al. 2020).

Atlantic sharpnose sharks are viviparous with a gestation period of 10 to 12 months; parturition occurs from May to July, and pups are about 30 cm in total length at birth (Parsons 1983; Loefer and Sedberry 2003; Hoffmayer et al. 2013). Parturition is thought to occur offshore (Drymon et al. 2010; Knip et al. 2010).

In addition, it has been documented that a shift in reproductive timing of Atlantic sharpnose sharks in the north-central Gulf of America to earlier in the spring occurred and corresponded with a 3.0°C increase in spring sea surface temperature over a 29-year period (Hoffmayer et al. 2010). Atlantic sharpnose sharks in the Gulf were also shown to exhibit a protracted reproductive cycle, similar to sharpnose species in tropical regions, and it was suggested it may be related to changing environmental conditions (i.e. elevated water temperatures) in the northern Gulf (Hoffmayer et al. 2013).

Atlantic sharpnose sharks are opportunistic predators that exhibit an ontogenetic shift in their diet (Hoffmayer and Parsons 2003; Bethea et al. 2006). Young-of-the-year consume mostly teleosts (sciaenids) and shrimps (Hoffmayer and Parsons 2003; Bethea et al. 2006); adults tend to feed on cephalopods, crustaceans, and teleosts (sciaenids), although diet composition may vary by locale (Hoffmayer and Parsons 2003; Bethea et al. 2006; Drymon et al. 2012; Higgs et al. 2013; Delorenzo et al. 2015; Plumlee and Wells 2016). Given that crustaceans are an important component of their diet, Atlantic sharpnose sharks may be sensitive to the effects of ocean acidification. However, since the Atlantic sharpnose shark is considered an opportunistic predator (Hoffmayer and Parsons 2003; Bethea et al. 2006), feeding on more abundantly available species in the area, it may be less susceptible to changes in prey groups in the face of climate change (Viana-Morayta et al. 2020).

Atlantic sharpnose sharks exhibit relatively rapid growth with von Bertalanffy growth coefficients of 0.63 for females and 0.85 for males. Females reach 50% maturity at 62.3 cm FL (1.3 years); maximum

observed age is reported to be 9.5 years and natural mortality is relatively low at 0.209 to 0.256 (Branstetter 1981; Parsons 1985; Carlson and Baremore 2003; Driggers et al. 2013; SEDAR 2013).

The last stock assessment for the Atlantic sharpnose sharks, Gulf of America stock was conducted in 2013 and found that the stock is not overfished and overfishing is not occurring ($SSF_{2011}/SSF_{MSY} = 1.01$; $F_{2011}/F_{MSY} = 0.57$) (SEDAR 2013). Both SSF_{2011}/SSF_{MSY} and F_{2011}/F_{MSY} values suggested that the Gulf of America stock had a more negative outlook compared to the Atlantic stock for Atlantic sharpnose sharks.

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Atlantic yellowfin tuna - *Thunnus albacares*Overall Vulnerability Rank = moderate ■ (91% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 92% of scores ≥ 2

<i>Thunnus albacares</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	2.9		Low Moderate High Very High
	Habitat Specificity	1.6	2.9		
	Mobility and Dispersal of Early Life Stages	2.1	2.2		
	Other Stressors	2.1	2.4		
	Population Growth Rate	2.2	2.3		
	Prey Specificity	1.5	2.9		
	Reproductive Cycle	1.6	2.6		
	Reproductive Strategy Sensitivity	2.1	2.2		
	Sensitivity to Ocean Acidification	2	1.8		
	Sensitivity to Temperature	2	2.7		
	Site Fidelity	1.6	2.2		
	Specificity in Early Life History Requirements	2.9	2.3		
	Stock Size Status	2.3	2.6		
	Sensitivity Score	low			
Exposure Factors	Mixed Layer Depth	1.3	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.7	-		
	Sea Surface Salinity	2.2	-		
	Sea Surface Temperature	3.8	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Atlantic yellowfin tuna (*Thunnus albacares*)

Contributors: J. Kneebone, T. Loughran, E. Orbesen, D. Snodgrass

Overall Climate Vulnerability Rank: **Moderate** (91% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.8), Sea Surface Oxygen (3.7), and pH (4.0). Yellowfin tuna are found near the surface of tropical and subtropical oceans occupying a wide range of temperatures (Hoolihan et al. 2014).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Specificity in Early Life History Requirements (2.9). Lethal thermal limits of larval yellowfin tuna were found to be less than 21°C and over 33°C, in addition to dissolved oxygen saturation of less than 34% (Cornic et al. 2018; Wexler et al. 2011).

Data Quality: 92% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Recent studies have described the impact of climate change to yellowfin tuna. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal shifts of yellowfin tuna throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Yellowfin tuna catch was estimated to have shifted poleward 7 ± 1 km for every 1°C increase, and an average of 2 ± 1 km/year for 2019 relative to 2002. Modeling efforts have shown similar poleward shifts. Erauskin-Extramiana et al. (2019) used species distribution models to examine the potential impact of a warming ocean on 22 different tuna stocks, and predicted that yellowfin tuna abundance will increase in most of their distributional area, particularly in their core tropical areas of the Atlantic. Yellowfin tuna distribution was also expected to shift southward and westward during the rest of the century. In contrast, Braun et al. 2023 found that yellowfin tuna may incur up to 20% annual habitat loss in the western North Atlantic and 35% habitat loss in the Gulf of America, primarily during winter months. The authors also estimated that the center of yellowfin tuna distribution will shift northeastward by >400 km in the western North Atlantic and around 40 km in the Gulf. A vertically expanding oxygen minimum zone may also limit the vertical distribution of other tropical tuna species and increase competition for prey resources, though this possibility remains to be verified (Dell’Apa et al. 2023; Stramma et al. 2012).

Yellowfin tuna larvae have a relatively high rate of natural mortality (0.33/d) due to starvation and predation (Lang et al. 1994, Pecoraro et al. 2017). These early life stages could become more vulnerable with climate variability as studies have shown a relationship between lower pH and organ damage in larval tuna resulting in decreased growth and survival (Frommel et al. 2016). Yellowfin tuna are limited to waters with dissolved oxygen concentrations of 3.5 ml/L, and low pH has been demonstrated to reduce the hemoglobin oxygen carrying capacity for yellowfin tuna (Graham and Dickson 2004), which means any reduction in ocean pH could reduce the habitat suitability especially along the eastern Atlantic. Additionally, lower oxygen availability in the Gulf of America may impact larval survival, further exacerbated by warmer sea surface temperatures (Dell’Apa et al. 2018; Dell’Apa et al. 2023; Wexler et

al. 2011). The average body size of yellowfin tuna is also projected to decrease by 15.1% by 2050 (Erauskin-Extramiana et al. 2023).

Life History Synopsis: Yellowfin tuna is a highly migratory species found across the tropical and subtropical Atlantic from 45°N to 40°S. Conventionally tagged individuals have exhibited transatlantic and transequatorial movements and it is currently assumed that yellowfin tuna in the Atlantic comprise one stock (SCRS 2023). However, there is some uncertainty of stock structure as a recent analysis using Single Nucleotide Polymorphism (SNPs) genotyping suggests that fish collected in the East and West Atlantic are genetically different (Pecoraro et al. 2018).

Yellowfin tuna spend the majority of their time in the epipelagic but occur over a broad range of temperatures and have been observed in water as warm as 32.1°C (Hoolihan et al. 2014). Adult yellowfin tuna are capable of short duration dives into the mesopelagic and bathypelagic, and can experience changes in temperature exceeding 24°C (Hoolihan et al. 2014, Orbesen unpublished data).

Yellowfin tuna are a batch spawner and form spawning aggregations (Schaefer 2001). In the West Atlantic, spawning has been reported to take place in the Gulf of America between May and August and in the southeastern Caribbean Sea between July and September (Arrocha et al. 2001, Pacicco et al. 2023). Spawning in the East Atlantic has been reported to occur from December through April. In the eastern Pacific, spawning occurs at night when temperatures are >24°C, primarily between 27°C and 29°C (Schaefer and Fuller 2022). The eggs develop rapidly and are highly temperature dependent with mean egg-stage durations ranging from 18 to 28 hours (Margulies et al. 2007). Juvenile yellowfin tuna undergo rapid growth that continues through adulthood reaching a maximum age of at least 18 years and over 200 cm fork length (Collette and Nauen 1983; Pacicco et al. 2021). Yellowfin tuna are opportunistic feeders that undergo an ontogenetic shift in diet with small juvenile fish being more restricted to the near surface mixed layer feeding on planktonic organisms, while larger fish (>45 cm) have improved endothermic capabilities allowing them to increase their vertical range and primarily feed on teleosts and mesopelagic species (Graham et al. 2007, Korsmeyer and Dewar 2001). Yellowfin tuna are reported to reach the spawning maturity threshold (L_{50}) at 2.36 years of age and 109.1 cm in the western Atlantic (Pacicco et al. 2023), and 124.6 cm in the eastern Atlantic (Diaha et al. 2016).

The last stock assessment on Atlantic yellowfin tuna was conducted in 2019 and found that the stock was not overfished or subject to overfishing ($B_{2018}/B_{MSY} = 1.17$ (0.75-1.62); $F_{2018}/F_{MSY} = 0.96$ (0.56-1.50)) (SCRS 2023). However there is some uncertainty in their status as it was also reported that there was a 24% probability of an overfished status, and a 43% probability of overfishing taking place (SCRS 2023).

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Basking shark - *Cetorhinus maximus*Overall Vulnerability Rank = moderate  (70% certainty)Biological Sensitivity = moderate Climate Exposure = high Distributional Vulnerability Rank = high  (100% certainty)

Directional Effect = negative (98% certainty)

Data Quality = 54% of scores ≥ 2

<i>Cetorhinus maximus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	3		Low Moderate High Very High
	Habitat Specificity	2.1	2.8		
	Mobility and Dispersal of Early Life Stages	1.4	1.6		
	Other Stressors	1.4	1.4		
	Population Growth Rate	3	2.2		
	Prey Specificity	2.4	3		
	Reproductive Cycle	1.4	2.3		
	Reproductive Strategy Sensitivity	1.7	1.6		
	Sensitivity to Ocean Acidification	3	1.4		
	Sensitivity to Temperature	1.8	2.4		
	Site Fidelity	2.5	1.5		
	Specificity in Early Life History Requirements	1	1.5		
	Stock Size Status	2.6	2.1		
	Sensitivity Score		moderate		
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.5	-		
	Sea Surface Salinity	2.6	-		
	Sea Surface Temperature	3.5	-		
	Exposure Score		high		
Overall Vulnerability Rank		moderate			

Basking shark (*Cetorhinus maximus*)

Contributors: T. Curtis, E. Gutierrez, E. Hoffmayer, T. Loughran, C. McCandless, G. Skomal

Overall Climate Vulnerability Rank: **Moderate** (70% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: pH (4.0), Sea Surface Oxygen (3.5) and Sea Surface Temperature (3.5). Basking sharks are found in subpolar to temperate seas globally, ranging from Newfoundland in the summer to the southeastern United States, Sargasso Sea, and Caribbean Sea to as far south as the southeast coast of Brazil in the winter months, occasionally venturing into the Gulf of America via the Loop Current (Braun et al. 2018; Hoffmayer et al. 2011; Kohler and Turner 2019; Skomal et al. 2009). The species occupies meso to bathypelagic depths during winter and epipelagic depths in the summer (Braun et al. 2018; Skomal et al. 2009).

Biological Sensitivity: **Moderate.** Three sensitivity attribute means scored above 2.5: Population Growth Rate (3.04), Sensitivity to Ocean Acidification (2.96), and Stock Size Status (2.6). Basking sharks are filter feeders, reliant on copepods and other small crustaceans which are susceptible to the impacts of ocean acidification (Castro 2011). Additionally, basking sharks have low population growth ($r=0.02$) and genetic diversity, which is partly mitigated by the rare trans-Atlantic migration by individuals (Johnston et al. 2019; Gore et al. 2008; Rus Hoelzel et al. 2006; Taylor 2022).

Data Quality: 54% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Basking sharks are sensitive to the effects of ocean acidification and the seasonality of ocean productivity due to their reliance on zooplankton as prey (Castro 2011; USGCRP 2018). Sea surface temperature and variations in the North Atlantic Oscillation and Gulf Stream were found to influence the abundance and distribution of zooplankton, directly contributing to the distribution and relative abundance of basking sharks (Cotton et al. 2005; Hoogenboom et al. 2015). Their low population growth and longevity (maximum age over 50 years) may exacerbate these stressors (Taylor 2022). For purposes of scoring, the body of scientific information available suggested that basking sharks were ectothermic. However, new literature has recently come available (Dolton et al. 2023) indicating that basking sharks may be regionally endothermic. While the timing of this publication precluded it from being included in the scoring and workshop process, it could be included in a future revision of this climate vulnerability assessment.

Life History Synopsis: Basking sharks are globally distributed throughout subpolar to temperate waters, occupying various depths depending on the season (Braun et al. 2018). Although movement data are poor for basking sharks, existing data does suggest this species can make large-scale movements (Braun et al. 2018; Doherty et al. 2017, Gore et al. 2008; Skomal et al. 2009). Such movement demonstrated strong seasonality with individuals moving to coastal shelf waters in higher latitudes as far north as Newfoundland during the summer and moving south to the southeastern North Atlantic, Gulf of America, Sargasso Sea, and Caribbean Sea to as far south as Brazil in the fall and winter (Braun et al. 2018; Hoffmayer et al. 2011; Kohler and Turner 2019; Skomal et al. 2009).

Basking sharks are filter feeders, using gill rakers to strain zooplankton through selective foraging on patches made up of copepods (Crowe et al. 2018; Sims and Merrett 2008). Seasonal fronts have been

found to influence basking shark presence; fronts play a significant role in forming zooplankton aggregations, which in turn attract basking sharks seeking feeding opportunities (Curtis et al. 2014; Miller et al. 2015; Sims et al. 2006). Large aggregations of basking sharks appear infrequently along the Northeastern United States and Canada coast (Crowe et al. 2018). Ten aggregations of 30 to almost 1,400 individuals appeared between 1980 and 2013 throughout summer and fall seasons (Crowe et al. 2018). While the cause of these aggregations is unknown, high concentrations of zooplankton were found throughout the aggregation areas (Crowe et al. 2018).

Little information is available regarding basking shark reproduction. Gestation is estimated as slightly over a year, ending with live birth to a litter of pups possibly 1.5 to 2 m long (Castro 2011). Basking sharks are long-lived and slow growing, reaching maturity at lengths of 8.1 to 9.8 m total length for females and 4.6 to 6.1 m total length for males (Castro 2011). The intrinsic rate of increase and von Bertalanffy growth coefficient for this species are 0.0197 and 0.062, respectively (Pauly 2002, Taylor 2022). Maximum age is unconfirmed, but estimated to be after individuals reach over 10 m total length (Pauly 2002). Natanson et al. (2008) determined the maximum number of band pairs for males and females were 30 and 27, respectively but the current evidence suggests that the basking shark deposit band pairs relative to structural morphology rather than time.

Landing of basking sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999). A Canadian assessment (DFO 2008), estimated population size to be approximately 10,000 individuals as of 2007, with no indications of declining relative abundance since 1979 in United States waters. The genetic diversity of the population is low, partially aided by infrequent trans-Atlantic migrations of individuals (Rus Hoelzel et al. 2006; Johnston et al. 2019). Basking sharks were targeted species for harpoon and net fisheries throughout the mid to late 1900s, with catches peaking and falling multiple times throughout the period (Rigby et al. 2021). This pattern may be indicative of cyclical stock depletion and recovery (Rigby et al. 2021). Basking shark population is estimated to be below 30% of historical abundance (Rigby et al. 2021).

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Bigeye sand tiger shark - *Odontaspis noronhai*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = moderate ☐Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☐ (99% certainty)

Directional Effect = neutral (70% certainty)

Data Quality = 0% of scores ≥ 2

<i>Odontaspis noronhai</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.7	0.6		Low Moderate High Very High
	Habitat Specificity	1.8	0.8		
	Mobility and Dispersal of Early Life Stages	1.4	0.4		
	Other Stressors	2	0.2		
	Population Growth Rate	2.9	0.5		
	Prey Specificity	1.9	0.8		
	Reproductive Cycle	1.6	0.2		
	Reproductive Strategy Sensitivity	1.9	0.2		
	Sensitivity to Ocean Acidification	1.3	0.4		
	Sensitivity to Temperature	1.9	1.2		
	Site Fidelity	2	0.2		
	Specificity in Early Life History Requirements	1	0.6		
	Stock Size Status	2.6	0.2		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Bigeye sand tiger shark (*Odontaspis noronhai*)

Contributors: T. Curtis, D. Kerstetter, T. Loughran, C. McCandless, D. Ortiz

Overall Climate Vulnerability Rank: **Not applicable.**

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (2.9) and Stock Size and Status (2.6). There is little information available regarding bigeye sand tiger stock status or population growth.

Data Quality: 0% of the data quality scores were 2 or greater. Little information is available regarding the species life history.

Climate Effects on Distribution and Abundance: No published studies have specifically examined the effect of climate factors on the population productivity or changes in distribution for bigeye sand tiger under climate change scenarios. Bigeye sand tiger sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: Very little is known about the life history of the bigeye sand tiger. Although the type specimen was caught off Madeira (Maul 1955), subsequent catch records suggest a broad global distribution in tropical and warm-temperate coastal and offshore waters (Ebert et al. 2013). Based on the limited number of catch records from commercial shallow-water fisheries, as well as its large eyes and dark coloration (Kerstetter and Taylor 2008), it is likely a diel vertically migrating species similar to bigeye thresher, spending day periods in the mesopelagic and rising to the lower epipelagic at night. An individual caught in the western Gulf of America had fish otoliths and a squid beak in its stomach, suggesting a generalist diet (Branstetter and McEachran 1986). Reproduction is likely similar to other mackerel sharks, with aplacental viviparity with oophagy (Dulvy and Reynolds 1997). Compagno (2001) described male maturity at 2.2 to 3.2 m total length and female maturity at *ca.* 3.2 m. No information is available regarding growth rates or population status (Castro 2011).

Landing of bigeye sand tiger sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Bigeye sixgill shark - *Hexanchus nakamurai*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (92% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 0% of scores ≥ 2

<i>Hexanchus nakamurai</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	2	1		Low Moderate High Very High
	Habitat Specificity	1.8	1.3		
	Mobility and Dispersal of Early Life Stages	1.4	1.1		
	Other Stressors	1.6	0.6		
	Population Growth Rate	3.4	1.1		
	Prey Specificity	1.7	1.4		
	Reproductive Cycle	1.4	1.5		
	Reproductive Strategy Sensitivity	1.7	0.3		
	Sensitivity to Ocean Acidification	1.4	0.7		
	Sensitivity to Temperature	1.6	1.4		
	Site Fidelity	2.2	0.5		
	Specificity in Early Life History Requirements	1	0.9		
	Stock Size Status	2.3	0.4		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Bigeye sixgill shark (*Hexanchus nakamurai*)

Contributors: T. Curtis, T. Daly-Engel, D. Grubbs, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable.**

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.4). There is little information available regarding population growth rate of bigeye sixgill sharks, apart from size at maturity: 123 to 157 cm total length (TL) for males, 142-178 cm TL for females (Barnett et al. 2012).

Data Quality: 0% of the data quality scores were 2 or greater, indicating low data quality across all sensitivity attributes.

Climate Effects on Distribution and Abundance: Bigeye sixgill depend heavily on the mesopelagic ocean zone from 100 to 700 m depth to feed on prey in the deep scattering layer, undertaking daily vertical migrations in the water column (Ebert et al. 2016). Though populations are currently stable, climate modeling studies have shown that the mesopelagic is likely to be the first oceanic zone to experience severe deoxygenation, which will negatively affect the food web that sustains bigeye sixgill (Gong et al. 2021). Brooks et al. (2015) demonstrated that the depth range of bigeye sixgill is larger than previously stated, with individuals occurring as deep as 701.5 m in the Bahamas versus 500 m (Castro 2011). The variance in vertical distribution of bigeye sixgill sharks may be driven by temperature, though further research is needed. Because of its limited range, dependence on vertical movement, and ectothermic nature, bigeye sixgill may shift deeper to avoid local warming, making it vulnerable to climate effects like deoxygenation.

Life History Synopsis: Recent literature suggests that bigeye sixgill in the Atlantic, currently listed as *H. nakamurai* in the Integrated Taxonomic Information System, is distinct from other *H. nakamurai* populations in the Pacific and Indian Oceans. Daly-Engel et al. (2019) recommends resurrecting the name *H. vitulus*, originally described in Springer and Waller (1969), for bigeye sixgill populations found in the western North Atlantic accompanied by the common name Atlantic sixgill. Bigeye sixgill is the smaller of the two members of genus *Hexanchus* in the western North Atlantic, reaching a observed maximum <180 cm TL compared with >480 cm for sixgill shark (*H. griseus*) (Castro 2011; Daly-Engel et al. 2019; Teng 1962). This species is known to occur in the western North Atlantic, including the Gulf of America and Caribbean, up to a maximum depth of 700 m (Brooks et al. 2015, Daly-Engel et al. 2019; Ebert et al. 2016; Teng 1962). Like the better-studied sixgill (*H. griseus*), bigeye sixgill undergoes diel vertical migrations, staying near the bottom during the day and moving up at night to feed on small prey of the mesopelagic layer (Compagno et al. 2005).

Daly-Engel et al. (2019) suggests that *H. vitulus* (the Atlantic sixgill) may be restricted to the western Atlantic, and that the *H. nakamurai* (bigeye sixgill) may be restricted to the eastern Atlantic. This finding highlights the low historical connectivity and dispersal reflected in this species. Genetic diversity has not been well-studied but is likely to be low in this species, which could impede rebound rates following depletion (Martin et al. 1992).

Intrinsic rates of increase and reproduction are low in bigeye sixgill, similar to many deep-water elasmobranchs (Kyne and Simpfendorfer 2010). This is likely due to energy allocation that maximizes reproductive output in a light-poor environment where resources are scarce, and the low metabolic rates of elasmobranchs in general and deep-sea species in particular (Kyne and Simpfendorfer 2010; Martin and Palumbi 1993). This species is ovoviviparous, giving birth to 13 to 26 young that average 40 cm TL (Ebert et al. 2016). Neonates with visible yolk bellies have been caught in the northern Gulf of America (Castro 2011 and R.D. Grubbs, pers. comm.)

This species has not been aged due to its uncalcified vertebrae (Castro 2011). Greenland sharks (*Somniosus microcephalus*), which belong to an ancestral genus with many physiological similarities to *Hexanchus*, have been suggested to live for >200 years (Nielsen et al. 2016); however, caution should be used when assuming a positive size/age relationship when correlating the carbon-14 derived ages to the measured fish lengths as age is often decoupled from maximum length (Natanson et al. 2018). Bigeye sixgill likely do not have the same extreme lifespan estimated for the Greenland shark (Greenland shark primary habitat is colder, and bigeye sixgill are much smaller in size), but its environment and metabolism suggest this species is likely to live longer than many other elasmobranchs (Cotton and Grubbs 2015; Martin and Palumbi 1993).

Exact population sizes of bigeye sixgill are unknown, but most of its range is currently unfished, it is encountered regularly in the Caribbean and Gulf of America, and was considered stable as recently as 2019 (Cotton et al. 2020; Gulak and Carlson 2013). Bigeye sixgill is known as occasional bycatch in trawl and bottom longline fisheries, though quantitative bycatch data for some fisheries are missing, and pressure is likely to increase as commercial fishing fleets expand into deeper water (Cotton et al. 2020; Cotton and Grubbs 2015; NFMS 2021). Landing of bigeye sixgill sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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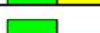
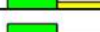
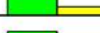
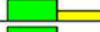
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Bigeye thresher shark - *Alopias superciliosus*Overall Vulnerability Rank = moderate  (99% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = very high  (82% certainty)

Directional Effect = neutral (70% certainty)

Data Quality = 23% of scores ≥ 2

<i>Alopias superciliosus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	1.8		Low Moderate High Very High
	Habitat Specificity	1.4	2.1		
	Mobility and Dispersal of Early Life Stages	1.1	2.3		
	Other Stressors	1.2	1.1		
	Population Growth Rate	3.9	1.8		
	Prey Specificity	1.5	2		
	Reproductive Cycle	1.1	1.5		
	Reproductive Strategy Sensitivity	1.6	1.2		
	Sensitivity to Ocean Acidification	1.3	1.2		
	Sensitivity to Temperature	1.3	1.8		
	Site Fidelity	1.3	1		
	Specificity in Early Life History Requirements	1	1.4		
	Stock Size Status	2	0.9		
	Sensitivity Score		low		
Exposure Factors	Mixed Layer Depth	1.5	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.3	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.7	-		
	Sea Surface Salinity	2.5	-		
	Sea Surface Temperature	3.7	-		
	Exposure Score		very high		
Overall Vulnerability Rank		moderate			

Bigeye thresher shark (*Alopias superciliosus*)

Contributors: T. Curtis, E. Gutierrez, D. Kerstetter, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (23% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: pH (4.0), Sea Surface Oxygen (3.7), and Sea Surface Temperature (3.7). The bigeye thresher is a pelagic shark, found throughout pelagic and neritic waters across the world (Coelho et al. 2015).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.8). There is little information available regarding the population growth rate of bigeye thresher sharks, though von Bertalanffy growth coefficient estimates were predicted as 0.092 and 0.088 for females and males, respectively (Liu et al., 1998).

Data Quality: 23% of the data quality scores were 2 or greater. The stock size and status and site fidelity attributes were scored the lowest in terms of data quality. There is no available information regarding stock size or status. Annual summer aggregations of bigeye thresher shark have been sighted off Cape Hatteras, North Carolina, though there is no supporting literature as to the cause of these aggregations (Castro 2011).

Climate Effects on Distribution and Abundance: No published studies have specifically examined the effect of climate factors on the population productivity or changes in distribution for bigeye thresher under climate change scenarios, although modeling by Hazen et al. (2012) and Braun et al. 2023 found negative changes to generalized “shark” pelagic habitat under such scenarios. Crear et al. (2023) identified temperature-influenced spatiotemporal shifts in *Alopias vulpinus* catches, so the bigeye thresher could experience comparable changes in distribution.

Bigeye thresher has the red myotomal musculature seen in other thresher species, but the arrangement pattern and lack of counter-current retes for metabolic heat conservation suggests that the bigeye thresher does not have systemic endothermy (Sepulveda et al. 2005). However, the bigeye thresher does have such retes around its brain and eyes to facilitate feeding in a broad range of temperature regimes seen during diel vertical migrations. Due to diel movements and physiological adaptations, as well as the high vagility of the species, temperature regime shifts under climate change are unlikely to affect bigeye thresher (Young et al. 2016). It is unclear how climate change will affect bigeye thresher abundance.

Life History Synopsis: The bigeye thresher is a relatively large thresher shark, reaching 488 cm total length (TL) and 363 kg (IGFA 2023), although lengths around 300 cm TL are more common (Cervigon et al. 1992). It is broadly distributed in tropical and subtropical waters from approximately 46°N to 40°S and 180°W to 180°E (FAO-FIGIS 2005) and is encountered in pelagic fisheries from coastal shelf margins and island seamounts to the high seas (Compagno 1984, Compagno 2001, Mundy 2005). Depth distribution is from 0 to 730 m (Nakano et al. 2003, McMillan et al. 2011), with diel differences seen in electronically tagged bigeye threshers in the eastern Tropical Pacific (Nakano et al. 2003, Kohin et al. 2006), Hawaii (Weng and Block 2004), the northern Gulf of America (Weng and Block 2004), tropical

Atlantic around the St. Peter and St. Paul Archipelago (Coelho et al. 2015), and southern California in the eastern Pacific (Sepulveda et al. 2019).

The vertical habitat overlap with the effective fishing depths of pelagic longline gear, especially when configured to fish at shallow depths during the night for targeting swordfish, enhances vulnerability of bigeye thresher as bycatch (Coelho et al. 2015). A study using electronic pop-up archival tags found a 86% rate of survival for 30 days following release from deep-set buoy gear in southern California (Sepulveda et al. 2019).

Reproduction is similar to other mackerel sharks, with aplacental viviparity with oophagy (Dulvy and Reynolds 1997). Fecundity is low, with two pups per litter, although up to four have been reported (Moreno and Morón 1992, Chen et al. 1997, Compagno 2001). Based on the presence of near-term fetuses, the area around the Straits of Gibraltar has been proposed as a potential nursery area (Moreno & Morón 1992). The presence of neonates (<80 cm PCL) captured mainly in the area between 10 and 15°N and 150 to 180 W south of the Main Hawaiian Islands also suggests a nursery ground (Matsunaga & Yokawa 2013). Captures of pregnant females also suggest nursery areas in the tropical East Atlantic and southwest Atlantic (Fernandez-Carvalho et al. 2015).

The species has a broad diet ranging from pelagic fishes and squids (Compagno 2001, Mundy 2005) to demersal fishes (Compagno 2001) using its long caudal fin to stun prey similar to other thresher sharks. A stomach content study of bigeye threshers captured in the southern California driftnet fishery found a very diverse diet including pelagic and midwater fishes, as well as some invertebrates, consistent with an opportunistic feeding predator ranging between several different water strata (Preti et al. 2008). In another stomach content study, bigeye threshers were found to be feeding on cephalopods off Ecuador up to 51% by diet weight (Galván-Magaña et al. 2013). Bigeye thresher were also found to be feeding on large fishes in waters in the Adaman Sea off Thailand (Chumchuen and Sukramongkol 2022).

Maximum lifespan for bigeye thresher is estimated at 20 years (Liu et al. 1998). Estimated growth coefficients (Atlantic: Fernandez-Carvalho et al. 2011; Pacific: Liu et al. 1998) indicate a slow-growing species with slow population growth rate (Young et al. 2016). Bigeye threshers also have a late age of maturity (12 years for females and 10 for males), and the gestation period has been estimated at 12 months (Liu et al. 1998).

Landing of bigeye thresher sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999). However, analysis of observer data from the pelagic longline fishery from 1992-2013 indicates the relative abundance in the western North Atlantic is stable (Young et al 2016).

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Bignose shark - *Carcharhinus altimus*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (84% certainty)

Directional Effect = neutral (89% certainty)

Data Quality = 0% of scores ≥ 2

<i>Carcharhinus altimus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	1.6		Low Moderate High Very High
	Habitat Specificity	1.6	1.5		
	Mobility and Dispersal of Early Life Stages	1.4	1.1		
	Other Stressors	1.8	0.5		
	Population Growth Rate	3.3	1.4		
	Prey Specificity	1.9	1.2		
	Reproductive Cycle	1.5	1.8		
	Reproductive Strategy Sensitivity	1.8	0.4		
	Sensitivity to Ocean Acidification	1.2	0.8		
	Sensitivity to Temperature	1.6	1.8		
	Site Fidelity	2	0.3		
	Specificity in Early Life History Requirements	1	0.6		
	Stock Size Status	2.4	0.6		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Bignose shark (*Carcharhinus altimus*)

Contributors: L. Crawford, D. Grubbs, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.3). Very little is known about the population growth of the bignose shark. The species is large-bodied and long lived (23 years), and has a late age at maturity (17 years) (Rigby et al. 2020). These life history characteristics may limit the population growth rate of the bignose shark.

Data Quality: None of the data quality scores were 2 or greater. The sensitivity attributes that scored below 1.0 were: other stressors (0.5), reproductive strategy sensitivity (0.4), sensitivity to ocean acidification (0.8), site fidelity (0.3), specificity in early life history requirements (0.6), and stock size status (0.6).

Climate Effects on Distribution and Abundance: Limited information is available regarding the effects of climate factors on bignose sharks. Diaz-Carbadillo et al. 2022 used ecological niche modeling (MaxEnt) and four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) to evaluate potential distributions of numerous carcharhinid species. Using RCP8.5, available habitat is projected to be maintained with loss along coastal states, particularly Florida. Additionally, bignose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021). It is unclear how climate change will affect bignose abundance.

Life History Synopsis: Bignose sharks are found globally throughout temperate to tropical waters, generally in 8 to 18.67°C (NMFS SEFSC, unpublished data; Rigby et al. 2020). Throughout the southwest Atlantic, they are distributed between Delaware Bay and Florida, the Gulf of America, and the Caribbean Sea (Kohler and Turner 2019). Seasonal latitudinal migrations occur annually between the extent of their range in the spring and southern extent in the fall (Kohler and Turner 2019). Bignose sharks are largely bottom dwelling but may migrate diurnally for foraging opportunities, residing near the seafloor during daylight hours and migrate vertically to surface waters at night (Anderson and Stevens 1996; Rigby et al. 2020).

Bignose sharks are yolk-sac placental viviparous and likely have a biennial reproductive cycle (Ebert et al. 2013; Rigby et al. 2020). Parturition occurs in summer resulting in litters of 3 to 15 pups measuring 60 to 90 cm (Ebert et al. 2013; Rigby et al. 2020). Female age at maturity is estimated at 17 years, and maximum age of the species is estimated to be 23 years (Jensen et al. 1996; Rigby et al. 2020). The generalist diet of bignose sharks may include chimaeras, smaller sharks, squids, and octopuses (Castro 2011).

Little information is available regarding population growth or status of bignose shark in the western North Atlantic. A possible 20 to 29% population reduction in the western North Atlantic over the course

of 60 years may have occurred due to fishing pressure (Pillans et al. 2009). Landing of bignose sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Blacknose shark - Atlantic stock (*Carcharhinus acronotus*)Overall Vulnerability Rank = moderate ■ (85% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (99% certainty)

Directional Effect = neutral (96% certainty)

Data Quality = 46% of scores ≥ 2

<i>Carcharhinus acronotus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	2		Low Moderate High Very High
	Habitat Specificity	1.8	2.7		
	Mobility and Dispersal of Early Life Stages	1.4	1.8		
	Other Stressors	1.9	1.8		
	Population Growth Rate	2.4	2.6		
	Prey Specificity	1.6	2.3		
	Reproductive Cycle	1.1	2.3		
	Reproductive Strategy Sensitivity	1.4	1.7		
	Sensitivity to Ocean Acidification	1.2	0.8		
	Sensitivity to Temperature	1.7	1.8		
	Site Fidelity	1.5	2.4		
	Specificity in Early Life History Requirements	1	1.8		
	Stock Size Status	2.8	2.3		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3.5	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1.3	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Blacknose shark - Atlantic stock (*Carcharhinus acronotus*)

Contributors: J. Carlson, W. Driggers, B. Frazier, T. Loughran, C. McCandless, C. Peterson

Overall Climate Vulnerability Rank: **Moderate** (85% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors contributed to this score: Sea Surface Oxygen (3.9), Bottom Temperature (3.9), Bottom Salinity (3.5), and pH (4.0). Atlantic blacknose sharks are present throughout the year in warm coastal waters ranging off the United States Atlantic Coast from Virginia through Florida (Castro 2011; Kohler and Turner 2019). They are primarily found outside of inlets, off Atlantic beaches, or further offshore over sand, shell, and rock bottoms, but can also be found in estuarine waters over mud bottoms, especially in the southern part of their range (McCandless pers comm; McCandless et al. 2007). This species' distribution is generally driven by salinity with a preference for more saline waters, but Atlantic blacknose sharks have been documented in salinities ranging from 16.9 to 38.6 psu, temperatures from 15.8 to 31.3°C, dissolved oxygen from 3.2 to 10.6 mg/L, pH from 6.5 to 8.5, and depths from 1.0 to 22m (C. McCandless pers comm; McCandless et al 2007; NMFS SEFSC, unpublished data).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Stock Size and Status (2.8). The relative biomass level of Atlantic blacknose is $SSF_{2009}/SSF_{MSY} = 0.43-0.64$ and the stock is overfished, and overfishing is occurring (SEDAR 2011).

Data Quality: 46% of the data quality scores were 2 or greater. One sensitivity attribute received a score below 1: Sensitivity to Ocean Acidification (0.8). Little information is available regarding Atlantic blacknose shark's sensitivity to ocean acidification, outside of having a non-calcium carbonate dependent diet.

Climate Effects on Distribution and Abundance: One study investigating the effect of climate factors on Atlantic blacknose is available at this time. Diaz-Carballido et al. (2021) evaluated potential shifts in distribution for carcharhinid sharks using ecological niche modeling (MaxEnt) and four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5). Under RCP8.5, Atlantic blacknose lost 15.9% of its global distribution and gained 5.6%, equalling a shift of -10.3%. The most noticeable loss was habitat throughout the northern Gulf of America, mid Atlantic, and the coasts of Florida and Cuba whereas the majority of habitat was maintained throughout the eastern Caribbean Sea and South American coast. The use of nearshore habitats may increase the vulnerability of Atlantic blacknose to rising sea surface temperatures. There is a possibility that negative effects to abundance could stem from the disruption of nearshore habitat, particularly as they relate to salinity (e.g., precipitation). Additionally, blacknose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: The blacknose shark is a relatively small (max. length approx. 101 cm fork length) coastal shark species (SEDAR 2011). During the last assessment it was determined that two stocks exist, one from the mid- to south Atlantic Coast of the United States and the other in the Gulf of America (SEDAR 2011). Adults are highly mobile and seasonally migrate throughout their range; a subset of adult

individuals reside off the coast of Florida throughout the year (Bangley and Rulifson 2017; Kohler and Turner 2019; Peterson and Grubbs 2020; Williams et al. 2019). A portion of the Atlantic stock migrates north during spring and summer months to nearshore waters, primarily off of North and South Carolina, and then migrates south off the east coast of Florida in early fall (Kohler and Turner, 2019). Use of Virginia's nearshore waters by adults has increased during the summer months over the last decade with two even documented within the estuarine waters of Virginia's Eastern Shore in the last five years (C. McCandless pers comm). Atlantic blacknose are generally found in coastal waters year round but may migrate farther offshore in winter (Driggers et al. 2010).

Female Atlantic blacknose are viviparous and in the Atlantic, have been found to have both annual and biennial reproductive cycles (J. Gelsleichter pers comm), giving birth to litters of 1 to 5 pups in June after a gestation period of 10 months (Driggers et al. 2004; Driggers et al. 2010; SEDAR 2011). In the Atlantic, blacknose sharks are found in higher salinity (rarely found in salinities less than 31 ppt) (Ulrich et al. 2007). Ontogenetic shifts in diet have not been recorded for Atlantic blacknose although the diet of young of the year blacknose sharks has not been studied. The species displays a generalist and opportunistic diet of available teleosts including mehaden, anchovies, whiting, tonguefish, mullet, and bluefish (Castro 2011).

Atlantic blacknose reach age at 50% maturity at 4.5 years and 4.3 years for females and males, respectively (Driggers et al. 2010). Von Bertalanffy growth coefficients are estimated as 0.30 for females and 0.36 for males, and the species reaches a maximum age of 20.5 years (SEDAR 2011, Frazier et al. 2015). The last stock assessment on the Atlantic stock of blacknose shark was conducted in 2010 and found that the stock is overfished and overfishing is occurring ($SSF_{2009}/SSF_{MSY} = 0.43-0.64$; $F_{2009}/F_{MSY} = 3.26-22.53$) (SEDAR 2011).

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Blacknose shark - Gulf of America stock (*Carcharhinus acronotus*)Overall Vulnerability Rank = moderate ■ (98% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (98% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 54% of scores ≥ 2

<i>Carcharhinus acronotus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	1.8		Low Moderate High Very High
	Habitat Specificity	1.8	2.6		
	Mobility and Dispersal of Early Life Stages	1.4	2		
	Other Stressors	1.8	1.8		
	Population Growth Rate	2.2	2.6		
	Prey Specificity	1.5	2.4		
	Reproductive Cycle	1.1	2.4		
	Reproductive Strategy Sensitivity	1.4	1.8		
	Sensitivity to Ocean Acidification	1.2	0.8		
	Sensitivity to Temperature	1.6	2		
	Site Fidelity	1.6	1.6		
	Specificity in Early Life History Requirements	1	1.4		
	Stock Size Status	2.7	2.1		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	2.9	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Blacknose shark - Gulf of America stock (*Carcharinus acronotus*)

Contributors: J. Carlson, W. Driggers, M. Drymon, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (98% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors contributed to this score: Sea Surface Oxygen (3.9), Bottom Temperature (3.9), and pH (4.0). Blacknose sharks are present throughout the year in the warm coastal waters of the Gulf of America, generally found in areas with sand, silt and clay, and seagrass bottoms (McCandless et al. 2007). This species' distribution is generally driven by salinity with a preference for more saline waters, but Gulf of America blacknose sharks have been documented in salinities ranging from 17 to 38 psu, temperatures from 17 to 34°C, dissolved oxygen from 2.0 to 8.7 mg/L, and depths of 0.6 to 30 m (J. Carlson pers comm; Castro 2011; McCandless et al. 2007; NMFS SEFSC, unpublished data).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Stock Size and Status (2.7). A stock assessment was conducted for blacknose shark Gulf of America stock in 2010, however due to high levels of uncertainty stock status could not be determined (SEDAR 2011). However, indices of abundance from the Gulf of America analyzed using a dynamic factor model suggests a decline in blacknose shark abundance within the Gulf from the 1990s (Peterson et al. 2017).

Data Quality: 54% of the data quality scores were 2 or greater. One sensitivity attribute received a score below 1: Sensitivity to Ocean Acidification (0.8). Little information is available regarding blacknose shark's sensitivity to ocean acidification, outside of not having a calcium carbonate dependent diet.

Climate Effects on Distribution and Abundance: One study investigating the effect of climate factors on blacknose is available at this time. Diaz-Carballido et al. (2021) used ecological niche modeling (MaxEnt) under four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) to evaluate potential shifts in distribution for carcharhinid sharks. Under RCP8.5, Atlantic blacknose lost 15.9% of its global distribution and gained 5.6%, equalling a shift of -10.3%. The most noticeable loss was habitat throughout the northern Gulf of America, mid Atlantic, and the coasts of Florida and Cuba whereas the majority of habitat was maintained throughout the eastern Caribbean Sea and South American coast. The use of nearshore habitats may increase the vulnerability of blacknose to rising sea surface temperatures, and may also increase the risk of exposure to harmful algal blooms, which have been found to impact other small coastal shark species (Flewelling et al. 2010). There is a possibility that negative effects to abundance could stem from the disruption of nearshore habitat, particularly as they relate to salinity (e.g., precipitation). Additionally, blacknose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: The blacknose shark is a relatively small (max. length approx. 101 cm fork length) coastal shark species (SEDAR 2011). During the last assessment it was determined that split into two stocks exist one from found throughout the Mid to South Atlantic and the other in the Gulf of America (SEDAR 2011). While the two stocks share many life history characteristics, there are marked differences

between the reproductive cycle and growth of blacknose sharks in the Atlantic and Gulf of America. Adults are highly mobile and show evidence of sexual segregation in distribution throughout the northern Gulf of America, with males showing more seasonality in their distribution and were partially influenced by salinity, favoring conditions of 30 to 33 psu (Drymon et al. 2020).

Blacknose sharks in the Gulf of America are viviparous and reproduce annually, giving birth to litters of 1 to 5 pups in June after a gestation period of 10 months (Driggers et al. 2004; Driggers et al. 2010; SEDAR, 2011; Sulikowski et al. 2007). This is a notable difference from the Atlantic stock of the same species which reproduces biennially (Driggers et al. 2010). Juveniles are found in nearshore and estuarine habitat characterized by live to sandy bottom, in temperatures of 17 to 33.6°C, salinities of 17 to 38 psu, and dissolved oxygen from 0.85 to 8.7 mg/L (J. Carlson pers comm; McCandless et al. 2007; Mullins et al. 2021; Peterson and Grubbs 2020). Ontogenetic shifts in diet have not been recorded for blacknose sharks. The species displays an opportunistic diet of available teleosts including mehaden, anchovies, whiting, tonguefish, mullet, and bluefish (Castro 2011).

Blacknose sharks reach age at 50% maturity at 6.63 years and 5.4 years for females and males, respectively (Driggers et al. 2010). Von Bertalanffy growth coefficients are estimated as 0.30 for females and 0.36 for males, and both sexes reach a maximum age of 20.5 years (SEDAR 2011). A stock assessment is not available for the Gulf of America stock of blacknose sharks due high uncertainty in the assessment model (SEDAR 2011).

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Blacktip shark - Atlantic stock (*Carcharhinus limbatus*)Overall Vulnerability Rank = moderate ■ (100% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (96% certainty)

Data Quality = 92% of scores ≥ 2

<i>Carcharhinus limbatus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.3	2.4		Low Moderate High Very High
	Habitat Specificity	1.8	2.8		
	Mobility and Dispersal of Early Life Stages	1.6	2.7		
	Other Stressors	1.7	2.4		
	Population Growth Rate	2.4	2.8		
	Prey Specificity	1.6	3		
	Reproductive Cycle	1.2	3		
	Reproductive Strategy Sensitivity	1.6	2.4		
	Sensitivity to Ocean Acidification	1.2	1.2		
	Sensitivity to Temperature	1.7	2.9		
	Site Fidelity	1.8	2		
	Specificity in Early Life History Requirements	1	2.3		
	Stock Size Status	2	2.7		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3	-		
	Bottom Temperature	3.7	-		
	Mixed Layer Depth	1.3	-		
	pH	4	-		
	Primary Production	1.3	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.7	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Blacktip shark - Atlantic stock (*Carcharhinus limbatus*)

Contributors: C. Bangley, D. Crear, B. Frazier, E. Gutierrez, T. Loughran, C. McCandless, C. Peterson

Overall Climate Vulnerability Rank: **Moderate** (85% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score for Atlantic blacktip: Bottom Temperature (3.7), pH (4.0) and Sea Surface Oxygen (3.7). Both juveniles and adults primarily reside over continental shelves, inhabiting various depths (13.35 to 66.75 m) within coastal to pelagic waters (Bethea et al. 2015; Ward-Paige et al. 2015; Matich et al. 2022; NMFS SEFSC, unpublished data).

Biological Sensitivity: **Low.** All sensitivity score means were scored below 2.5.

Data Quality: 92% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that blacktip sharks will gain suitable habitat across the northward extent of their range in the western North Atlantic. In United States waters, the Atlantic blacktip stock may gain habitat around New England (Diaz-Carballido et al. 2022). These results are further supported by Bowers and Kajiura (2023), which documented shifts in the distribution and migration patterns of blacktip sharks in the Atlantic (East Coast of the United States) using acoustic telemetry and satellite telemetry. These shifts have been attributed to warmer water temperatures (Bowers and Kajiura 2023). Blacktip sharks were observed over the multi-year study to shift their winter range northward, as well as migrating further north than previously observed. Additionally, blacktip sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Neonates are reliant on coastal waters and estuarine areas for nursery grounds and to avoid predators, therefore there is a possibility that negative effects to abundance could stem from the disruption of this habitat (Matich et al. 2022). Blacktip sharks may experience hypoxic conditions in summer months in nearshore waters occupied by younger life stages. These life stages were found to be vulnerable to hypoxic conditions and unable to sustainably use active ventilation or rest under low oxygen conditions (Andres 2022).

Life History Synopsis: Blacktip sharks are a common large coastal shark found from New England, south through the Florida Keys, into the Gulf of America and Caribbean Sea, occurring as far south as Brazil. In the Atlantic (East Coast of the United States), they were considered a rare stray north of Cape Hatteras prior to the 1990s; however they have been found in abundance as far north as New York in recent years (Castro 2011; Bowers and Kajiura 2023). Atlantic blacktip sharks make seasonal migrations and are known to display sexual segregation with males migrating further north into cooler waters than females (Bowers and Kajiura 2023). Both sexes overwinter off of Florida, often forming large schools found close to shore (Kajiura and Tellman 2016). Blacktip sharks in the Atlantic are generally found in salinities near 35 psu and temperatures between 17 and 28°C (NMFS SEFSC, unpublished data).

Blacktip sharks are a placental, viviparous species with a biennial reproductive periodicity and a gestation period of one year (SEDAR 2020). Male blacktip sharks take 5.3 years (115.15 cm fork length) and females 6.7 years (123.05 cm FL) to reach 50% maturity, and average brood size is 4.09 pups (range 1 to 7, Natanson et al. 2019). Maximum estimated longevity is 17.5 years (Deacy and Moncreif-Cox 2019). Northern seasonal migrations appear to be based on water temperature cues; females give birth in estuarine and sound waters of Georgia and South Carolina in May and June, with mating occurring concurrent to parturition. After mating, most males migrate further north, while females remain off of the Carolinas. Blacktip sharks of all life stages appear to be generalist feeders and do undergo an ontogenetic shift in diet, with adult blacktips consuming smaller elasmobranch prey in addition to teleost fishes consumed by young of year and juvenile blacktip sharks (Castro 1996).

Atlantic blacktips are considered a distinct stock, with tag and recapture, telemetry, and genetic results indicating there is no mixing between Atlantic, and Gulf of America (eastern and western) populations (Kohler and Turner 2001; Swift et al. 2023, Bowers and Kajiura 2023). The last stock assessment on the Atlantic stock of blacktip shark was conducted in 2020 and found that the stock was not overfished and not subject to overfishing ($SSF_{2018}/SSF_{MSY} = 1.16$; $F_{2018}/F_{MSY} = 0.51$) (SEDAR 2020).

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Blacktip shark - Gulf of America stock (*Carcharhinus limbatus*)Overall Vulnerability Rank = moderate ■ (50% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 92% of scores ≥ 2

<i>Carcharhinus limbatus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	Low Moderate High Very High
Sensitivity Attributes	Adult Mobility	1.5	2.8	</	

Blacktip shark - Gulf of America stock (*Carcharhinus limbatus*)

Contributors: J. Carlson, L. Crawford, W. Driggers, M. Drymon, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (50% certainty from bootstrap analysis).

Climate Exposure: **Very High** Four exposure factors contributed to this score: Bottom salinity (3.1), Bottom Temperature (3.8), pH (4), and Sea Surface Oxygen (3.9). Gulf of America blacktip sharks are found in tropical and subtropical waters and are abundant in the Gulf. Both juveniles and adults tend to inhabit various depths (9 to 102 m) within coastal to pelagic waters (Bethea et al. 2015; Ward-Paige et al. 2015; Matich et al. 2022; NMFS SEFSC, unpublished data).

Biological Sensitivity: **Low.** All sensitivity score means were scored below 2.5.

Data Quality: 92% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that blacktip sharks will gain suitable habitat across the northward extent of their range in the western North Atlantic. While the Atlantic stock of blacktip sharks may gain habitat around New England, the Gulf of America stock may lose habitat in the eastern Gulf of America along the coast of Florida under the least and most extreme scenarios (Diaz-Carbadillo et al. 2022). Additionally, blacktip sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Neonates are reliant on coastal waters and estuarine areas for nursery grounds and to avoid predators, therefore there is a possibility that negative effects to abundance could stem from the disruption of this habitat (Matich et al. 2022). Blacktip sharks may experience hypoxic conditions in summer months in nearshore waters occupied by younger life stages. These life stages were found to be vulnerable to hypoxic conditions and unable to sustainably use active ventilation or rest under low oxygen conditions (Andres 2022).

Life History Synopsis: The Gulf of America stock of blacktip sharks are found throughout the Gulf. Blacktip sharks utilize deeper waters during daytime, and feed in shallower waters at night (Diaz-Carballido et al. 2022; Pickens et al. 2022). Adults migrate south during fall months, reaching offshore waters or coastal waters off of Florida. Adult females prefer inshore habitat, while suitable habitat for males extends across the continental shelf (Drymon et al. 2020; Kohler and Turner 2019). Blacktip sharks migrate from overwintering grounds in the southern Gulf of America to mating and nursery grounds in the northern Gulf. The species is known to form migratory aggregations (Castro 2011).

Blacktip sharks are a viviparous species with a biennial reproductive cycle (Baremore and Passerotti 2013). Gestation lasts approximately one year, with an average litter size of 4.5 pups (38 cm fork length (FL)) (Baremore and Passerotti 2013). Mating occurs in May and June, and birth occurs the following May and June (Baremore and Passerotti 2013). Neonates are reliant on coastal waters and estuarine

areas for nursery grounds and are generally found in salinities between 16.6 to 38 psu and temperatures of 20.8 to 32.6°C (J. Carlson pers comm; Matich et al. 2022). Juveniles are found in salinities between 16.6 to 38 psu and temperatures of 19.2 to 32.9°C (J. Carlson pers comm).

Juvenile Gulf of America blacktip shark diet primarily consists of Gulf menhaden and Atlantic croaker (Bethea et al. 2004; Matich et al. 2021a). Adult blacktip sharks feed primarily on small schooling fish species, and to a lesser extent bluefish, ladyfish, hardhead catfish, mullet, grunts and stingrays (Castro 2011, Matich et al. 2021b).

Blacktip sharks are reported to be a relatively short lived species compared to most large coastal sharks: the maximum age is estimated as 12.5 years and 11.5 years for females and males respectively (SEDAR 2012), however, recent aging efforts found 15 to 18 year old blacktip sharks in the Gulf (Wells and Drymon, unpublished data). Males reach 50% maturity at 4.8 years (105.8 cm FL) and females 6.3 years (119.2 cm FL) (Baremore and Passerotti 2013). The growth rate of the Gulf of America blacktip shark is 0.16 for females and 0.21 for males (Passerotti and Baremore 2012).

The last stock assessment on the Gulf of America stock of blacktip shark was conducted in 2018 and found that the stock was not overfished or subject to overfishing ($SSF_{2016}/SSF_{MSY} = 2.73$; $F_{2016}/F_{MSY} = 0.023$) (SCRS 2018). Catch estimates for Gulf of America blacktip sharks peaked in the late 1980s and have steadily declined with interannual variability. Abundance declined slowly between the 1980s and the mid-1990s and reached an inflection point in the mid-1990s and has been slowly increasing since (SEDAR 2018).

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Blue marlin - *Makaira nigricans*

Overall Vulnerability Rank = high (86% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (100% certainty)

Directional Effect = neutral (94% certainty)

Data Quality = 46% of scores ≥ 2

<i>Makaira nigricans</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	2.8		Low Moderate High Very High
	Habitat Specificity	1.5	2.9		
	Mobility and Dispersal of Early Life Stages	2.1	1.6		
	Other Stressors	1.8	1.9		
	Population Growth Rate	2.6	1.5		
	Prey Specificity	1.6	2.1		
	Reproductive Cycle	2.2	2.2		
	Reproductive Strategy Sensitivity	1.8	1.2		
	Sensitivity to Ocean Acidification	1.8	1.2		
	Sensitivity to Temperature	2.5	2.6		
	Site Fidelity	1.9	1.7		
	Specificity in Early Life History Requirements	2.1	1.2		
	Stock Size Status	3.1	2.6		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.5	-		
	Sea Surface Salinity	2.3	-		
	Sea Surface Temperature	3.6	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Blue marlin (*Makaira nigricans*)

Contributors: J. Graves, R. Logan, T. Loughran, E. Orbesen, D. Snodgrass

Overall Climate Vulnerability Rank: **High** (86% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Sea Surface Oxygen (3.5), pH (4.0) and Sea Surface Temperature (3.6). Blue marlin have pelagic egg and larval stages that are spent in surface waters, and adults spend the majority of time in the upper 25 m of surface waters with a maximum diving depth of over 800m (Goodyear et al. 2006).

Biological Sensitivity: **Moderate.** Three sensitivity attribute means scored above 2.5: Population Growth Rate (2.6), Stock Size and Status (3.1), and Sensitivity to Temperature (2.5). Blue marlin are found in epipelagic waters with surface temperatures between 22 and 31°C, generally in the tropics within the 24°C isotherm (Arocha and Ortiz 2006; Rooker et al. 2019). Blue marlin are overfished, and overfishing is occurring (SCRS 2023).

Data Quality: 46% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Blue marlin are primarily associated with the surface mixed layer but are capable of diving over 800m where temperatures are below 10°C (Goodyear et al. 2006). While blue marlin have a high thermal tolerance, they are primarily found in waters with sea surface temperatures between 19° and 30°C, and blue marlin habitat suitability is highly correlated with sea surface temperatures (Kraus and Rooker 2007; Goodyear et al. 2008), and temperatures during dives generally remain within 8°C of surface temperatures (Goodyear et al. 2006). Dale et al. (2022) reported that during a 16-year period (2000-2016) 96% of the core habitat for blue marlin declined in suitability, which was accompanied by a poleward shift in suitability. These findings are in contrast to Braun et al. (2023), which projected minor (<20%) increases in habitat suitability for blue marlin in the Gulf of America and western North Atlantic. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal and temporal shifts of blue marlin throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Since 2002, blue marlin catch was estimated to have shifted poleward 7 ± 2 km for every 1°C increase, and an average of 1 ± 1 km/year. Timing of the first 25% of blue marlin catch was estimated to have shifted 27 days earlier off New York for 2019 relative to 2002.

In the eastern Atlantic, blue marlin have exhibited a restriction in vertical movement due to limited oxygen at depth, and a continued expansion in oxygen minimum zones (dissolved oxygen ≤ 3.5 ml/l) could further restrict their vertical movements making them more vulnerable to fishing gear (Stramma et al. 2012, Prince et al. 2010). Elevated sea surface water temperatures may also lead to blue marlin occupying deeper depths in order to follow forage opportunities, particularly in the Gulf of America (Dell'Apa et al. 2018; Dell'Apa et al. 2023). It is unclear how climate change will affect blue marlin abundance, however, if the rate of post-release recovery is altered due to increasing temperatures or oxygen availability, recreational fisheries releasing blue marlin may contribute to population decline (Logan et al. 2022).

Life History Synopsis: Blue marlin are a highly prized game fish targeted by recreational fisherman in a mostly catch and release fishery, and by some small scale artisanal fisheries. However, the greatest source of fishing mortality results from bycatch in commercial tuna and swordfish pelagic longline fisheries internationally. Blue marlin are widely distributed throughout the Atlantic from 45°N to 35°S, and are capable of large scale movements. Of all the conventionally tagged HMS species, the blue marlin has exhibited the longest movement, from the northeastern coast of the United States, to Mauritius in the Indian Ocean (Orbesen et al. 2008). A recent study also observed trans-Atlantic and trans-equatorial movements for blue marlin electronically tagged in the eastern and western Atlantic (Andrzejczek et al. 2023). These tagging studies, coupled with a genetic analysis of stock structure (McDowell et al. 2007), support the current International Commission for the Conservation of Atlantic Tunas' management assumption of one Atlantic-wide stock.

There is a high level of uncertainty associated with the biology and early life history of blue marlin including age and growth and mortality estimates (Drew et al. 2006). Spawning has been reported throughout the western North Atlantic including the Yucatan (June-August), Caribbean (Bahamas, Puerto Rico, Virgin Islands, and Dominican Republic, May-November), the Florida Straits (May-September), and Bermuda (July) (Erdman 1968, Richardson et al. 2009, Prince et al. 2005, Luckhurst et al. 2006, Serafy et al. 2003, and Brown-Peterson et al. 2008). In the South Atlantic, blue marlin have been reported to spawn in January and February between 17 and 23°S (Amorim et al. 1998). Larval stages of blue marlin feed on plankton, including those within sargassum beds, which is also used for sheltering purposes (Wells and Rooker 2009). Blue marlin diet consists of squid and pelagic fish species (Gempylidae, Scombridae, Exocoetidae, Alepisauridae) (Satoh et al. 2004).

Blue marlin exhibit rapid growth and development reaching 24cm lower jaw fork length in 40 days, and could weigh as much as 47 kg by age 1 (Prince et al. 1991). Blue marlin exhibit sexual dimorphism beginning at approximately 140 cm with females reaching sizes exceeding 680.4 kg while males do not exceed 158.8 kg (Prince et al. 1991). Blue marlin are estimated to reach maturity between 2 and 4 years of age (De Sylva and Breder 1997). Aging of blue marlin can be difficult due to the small size of their otoliths, and the vascularization of dorsal and anal spines exhibited in older individuals. Maximum ages have been estimated to be anywhere between 20 and 30 years (Hill et al. 1989) to as high as 43.5 years (Hoolihan et al. 2019).

The last stock assessment on blue marlin was conducted in 2018 and found that the stock is overfished and subject to overfishing ($SSB_{2016}/SSB_{MSY} = 0.69 (0.52-0.91)$; $F_{2016}/F_{MSY} = 1.03 (0.74-1.50)$) (SCRS 2023). Recommendation 19-05 of the International Commission for the Conservation for Atlantic tunas states that blue marlin shall be released live when possible (ICCAT 2019). The current landing limit is 1,670 t, below the recommended 1,750 t that provides at least a 50% chance of rebuilding by 2028 (SCRS 2023).

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Bonnethead shark - Atlantic stock (*Sphyrna tiburo*)Overall Vulnerability Rank = very high ■ (96% certainty)Biological Sensitivity = high ■Climate Exposure = very high ■Distributional Vulnerability Rank = moderate ■ (65% certainty)

Directional Effect = neutral (44% certainty)

Data Quality = 77% of scores ≥ 2

<i>Sphyrna tiburo</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	2.1	2.3		Low Moderate High Very High
	Habitat Specificity	2.2	2.7		
	Mobility and Dispersal of Early Life Stages	1.5	2		
	Other Stressors	2.1	2		
	Population Growth Rate	2.6	2.3		
	Prey Specificity	3.4	2.8		
	Reproductive Cycle	1.5	2.7		
	Reproductive Strategy Sensitivity	1.8	2.3		
	Sensitivity to Ocean Acidification	2.2	1.7		
	Sensitivity to Temperature	2.2	2.3		
	Site Fidelity	3.2	2.2		
	Specificity in Early Life History Requirements	1.1	2.1		
	Stock Size Status	2	1.3		
	Sensitivity Score	high			
Exposure Factors	Bottom Salinity	3.8	-		
	Bottom Temperature	4	-		
	Mixed Layer Depth	1.3	-		
	pH	4	-		
	Primary Production	1.3	-		
	Precipitation	1	-		
	Sea Surface Oxygen	4	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		very high			

Bonnethead shark - Atlantic stock (*Sphyrna tiburo*)

Contributors: J. Carlson, B. Frazier, B. Keller, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Very High** (96% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors contributed to this score: Bottom Temperature (4.0), Bottom Salinity (3.8), Sea Surface Oxygen (4), and pH (4.0). Atlantic bonnethead sharks are found in warm, coastal waters in the southeastern United States, beginning south of Chesapeake Bay. Habitat characteristics vary from nearshore to estuarine waters over sand, shell, and mud bottoms, with temperatures between 15.1 to 33.3°C, salinities between 105 and 38.6 psu, dissolved oxygen between 1.4 and 9.2 mg/L, pH between 6.7 and 8.6, and depths less than 13.1m (C. McCandless pers comm; McCandless et al. 2007; Ulrich et al. 2007).

Biological Sensitivity: **High.** Two sensitivity attribute means scored above 3.0: Prey Specificity (3.4) and Site Fidelity (3.2). Atlantic bonnethead sharks are dietary specialists, primarily feeding on blue crab and other invertebrates such as shrimp, stone crabs, and cephalopods (Branham et al. 2022; Kroetz et al. 2017). There are ongoing studies that demonstrate strong site fidelity of Atlantic bonnethead sharks throughout the southeastern United States Atlantic coast (Driggers et al. 2014; B. Keller unpublished data).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Limited information is available about the potential effects of climate on the distribution of bonnethead sharks. To date, range shifts have not been documented in this species. Climatic migrations have been observed in bonnethead sharks; individuals were observed to migrate further south during cold winters versus warm winters (Keller 2020). Outmigration was based on photoperiod; however, migration to spring and summer habitat was based on sea surface temperature. Over the duration of the Keller 2020 study (7+ years), monitored individuals returned to the same river each year displaying strong site fidelity (Keller 2020; Driggers et al. 2014). It is unclear whether bonnethead sharks will eventually move further north with warming temperatures as additional habitat should become available. An upper thermal tolerance for this species has not been identified; therefore, they may continue to utilize the same estuaries under moderate increases in water temperature, despite availability of new habitat. Additionally, bonnethead sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Similar to the Gulf of America stock of bonnethead sharks, the Atlantic stock of bonnethead sharks may be vulnerable to anthropogenic activities that degrade estuarine habitats and overharvesting of their primary prey, blue crabs (Giltz and Taylor 2017; Waycott et al. 2009; Zhang et al. 2013). Bonnethead sharks are dependent upon shallow, coastal habitats and a crustacean-centric diet (Cortés et al. 1996; Bethea et al. 2007), exposing them to thermal and ocean acidification stressors throughout all life stages (Mullins et al. 2021; Ward-Paige et al. 2014). Blue crabs, a significant prey item of bonnethead sharks, are at particular risk of larval mortality under ocean acidification conditions (Giltz and Taylor 2017).

There is a possibility that negative effects to abundance could stem from the disruption of this habitat and specialist prey species.

Life History Synopsis: Atlantic bonnethead sharks (southeastern United States coastal population) are an estuarine dependent species primarily found south of the Chesapeake Bay, Virginia to the Florida Keys (Castro 2011). When waters warm, bonnethead sharks migrate north to estuaries along the United States Southeast Atlantic coast displaying strong site fidelity to these habitats with residency occurring from April to early November (Driggers et al. 2014; Keller 2020). The mechanism for this site fidelity is not understood, however research has shown bonnethead sharks may use the earth's magnetic field to migrate to their estuary of residence (Keller et al. 2021). It is believed these estuaries serve as summer foraging grounds, particularly for gestating females (Driggers et al. 2014).

Bonnethead sharks are a placental, viviparous species, with a gestation period of 4 to 5 months resulting in the birth of approximately 9 pups in the late summer/early fall (Gonzales de Acevado 2020). Mating occurs shortly after parturition, and sperm is stored until ovulation occurs in late spring. Parturition occurs in nearshore waters in mid- late September, with young of year found in nearshore waters off of Florida and Georgia. Bonnethead sharks take 4 to 6 years to mature with an estimated maximum longevity of 18 years (Frazier et al. 2014). Juveniles and adults undergo seasonal climatic migrations, with overwintering occurring off of Florida and Georgia (McCandless et al. 2007). Bonnethead sharks are considered dietary specialists, with portunids (specifically blue crabs, *Callinectes sapidus*) comprising most of their diet. Other prey items may include penaeid shrimp, stomatopods and teleost fishes (Cortés et al. 1996; Bethea et al. 2007; Branham et al. 2022).

Bonnethead sharks in the Atlantic are considered a distinct stock, with tagging and genetic studies indicating no mixing between stocks in the Gulf of America and the Atlantic (Kohler and Turner 2019; Díaz-Jaimes et al. 2021). Bonnethead shark stock status of the Atlantic population is currently considered "unknown". The stock was last assessed in 2013, with the assessment indicating that the species as one stock was not overfished, and no overfishing was occurring; however, because the genetics indicated the species should be split into two stocks (one in the Atlantic and one in the Gulf of America) the result could not be used to determine the status of each individual stock (SEDAR 2013a; SEDAR 2013b).

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Bonnethead shark - Gulf of America stock (*Sphyrna tiburo*)

Overall Vulnerability Rank = high (78% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (86% certainty)

Directional Effect = negative (83% certainty)

Data Quality = 85% of scores ≥ 2

<i>Sphyrna tiburo</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	3		Low Moderate High Very High
	Habitat Specificity	2.5	2.9		
	Mobility and Dispersal of Early Life Stages	1.8	2.3		
	Other Stressors	1.6	2		
	Population Growth Rate	2.4	2.8		
	Prey Specificity	3.2	3		
	Reproductive Cycle	1.9	3		
	Reproductive Strategy Sensitivity	2	2.3		
	Sensitivity to Ocean Acidification	2.5	1.5		
	Sensitivity to Temperature	1.6	2.6		
	Site Fidelity	2.9	2.8		
	Specificity in Early Life History Requirements	1.1	2.4		
	Stock Size Status	1.8	1.8		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	2.8	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Bonnethead shark - Gulf of America stock (*Sphyrna tiburo*)

Contributors: B. Campbell, J. Carlson, B. Frazier, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (78% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), Sea Surface Oxygen (3.8), and pH (4.0). Bonnethead sharks are found in warm, coastal waters throughout the Gulf of America. Habitat characteristics vary from seagrass beds to estuarine waters over sand, shell, and mud bottoms, with temperatures between 15.98 and 33.6°C, dissolved oxygen between 1.9-10.5 mg/L, depths less than 8.5m, and salinities between 14.4 to 41.7 psu (J. Carlson pers comm, Castro 2011; Crear et al. 2023; Hueter and Tyminski 2007; McCandless et al. 2007; Mullins et al. 2021; Steiner et al. 2007; Ward-Paige et al. 2014).

Biological Sensitivity: **Moderate.** Three sensitivity attribute means scored above 2.5: Prey Specificity (3.2), Sensitivity to Ocean Acidification (2.52), and Site Fidelity (2.88). Bonnethead sharks in the Gulf of America are dietary specialists, primarily feeding on blue crab and other invertebrates such as shrimp, stone crabs, and cephalopods (Bethea et al. 2007; Branham et al. 2022; Kroetz et al. 2017). This diet specialization of invertebrates may increase bonnethead sharks' vulnerability to the impacts of ocean acidification.

Data Quality: 85% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Limited information is available about the potential effects of climate on the distribution of bonnethead sharks. To date, no range shifts have been observed or documented in this species. Climatic migrations have been observed in bonnethead sharks in the Atlantic (southeastern United States coast) population; individuals were observed to migrate further south during cold winters versus warm winters (Driggers et al. 2014; Keller 2020). Outmigration was based on photoperiod; however, migration to spring and summer habitat was based on sea surface temperature. Movement patterns in the Gulf of America are poorly understood; several studies examined habitat use over discrete areas and found evidence of site fidelity (Heupel et al. 2006; Kroetz et al. 2015; Dawdy et al. 2022). In the Gulf, northward movement of bonnethead sharks is not possible, therefore any potential shifts in range may be into nearshore waters if an upper thermal tolerance exists. Additionally, bonnethead sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

The Gulf of America stock of bonnethead sharks may be vulnerable to anthropogenic activities that degrade estuarine habitats and overharvesting of their primary prey, blue crabs (Giltz and Taylor 2017; Waycott et al. 2009; Zhang et al. 2013). Bonnethead sharks are dependent upon shallow, coastal habitats and a crustacean-centric diet, exposing them to thermal and ocean acidification stressors throughout all life stages (Mullins et al. 2021; Ward-Paige et al. 2014). Blue crabs, a significant prey item of bonnethead sharks, are at particular risk of larval mortality under ocean acidification conditions (Giltz and Taylor 2017). Bonnethead sharks are also reliant on vulnerable seagrass habitats which are declining globally (Waycott et al. 2009). Bonnethead sharks utilize seagrass habitats for multiple purposes,

including foraging. Recent studies have shown that bonnethead sharks are able to digest and uptake nutrients from seagrass, reclassifying them as an omnivorous predator (Leigh et al. 2018). There is a possibility that negative effects to abundance could stem from the disruption of this habitat and specialist prey species.

Life History Synopsis: Bonnethead sharks are an estuarine dependent species commonly found in shallow coastal waters throughout the Gulf of America and Caribbean Sea (Castro 2011), with evidence of cryptic speciation occurring in the southern Gulf (Fields et al. 2016; Gonzales et al. 2019). Little is known about bonnethead movement ecology in the Gulf. Several studies looked at movement ecology in the eastern Gulf using acoustic telemetry and found seasonal site fidelity to study areas, however broader movement patterns outside of those estuaries weren't examined (Heupel et al. 2006; Kroetz et al. 2015; Dawdy et al. 2022). In the eastern Gulf, conventional tagging data suggests seasonal migrations occur. Bonnethead sharks captured in Tampa Bay made southerly movements to the Florida Keys in the winter (Tyminski et al. 2007); however, bonnethead sharks may be found year round in the Big Bend, Florida area (D. Grubbs pers comm). Movement data are particularly lacking for the western Gulf where only a preliminary study has been conducted and limited conventional tagging data exist (Hueter and Tyminski 2007; Tyminski et al. 2007; Kohler and Turner 2019).

Bonnethead sharks are a placental, viviparous species, with a gestation period of 4 to 5 months resulting in the birth of approximately 10 pups in the late summer (Lombardi-Carlson 2007; Parsons 1993). Mating occurs shortly after parturition, and sperm is stored until ovulation occurs in late spring (Gonzalez De Acevedo 2020). There is limited data that suggest parturition may occur in nearshore waters, with young of year found in nearshore waters off of Texas in the early fall (E. Hoffmayer pers comm). Bonnethead sharks take 3.5 to 5.3 years to mature with an estimated maximum longevity of 17.1 years (Frazier et al. 2023). Differences in growth and reproduction have been documented in the eastern and western Gulf of America, with bonnethead sharks in the eastern Gulf displaying latitudinal variation in growth, and generally growing faster and to smaller asymptotic lengths than those from the western Gulf; however, longevity is similar between the two populations (Frazier et al. 2023). In the Gulf of America, bonnethead sharks are considered dietary specialists, with portunids (specifically blue crabs, *Callinectes sapidus*) comprising most of their diet, other prey items include penaeid shrimp, stomatopods, teleost fish, and seagrasses (Bethea et al. 2007; Branham et al. 2022; Kroetz et al. 2017).

Bonnethead sharks in the Gulf of America are considered a distinct stock, with tagging and genetic studies indicating no mixing between stocks in the Gulf and the Atlantic (Kohler and Turner 2019; Díaz-Jaimes et al. 2021). Furthermore, unpublished data (A. Fields) suggest multiple stocks exist within the Gulf. Bonnethead shark stock status of the Gulf of America population is currently considered "unknown". The stock was last assessed in 2013, with the assessment indicating that the species as one stock was not overfished, and no overfishing was occurring; however, because the genetics indicated the species should be split into two stocks (one in the Atlantic and one in the Gulf of America) the result could not be used to determine the status of each individual stock (SEDAR 2013a; SEDAR 2013b).

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Bull shark - *Carcharhinus leucas*Overall Vulnerability Rank = moderate  (95% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = high  (100% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 77% of scores ≥ 2

<i>Carcharhinus leucas</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	 Low  Moderate  High  Very High
Sensitivity Attributes	Adult Mobility	1.1	2.8		
	Habitat Specificity	2	2.8		
	Mobility and Dispersal of Early Life Stages	1.4	2.5		
	Other Stressors	2.2	2.1		
	Population Growth Rate	3	2.4		
	Prey Specificity	1.1	2.9		
	Reproductive Cycle	1.2	2.8		
	Reproductive Strategy Sensitivity	1.9	1.6		
	Sensitivity to Ocean Acidification	1.2	1.1		
	Sensitivity to Temperature	2	2.6		
	Site Fidelity	1.8	2.8		
	Specificity in Early Life History Requirements	1	2.1		
	Stock Size Status	2.1	1.3		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3.1	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1.1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Bull shark (*Carcharhinus leucas*)

Contributors: C. Bangley, J. Carlson, T. Daly-Engel, E. Gutierrez, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (95% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), pH (4.0) and Sea Surface Oxygen (3.9). Bull sharks are found in cosmopolitan, tropical and subtropical waters, generally in temperatures of 18.81 to 34.1°C and 2.1 to 39.5 psu (J. Carlson pers comms; Kohler and Turner 2019; NMFS SEFSC, unpublished data).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Population Growth Rate (3.0). Cortes 2016 estimated bull shark intrinsic rate of increase as 0.031 to 0.135, though this estimate is specific to Gulf of America populations.

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Bull sharks are euryhaline habitat generalists, occupying a variety of freshwater and marine environments globally (Gausmann 2021). Bull shark distribution within the United States is concentrated within the mid to south Atlantic and Gulf of America. A study using ecological niche modeling (MaxEnt) and four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) recently evaluated changes in suitable habitat for several carcharhinid species, including bull sharks (Diaz-Carballido et al. 2022). Much of the current coastal distribution of bull sharks throughout the Gulf of America, south Atlantic, and Caribbean Sea is projected to be maintained with small gains in suitable habitat. Habitat loss is largely projected along the western coast of Florida, which is used heavily as nursery habitat by bull sharks (Heupel and Simpfendorfer 2011). Increased freshwater inputs and elevated sea surface temperatures were also found to influence local distribution of juvenile bull sharks along Australian coastlines, with some indication of similar freshwater preferences in the Atlantic, meaning that higher frequency and intensity of storm events and rising sea surface temperatures may lead to long-term changes in bull shark distribution (Curtis et al. 2011; Werry et al. 2018; Balaguru et al. 2016). Additionally, bull sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Given the modest gains in suitable habitat and highly migratory nature of bull sharks, distribution expansion may alleviate habitat loss due to unsuitable environmental conditions. Bangley et al. (2018) found evidence of a new bull shark nursery forming within Pamlico Sound, North Carolina, due to rising water temperatures and salinity. The ability of bull sharks to establish new nursery grounds in response to suitable environmental conditions, despite strong site fidelity and philopatry shown throughout global populations, provides evidence of possible resiliency of this species to losses in suitable habitat (Bangley et al. 2018).

Bull shark abundance may be impacted by habitat degradation due to their reliance on nearshore and estuarine nursery grounds (Curtis et al. 2013). Early life stages are also reliant on vulnerable seagrass habitats which are declining globally (Waycott et al. 2009). Immature bull sharks were found in abundance throughout freshwater creeks that had been degraded through anthropogenic development,

exposing them to elevated levels of contaminants (Curtis et al. 2011; Curtis et al. 2013; Johnson-Restrepo 2008; Johnson-Restrepo 2005). Bull sharks are also exposed to phycotoxins as a result of harmful algal blooms throughout nearshore habitats and nursery grounds (Edwards et al. 2023). The effect of contaminant bioaccumulation, habitat degradation, and harmful algal blooms to abundance is uncertain and requires further research (Curtis et al. 2013; Edwards et al. 2023).

Life History Synopsis: Bull sharks are distributed throughout tropical to subtropical coastal waters between Chesapeake Bay, Virginia and the southeastern United States, including the Caribbean Sea and Gulf of America (Kohler and Turner 2019). Bull sharks are found in depths of a few meters to over 100 m, but are often present in shallower waters; they can also tolerate a low salinity and be found in freshwater systems throughout their global distribution (Castro 2011; Carlson et al. 2010; Gausmann 2021; Livernois et al. 2021; NMFS SEFSC, unpublished data). Bull shark populations throughout the western Atlantic have demonstrated site fidelity to coastal waters, and seasonal latitudinal migrations (Altobelli et al. 2020; Hammerschlag et al. 2012; Kohler and Turner 2019; Rider et al. 2021). Migrations take place between the Gulf of America and Atlantic Coast of the United States (Carlson et al. 2010; Kohler and Turner 2019; Altobelli et al. 2020; Rider et al. 2021). Further research is needed regarding seasonal migrations between the southeastern United States and Caribbean Sea (Brunnschweiler et al. 2010). Philopatry of female bull sharks to nursery estuaries has been detected throughout the Gulf of America and southeastern United States via tagging and tracking (Sandoval Laurrabaquio-A et al. 2019), while evidence of male philopatry within the same range has been detected using genetics (Barry et al. 2022). Bull sharks are habitat generalists, utilizing habitats ranging from coastal waters to nearshore inlets, estuaries, freshwater rivers, and live to sandy and muddy bottoms (Castro 2011; Curtis et al. 2011).

Bull sharks are a viviparous species with a biennial reproductive cycle (Castro 2011). Parturition of 7 to 12 pups occurs in summer months (average 75 cm total length (TL)) (Castro 2011). Gestation is predicted to last 12 months (Castro 2011). Neonates utilize nearshore estuarine habitats as nursery grounds, occupying these areas from summer (birth) through fall (Curtis et al. 2011). Immature and juvenile stages rely on nursery habitats for the first 9 years of life, and adult sharks continue to return to nursery habitats and have displayed site fidelity to specific habitats within larger estuarine complexes (Curtis et al. 2011; Curtis et al. 2013). The Indian River Lagoon, Florida has been identified as one of the most important bull shark nurseries within the Atlantic Coast of the United States (Curtis et al. 2011; Edwards et al. 2022). Within the Indian River Lagoon, neonates were detected throughout shallow habitats characterized by temperatures of $30.4 \pm 1.8^{\circ}\text{C}$, salinities of $17.0 \pm 7.7\text{‰}$, dissolved oxygen of 4.8 ± 1.5 mg/L, and adequate prey availability (Curtis et al. 2011). Pamlico Sound, North Carolina, has also been designated as a bull shark nursery after rising water temperature and salinities created favorable conditions for neonates (Bangley et al. 2018). Bull sharks reach maturity at 15 to 17 (210 to 220 cm TL) and 15 years of age (225 cm TL) for males and females, respectively (Natanson et al. 2014). Ontogenetic diet shifts occur as immature stages leave estuarine and freshwater conditions to marine habitats (seagrass beds) (Belicka et al. 2012). Bull sharks increasingly prey on higher trophic level prey as they mature (Daly et al. 2013). Prey items include freshwater, estuarine, and marine teleosts, elasmobranchs, and mollusks (Castro 2011).

Bull sharks are characterized by slow growth and low productivity, similar to other shark species. The intrinsic rate of increase for bull sharks is estimated to be low (0.031 to 0.135/yr), and von Bertalanffy growth coefficient is calculated as 0.071 (Cortes 2016; Cruz-Martinez et al. 2004; Heupel and Simpfendorfer 2011). Maximum age is estimated as 25 years for males and 27 years for females (Natanson et al. 2014).

Populations of bull sharks throughout the Gulf of America and Atlantic have been reportedly increasing since management measures were implemented in 1993 (Carlson et al. 2012; Froeschke et al. 2012; Rigby et al. 2021). Threats to bull shark populations include habitat degradation throughout nearshore nursery grounds, accumulation of contaminants from anthropogenic development and harmful algal blooms, and fishing of other global populations (Curtis et al. 2011; Rigby et al. 2021). No stock assessment is available for bull sharks.

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Caribbean reef shark - *Carcharhinus perezii*

Overall Vulnerability Rank = very high ■ (100% certainty)

Biological Sensitivity = high ■

Climate Exposure = very high ■

Distributional Vulnerability Rank = high ■ (98% certainty)

Directional Effect = negative (100% certainty)

Data Quality = 69% of scores ≥ 2

<i>Carcharhinus perezii</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.7	2.5		■ Low ■ Moderate ■ High ■ Very High
	Habitat Specificity	3.5	3		
	Mobility and Dispersal of Early Life Stages	1.5	2.3		
	Other Stressors	2.6	1.8		
	Population Growth Rate	3	2.6		
	Prey Specificity	1.4	2.9		
	Reproductive Cycle	1.4	2.2		
	Reproductive Strategy Sensitivity	1.9	1.4		
	Sensitivity to Ocean Acidification	3.4	1.6		
	Sensitivity to Temperature	2	2.4		
	Site Fidelity	3.6	3		
	Specificity in Early Life History Requirements	1	2.2		
	Stock Size Status	2.8	1.8		
	Sensitivity Score	high			
Exposure Factors	Bottom Salinity	3	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1.2	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		very high			

Caribbean reef shark (*Carcharhinus perezii*)

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Overall Climate Vulnerability Rank: **Very High** (100% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), pH (4.0) and Sea Surface Oxygen (3.9). Caribbean reef sharks are a reef-associated species found throughout tropical waters in the southeastern United States, southern Gulf of America, Caribbean Sea, and South America (Castro 2011; Driggers et al. 2011; Stoffers et al. 2021). Caribbean reef sharks exhibit ontogenetic shifts in habitat use, moving from shallow reefs to deeper shelf break habitats as they approach maturity (Brooks et al. 2013; Casselberry et al. 2020; Chapman et al. 2007; Garla et al. 2006a; Pikitch et al. 2005).

Biological Sensitivity: **High.** Four sensitivity attribute means scored above 3.0: Habitat Specificity (3.48), Population Growth Rate (3.0), Sensitivity to Ocean Acidification (3.36), and Site Fidelity (3.64). All life stages of Caribbean reef sharks are dependent on coral reefs, utilizing shallow and deep habitats for foraging opportunities and shelter (Chapman et al. 2007; Pikitch et al. 2005). Certain populations of Caribbean reef sharks display high site fidelity to natal reefs year-round while others may move up to 50 km between isolated reef habitats across open water (Bond et al. 2012; Chapman et al. 2005; Gallagher et al. 2021). This sensitivity is further exacerbated by a low population growth rate, informed by an intrinsic rate of increase of 0.013 and late age of maturity (4 to 9 years) (Mejia-Falla et al. 2019; Talwar et al. 2022).

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Recent studies have highlighted the vulnerability of reef shark species to climate change and anthropogenic stressors, further exacerbated by population declines (Heupel et al. 2019; Simpfendorfer et al. 2023). Caribbean reef shark reliance on and site fidelity to coral reefs increases its sensitivity to ocean acidification and rising sea surface temperatures, both of which have contributed to coral habitat degradation throughout the Greater Caribbean (Gledhill et al. 2008; Johnson et al. 2019; Muñiz-Castillo et al. 2019). Habitat loss throughout the Gulf of America and Caribbean Sea was projected using ecological niche modeling (MaxEnt) and four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5). Modest increases in suitable habitat were observed in all other RCPs except RCP8.5. Given site fidelity to coral habitats throughout all life stages of Caribbean reef sharks, and their ectothermic nature, it is uncertain how this species would respond to shifts in core habitat (Dell’Apa et al. 2023; Diaz-Carbadillo et al. 2022; Heupel et al. 2019).

These sensitivities may be further exacerbated by anthropogenic activity throughout reef shark distributions. Simpfendorfer et al. (2023) evaluated global declines in resident reef shark species and found a 41.3% deficit in Caribbean reef shark presence throughout its distribution (31 of 75 reefs) and a $64.8\% \pm 42.0$ to 87.5% decline in Caribbean reef shark populations. The habitat specificity of Caribbean reef sharks, paired with the reliance on vulnerable habitats, low reproductive output, and fishing

pressure increases the vulnerability of this species to the effects of climate change (Simpfendorfer et al. 2023).

Life History Synopsis: Caribbean reef sharks are found throughout the southeastern United States, southern Gulf of America, and Caribbean Sea (Castro 2011; Driggers et al. 2011; Stoffers et al. 2021). Adults use an array of shallow, near-island, lagoon habitat and shallow to deep reef habitat, primarily near shelf breaks in clear waters. Caribbean reef sharks are found across a range of depths, from 2 to 3 m to 436 m during diving behavior (Shipley et al. 2017). Adults make frequent nighttime excursions to shallower waters for foraging purposes (Shipley et al. 2017; Shipley et al. 2018).

Ontogenetic shifts in habitat use are well studied within Caribbean reef shark life history (Brooks et al. 2013; Casselberry et al. 2020; Chapman et al. 2007; Garla et al. 2006a; Pikitch et al. 2005; Stoffers et al. 2021). Young of year and early juveniles are found in shallower depths than adults, staying nearshore in reef habitat above 30 m; juveniles (3 to 4 years) approaching maturity (5 years) increase use of deeper reefs, bordering shelf break habitats (Casselberry et al. 2020; Stoffers et al. 2021; Tavares, 2009). Adults occupy deep reef habitat primarily near shelf breaks, though still utilize an array of shallow, near-island, lagoon and reef habitat (Pikitch et al. 2005, Chapman et al. 2007; Casselberry et al. 2020; Shipley et al. 2018). Caribbean reef sharks display varying degrees of site fidelity to natal reefs, which may limit their proclivity to move between reefs (Bond et al. 2012; Chapman et al. 2005).

Reproduction likely follows a biennial cycle (Castro 2011). In the Fernando de Noronha archipelago in Brazil, parturition is thought to occur between February and April (Garla et al. 2006b; Rangel et al. 2022). An additional parturition site in Venezuela recorded a parturition period of August to September (Tavares 2009). Average litter sizes have been recorded as 4 pups (Talwar et al. 2022). Caribbean reef shark diet encompasses a range of teleost species, cephalopods, crustaceans, and batoids (Bond et al. 2018; Tavares 2009). Ontogenetic shifts in diet were not found across age classes, though increased predation on rays and other prey items as individuals grow is prevalent (Bond et al. 2018). A trophic position of 3 to 3.5 is estimated for Caribbean reef sharks (Bond et al. 2018).

Age at maturity and maximum age are estimated as 14.8 and 43.3 to 57.8 years, respectively (Talwar et al. 2022). The intrinsic rate of increase for Caribbean reef sharks is 0.013, and the von Bertalanffy growth coefficient is 0.06 (Maija-Falla et al. 2019; Talwar et al. 2022).

Landing of Caribbean reef sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999). However, Simpfendorfer et al. (2023) found a 41.3% deficit in Caribbean reef shark presence throughout its distribution and Bernard et al. (2017) found low genetic diversity throughout Caribbean reef sharks, with sparse connectivity between populations.

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Caribbean sharpnose shark - *Rhizoprionodon porosus*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (99% certainty)

Directional Effect = neutral (83% certainty)

Data Quality = 62% of scores ≥ 2

<i>Rhizoprionodon porosus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	Low Moderate High Very High
Sensitivity Attributes	Adult Mobility	1.2	2.2		
	Habitat Specificity	1.8	2.2		
	Mobility and Dispersal of Early Life Stages	1.6	1.4		
	Other Stressors	1.9	2		
	Population Growth Rate	2.1	2		
	Prey Specificity	1.4	3		
	Reproductive Cycle	1.4	2.4		
	Reproductive Strategy Sensitivity	1.5	1.9		
	Sensitivity to Ocean Acidification	1.4	1.6		
	Sensitivity to Temperature	1.6	2		
	Site Fidelity	1.8	1.2		
	Specificity in Early Life History Requirements	1.1	2.2		
	Stock Size Status	2.2	1		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Caribbean sharpnose shark (*Rhizoprionodon porosus*)

Contributors: T. Loughran, C. McCandless, D. Ortiz

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low**. No sensitivity attribute means scored above 2.5.

Data Quality: 62% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Limited information is available regarding the effects of climate factors on Caribbean sharpnose. Use of nearshore nursery grounds and vulnerable habitats, including reefs, may increase the risk of habitat degradation due to anthropogenic development and ocean acidification (Pikitch et al. 2005; Yokota and Lessa 2006). Nearshore distribution may also increase the risk of exposure to harmful algal blooms, which have been found to impact Atlantic sharpnose (*Rhizoprionodon terraenovae*) (Flewelling et al. 2010). Atlantic sharpnose have also been found to shift reproductive timing due to an 3.0°C increase in spring sea surface temperature over a 29-year period (Hoffmayer et al. 2010). It is uncertain whether Caribbean sharpnose would exhibit the same behavior under rising sea surface temperatures. Additionally, sharpnose sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: Caribbean sharpnose shark is a relatively small (max. length approx. 110 cm total length (TL)) coastal species with a distribution ranging further than previously thought, going beyond the Caribbean Sea to the Mid to South Atlantic and Gulf of America (Davis et al. 2018; UWI 2015). This species displays broad habitat specificity, often found in shallow, sandy, estuarine, and reef habitat to offshore waters at depths of 500 m (UWI 2015).

Similar to other *Rhizoprionodon* species, Caribbean sharpnose are viviparous with a gestation period of 10 to 11 months and give birth to 2 to 6 pups (31 to 39 cm TL at birth) in spring and early summer (UWI, 2015). Lagoons and other nearshore features, including Glover’s Reef in Belize and Caiçara do Norte in Brazil, are used as nursery grounds by early life stages (Pikitch et al. 2005; Yokota and Lessa 2006). Evidence of sexual segregation outside of peak copulation season (late winter to early summer) exists, though it is unclear if this is related to reproductive migrations of gravid females to nursery grounds (Mattos et al. 2001). Other species of *Rhizoprionodon* have displayed sexual segregation outside of peak copulation times with females occurring in deeper water, and males aggregating in nearshore shallow waters (Castro and Wourms 1993; Drymon et al. 2010; Mattos et al. 2001; Parsons and Hoffmayer 2005). Unlike other species of *Rhizoprionodon*, Caribbean sharpnose sharks are opportunistic predators that do not exhibit an ontogenetic shift in their diet (Shibuya and de Souza, 2019). Diet composition depends on what is available in the environment, but may include teleosts, molluscs, cephalopods, and crustaceans (UWI, 2015).

Caribbean sharpnose display slower growth than other *Rhizoprionodon* species, with von Bertalanffy growth coefficient of 0.31 for females and 0.41 for males (Corrso et al. 2020). Age of maturity of approximately 2 to 3 years and maximum age is 5 to 8 years for males and females, respectively (UWI 2015; Yokota and Lessa 2006). Natural mortality is 0.107 for females and 0.151 for males (de Mattos and Maynou 2009).

Landing of Caribbean sharpnose sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999). Stock status is highly uncertain given mixing with congener Atlantic sharpnose. Caribbean sharpnose is estimated to have a population reduction of 30 to 49% over the past two decades primarily outside United States waters (Carlson et al. 2021).

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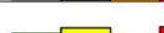
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Common thresher shark (*Alopias vulpinus*)Overall Vulnerability Rank = moderate  (96% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = high  (82% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 69% of scores ≥ 2

<i>Alopias vulpinus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.5	2.3		Low Moderate High Very High
	Habitat Specificity	1.6	2.5		
	Mobility and Dispersal of Early Life Stages	1.3	2.7		
	Other Stressors	1.6	1.8		
	Population Growth Rate	3.3	2.6		
	Prey Specificity	1.6	2.7		
	Reproductive Cycle	1.2	2.2		
	Reproductive Strategy Sensitivity	1.8	2.2		
	Sensitivity to Ocean Acidification	1.4	0.9		
	Sensitivity to Temperature	1.4	2.4		
	Site Fidelity	1.8	1.2		
	Specificity in Early Life History Requirements	1.1	2.2		
	Stock Size Status	2.2	0.7		
	Sensitivity Score		low		
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.7	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.7	-		
	Sea Surface Salinity	2.7	-		
	Sea Surface Temperature	3.5	-		
	Exposure Score		very high		
Overall Vulnerability Rank		moderate			

Common thresher shark (*Alopias vulpinus*)

Contributors: T. Curtis, E. Gutierrez, J. Kneebone, T. Loughran, C. McCandless, G. Skomal

Overall Climate Vulnerability Rank: **Moderate** (96% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: pH (4.0), Sea Surface Oxygen (3.7) and Sea Surface Temperature (3.5). Common thresher sharks are found in coastal, temperate waters, on continental shelves and epipelagic waters ranging from 6 to 31°C (Kneebone et al. 2020). They are less adaptable to hypoxia than congener species (Heberer et al. 2010; Wootton et al. 2015; Morrison et al. 2023).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Population Growth Rate (3.3). The intrinsic rate of population increase in the northwest Atlantic has been best-estimated at $r_{\max} = 0.077$ (Cortes 2016).

Data Quality: 69% of the data quality scores were 2 or greater. The sensitivity to ocean acidification and stock size and status attributes received scores below 1, as there is little information regarding the current relative biomass level or effect of ocean acidification on common thresher sharks available.

Climate Effects on Distribution and Abundance: There is minimal information on the effects of climate on common thresher shark abundance or distribution. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Common thresher catch was found to shift poleward 5 km/year, and 33 km for every 1°C increase between 2002 and 2019. Given this trend, it is possible that the species' suitable habitat will decrease in the Gulf of America and western North Atlantic by the end of the century; as has been forecast for shortfin mako (e.g., Braun et al. 2023), a species with a similar distribution and physiology. Temperature is an important influence on seasonal distribution patterns (Kneebone et al. 2020).

Common thresher sharks are regionally endothermic (Bernal and Sepulveda 2005; Sepulveda et al. 2005). The species can regulate its body temperature due to its high metabolic rate and a system of blood vessels called *rete mirabile* that function as a countercurrent heat exchange system to maintain heat within tissues and prevent cooling of the blood through the gills. The advantages of this physiological adaptation when encountering elevated ocean temperatures due to climate change remains uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Climate impacts on preferred prey species, particularly Atlantic menhaden (*Brevoortia tyrannus*), a species whose population dynamics are strongly influenced by climate effects (Buchheister et al. 2016), may also affect common thresher shark distribution along the coast of North America. It is unclear how climate change will affect common thresher shark abundance.

Life History Synopsis: The common thresher shark is distributed throughout the northwest Atlantic from the Gulf of America north to the Grand Banks of Canada (Compagno 1984; Kneebone et al. 2020). It has been documented to occupy depths from the surface to 2000 m and water temperatures from 1.2 to 25.6 C, but spends the majority of time in depths 10 to 75 m and temperatures 14 to 22°C (J. Kneebone,

unpublished data). Common thresher sharks are highly migratory throughout their life, making seasonal north-south migrations along the east coast of North America ranging from Florida to Canada (J. Kneebone, unpublished data). Inshore-offshore migrations are also evident in sub-adult and adult life stages, particularly during the winter when offshore movements in relation to the northern edge of the Gulf Stream are common (J. Kneebone, unpublished data). Young juveniles predominantly occur in coastal waters <50 m in depth; deeper depths are utilized more as individuals increase in size (Kneebone et al. 2020).

Common thresher sharks feed primarily on schooling fishes and cephalopods such as menhaden, herring, Atlantic mackerel, Chub mackerel, Atlantic saury, bluefish, sand lance, bonito, butterfish, and squid (Branstetter 2002). In the western North Atlantic, common thresher sharks have a longevity of around 38 years (Natanson et al. 2016) with 50% maturity achieved around 8 years in males and 13 years in females (Gervelis and Natanson 2013; Natanson et al. 2016). Mature females produce 2 to 4 pups (average of 3.7) bi- or triennially, with a 1 or 2 year resting period between pregnancies (Natanson and Gervelis 2013).

The population status of common thresher sharks in the northwest Atlantic is unknown, with evidence of substantial decline from 1987 to 2013 based on pelagic longline logbook data (Lynch et al. 2018). However, increasing CPUE was found in recreational sportfishing tournaments off the Northeast United States from 1978 to 2014 and a stable trend in relative abundance from 1992 to 2013 based on observer data from the pelagic longline fishery (Young et al. 2016). The species remains a target of both commercial and recreational fisheries in United States Atlantic waters and is currently the only species that can be legally harvested in the pelagic sharks quota of the HMS Fishery Management Plan. It is also captured throughout the western North Atlantic by high seas International Commission for the Conservation of Atlantic Tunas fisheries (Young et al. 2016).

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Dusky shark - *Carcharhinus obscurus*Overall Vulnerability Rank = very high ■ (99% certainty)Biological Sensitivity = high ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = negative (80% certainty)

Data Quality = 62% of scores ≥ 2

<i>Carcharhinus obscurus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.7		Low Moderate High Very High
	Habitat Specificity	1.9	2.8		
	Mobility and Dispersal of Early Life Stages	1.3	2.7		
	Other Stressors	1.7	1.6		
	Population Growth Rate	3.6	2.8		
	Prey Specificity	1.2	2.6		
	Reproductive Cycle	1.6	2.7		
	Reproductive Strategy Sensitivity	1.6	1.6		
	Sensitivity to Ocean Acidification	1.2	1.3		
	Sensitivity to Temperature	1.4	2.4		
	Site Fidelity	1.7	1.8		
	Specificity in Early Life History Requirements	1	1.7		
	Stock Size Status	3.4	2.9		
	Sensitivity Score	high			
Exposure Factors	Bottom Salinity	3.2	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		very high			

Dusky shark (*Carcharhinus obscurus*)

Contributors: C. Bangley, D. Crear, E. Hoffmayer, T. Loughran, H. Marshall, C. McCandless

Overall Climate Vulnerability Rank: **Very High** (99% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), pH (4.0), and Sea Surface Oxygen (3.9). Dusky sharks are found in depths up to 400 m, spending the majority of time in 20 to 140 m waters ranging in temperatures between 14 and 30°C (McCandless et al. 2014; NMFS SEFSC, unpublished data).

Biological Sensitivity: **High.** Two sensitivity attribute means scored above 3.0: Population Growth Rate (3.6) and Stock Size and Status (3.4). Dusky shark has a low intrinsic rate of increase, estimated at 0.028/yr (Cortes 1998), and the stock is currently overfished and overfishing is occurring (SEDAR 2016).

Data Quality: 62% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Dusky sharks are largely vulnerable due to their stock status of overfished and overfishing, further exacerbated by a low population growth rate and slow maturity (Romine et al. 2009). Several studies have described the distribution and abundance of dusky sharks throughout the western North Atlantic Ocean (Compagno 1984, Hoffmayer et al. 2014, Bangley et al. 2020); however, only a few have addressed the topic of climate change. Hare et al. 2016 conducted a vulnerability assessment of climate responses in fishery management-relevant species across the Northeastern United States, and determined that dusky sharks are among the most likely of evaluated species to shift distribution in response to climate change. Other temperate mid-Atlantic shark species have already shown distributional shifts poleward in response to changing thermal conditions (Kleisner et al. 2016; Pinsky et al. 2013; Bangley et al. 2018). Bangley et al. (2020) suggested that if thermal and productivity patterns are altered by climate change, then dusky sharks were likely to shift their distributions poleward in response. Additionally, dusky sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that dusky sharks are projected to lose a small amount of suitable habitat (3.4 to 9.5%) compared to 24 other Carcharhinid species that inhabit Mexican waters (Diaz-Carballido et al. 2022). The largest reduction in its western Atlantic distribution occurred in the eastern Gulf of America along the west Florida shelf and secondarily in nearshore waters along the Mid-Atlantic Bight under RCP2.6 and RCP8.5 (Diaz-Carballido et al. 2022). It is unclear how climate change will affect dusky shark abundance.

Life History Synopsis: The dusky shark is a large coastal-pelagic shark species found in subtropical continental shelf waters of the United States Atlantic coast from southern Massachusetts to Florida, including the Bahamas and the Gulf of America (Bigelow and Schroeder 1948, Castro 2011). Juvenile dusky sharks generally avoid low salinities but have been found in shallow estuarine areas along the southeastern United States coast (Musick et al. 1993, Castro 1993, Bangley et al. 2020). Adults are highly

migratory and occupy habitats from the surf zone out to depths of 400 m (Compagno 1984; Hoffmayer et al. 2014). Galapagos (*Carcharhinus galapagensis*) and dusky sharks are closely related and morphologically cryptic, but are confirmed as separate species with no evidence of hybridization (Corrigan et al. 2017).

The dusky shark demonstrates seasonal north-south movements along the Atlantic coast of the United States (Compagno 1984, Kohler et al. 1998, Kohler and Turner 2019). Mark recapture data show a maximum straight-line distance traveled of 3800 km, with a mean distance traveled of 1060 km (Kohler and Turner 2019). Movements from the United States Atlantic coast to the Gulf of America was common, demonstrating the ability for this species to make long distance movements (Kohler and Turner 2019). Satellite tag data revealed that larger juvenile and subadult dusky sharks utilized offshore continental shelf waters of the United States Atlantic Coast and spent the majority of their time between 0 and 40m depths and between temperatures of 20 to 26°C (Kroetz et al. 2021). Additionally, adult dusky sharks in the northern Gulf of America primarily utilized offshore waters of the continental shelf edge during the summer and fall and spent the majority of their time between 20 to 125 m depth and between 23 and 30°C (Hoffmayer et al. 2014).

Dusky sharks exhibit placental viviparity, giving live birth to a litter of between 3 to 12 pups (mean 7), with a gestation period of 24 months, a 1-year resting period between reproductive cycles, and a size at birth of 85 to 100 cm (Branstetter and Burgess 1996, Romine et al. 2009, Castro 2009). Dusky sharks mature very late with females maturing between 17 to 19 years and males between 17 to 21 years (Natanson et al. 1995, Romine et al. 2009, Natanson et al. 2014). Bangle et al. (2020) found an association with a temperature range between 16 and 26°C for tagged juveniles.

The dusky shark is an apex predator with a relatively high trophic level (4.2) and a diverse diet including a wide variety of reef, bottom, and pelagic bony fishes, as well as other elasmobranchs, crustaceans, octopi, cuttlefish, squid, starfish, barnacles, bryozoans, and occasional marine mammals (Cortés 1999). Juvenile dusky shark diets consist predominantly of small pelagic teleosts and squid, and it is thought they are somewhat generalist, able to switch to available prey (Gelsleichter et al. 1999). Adult diets are diverse, including a wide variety of teleosts, elasmobranchs, cephalopods, and crustaceans (Simpfendorfer et al. 2001, Dudley et al. 2005, Rogers et al. 2012). Hussey et al. (2011) used stable isotope analysis to reveal a shift to continental shelf edge foraging in large dusky sharks, compared to smaller life stages.

Dusky sharks have a very slow maximum intrinsic rate of increase (0.02-0.04) and would thus likely be slow to recover from population depletions such as overfishing (Romine et al. 2009). Generation length for dusky sharks is calculated at 27 to 28 years (Romine et al. 2009). Based on these life history characteristics, dusky sharks are one of the most vulnerable shark species to fishing pressure in the western North Atlantic (Romine et al. 2009, Natanson et al. 2014).

While landing of dusky sharks through the commercial or recreational fishery has been prohibited since the year 2000 (NMFS 1999), the most recent stock assessment update in 2016 found dusky sharks continued to be overfished ($SSF_{2015}/SSF_{MSY} = 0.41-0.64$) and undergoing overfishing ($F_{2015}/F_{MSY} = 1.08-2.92$) (SEDAR 2016). While this stock assessment update indicated that the biomass had likely increased

since the previous assessment in 2011 and significant progress had been made in reducing overfishing, it also indicated that overfishing had not yet been completely ended. Additional measures were taken in Amendment 5b to end overfishing (NMFS 2017). The stock should be rebuilt by the year 2107.

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Finetooth shark - *Carcharhinus isodon*

Overall Vulnerability Rank = high (57% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (94% certainty)

Directional Effect = neutral (44% certainty)

Data Quality = 85% of scores ≥ 2

<i>Carcharhinus isodon</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	2.6		Low Moderate High Very High
	Habitat Specificity	2.2	2.5		
	Mobility and Dispersal of Early Life Stages	1.8	2.4		
	Other Stressors	1.7	2		
	Population Growth Rate	3	3		
	Prey Specificity	3.2	2.8		
	Reproductive Cycle	1.5	2.8		
	Reproductive Strategy Sensitivity	1.7	2.1		
	Sensitivity to Ocean Acidification	1.4	1.4		
	Sensitivity to Temperature	1.6	2.4		
	Site Fidelity	2.2	1.4		
	Specificity in Early Life History Requirements	1.1	2.2		
	Stock Size Status	1.3	2.8		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	2.6	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1.1	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Finetooth shark (*Carcharhinus isodon*)

Contributors: D. Crear, W. Driggers, M. Drymon, B. Frazier, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (57% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), Sea Surface Oxygen (3.8) and pH (4.0). Finetooth sharks are found in coastal waters in the western Atlantic from North Carolina to Brazil, occupying shallow coastal waters in the summer throughout the southeast U.S. coast from North Carolina to the Gulf of America (Castro 2011; Kohler and Turner 2019; Roskar et al. 2021). They are generally found in temperatures of 23.64 to 30.05°C and depths between 9 and 51 m (NMFS SEFSC, unpublished data).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored equal to or above 2.5: Prey Specificity (3.2) and Population Growth Rate (3.0). Finetooth sharks show a higher degree of prey specialization, primarily feeding on menhaden and showing no ontogenetic shifts in diet as they reach adult stages (Bethea et al. 2004; Higgs 2016). In terms of population growth rate, finetooth sharks have an intrinsic rate of increase of 0.06/yr according to the most recent stock assessment (SEDAR 2007).

Data Quality: 85% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Little to no information is available about the potential effects of climate on distribution and abundance of finetooth sharks. To date, no range shifts have been observed or documented in this species; however, data are limited to fishery-independent data, as well as tag and recapture data. There have not been any published telemetry data from this species to date. Finetooth sharks have a relatively small distribution compared to other coastal species, and are heavily dependent on estuarine habitat; they are also dietary specialists heavily dependent on menhaden. Given the limited distribution and reliance (primarily) on a single prey source, distribution and abundance of finetooth shark may be negatively impacted by climate change. Additionally, finetooth sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: Finetooth sharks are found in coastal waters of the western North Atlantic from North Carolina south through the Gulf of America. While also reported as far south as Brazil, the occurrence in other locales outside the United States is likely rare (Castro 2011). Finetooth sharks overwinter off south Florida in the Atlantic; in the Gulf of America, overwintering grounds are poorly defined. In the western North Atlantic, finetooth sharks appear to begin their spring migration north when water temperatures reach 20°C, leaving south Florida to migrate to estuarine and nearshore waters off of the Carolinas (Castro 2011). Finetooth sharks are generally found in waters from the surf zone to 5 m depth with seagrass to sandy and muddy bottoms (Ulrich et al. 2007).

Finetooth sharks are a placental, viviparous species with an annual or biennial reproductive periodicity (solely biennial in the Atlantic) resulting in the birth of an average of approximately 4 pups after a approximately 1 year gestation period (Driggers and Hoffmayer 2009; Brown et al. 2020). Finetooth sharks take 4 to 6.8 years to reach 50% maturity and have an estimated maximum longevity of 22.3

years (Vinyard et al. 2019). Parturition occurs in late May/June in nearshore and estuarine waters with young of year utilizing estuaries in South Carolina in the Atlantic, and northwestern Florida, Mississippi, Alabama and Texas during summer months. Finetooth sharks are a dietary specialist, with stomach content analysis studies showing a diet of >97% of menhaden (*Brevoortia* sp., Bethea et al. 2004; Shaw et al. 2016).

Recent studies have shown distinct populations of finetooth sharks in the Atlantic and the Gulf of America, with up to three groups identified within the Gulf (Portnoy et al. 2016). The last stock assessment on finetooth shark was conducted in 2007 and found that the stock was not overfished or subject to overfishing ($N_{2005}/N_{MSY} = 1.80$; $F_{2005}/F_{MSY} = 0.17$) (SEDAR 2007).

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Florida smoothhound - Gulf of America complex stock (*Mustelus norrisi*)Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (100% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 38% of scores ≥ 2

<i>Mustelus norrisi</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	Low Moderate High Very High
Sensitivity Attributes	Adult Mobility	1.4	1.8		
	Habitat Specificity	1.8	2.2		
	Mobility and Dispersal of Early Life Stages	1.7	1.2		
	Other Stressors	1.4	1.4		
	Population Growth Rate	2	2		
	Prey Specificity	1.7	2		
	Reproductive Cycle	1.4	2.2		
	Reproductive Strategy Sensitivity	1.6	0.5		
	Sensitivity to Ocean Acidification	1.7	1		
	Sensitivity to Temperature	1.9	1.2		
	Site Fidelity	1.8	1.2		
	Specificity in Early Life History Requirements	1.1	1.4		
	Stock Size Status	1.3	2.2		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Florida smoothhound - Gulf of America complex stock (*Mustelus norrisi*)

Contributors: D. Grubbs, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low**. No attribute means scored above 2.5.

Data Quality: 38% of the data quality scores were 2 or greater. One attribute scored below 1.0: Reproductive Strategy Sensitivity (0.5). The Florida smoothhound is a placental, viviparous species that reproduces annually (Jones et al. 2014; Tagliafico et al. 2017).

Climate Effects on Distribution and Abundance: Little to no information regarding the effect of climate factors on Florida smoothhound is available, though studies regarding hypoxia lend insight to possible stressors. Florida smoothhound sharks responded to hypoxic conditions by slowing their movement, whereas other shark species increased their swimming speed (Carlson and Parsons 2001). Smoothhound sharks are buccal pumpers, allowing them to remain motionless while breathing versus ram ventilators requiring swimming to pass oxygen through gills. By slowing activity, buccal pumpers may conserve energy and oxygen when encountering hypoxic conditions (Carlson and Parsons 2001). This may become advantageous if hypoxic areas increase in prevalence under future conditions.

Additionally, smooth dogfish (*Mustelus canis*) was identified by Chueng et al. 2018 as an at-risk species due to projected effects of fishing and climate change. *M. canis* is similar to Florida smoothhound despite its larger distribution within the Gulf of America and the Atlantic Coast. As a shelf-associated species, *Mustelus* spp. in the Gulf of America are generally found in depths between 50 to 330 m and temperatures of 11 to 25°C (Drymon et al. 2020; Giresi et al. 2015; Jones et al. 2014; NMFS SEFSC, unpublished data). Populations of *M. canis* responded negatively to warmer winter temperatures, which may restrict seasonal movements and lead to habitat compression (Morely et al. 2017; Pinsky et al. 2021). Rising water temperatures are of particular concern for species that spend part or all of their life history in the Gulf of America, where the upper 50 m of surface waters are warming at twice the global rate (Wang et al. 2023). Smoothhound sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021). *M. canis* also displayed impaired sensory abilities when exposed to projected ocean acidification conditions for the year 2100 (0.3 to 0.4 decline in pH) (Dixson et al. 2015). Limited information is available regarding other stressors smoothhound species may encounter, particularly in the Gulf of America and Caribbean Sea.

Life History Synopsis: Florida smoothhound are a small (max. length approx. 150 cm total length) shelf-associated species found in the Gulf of America and Caribbean Sea at depths up to 160 m (Castro 2011; Giresi et al. 2015; Jones et al. 2014; Tagliafico et al. 2017). Limited species-specific information is available due to difficulty distinguishing between similar species, *M. canis* and *M. sinusmexicanus*. Florida smoothhound are seasonally abundant in the eastern Gulf of America, occupying deeper waters in summer months and moving inshore along the western coast of Florida during winter and spring

(Castro 2011). Analysis of genetically verified *Mustelus* spp. indicated that Florida smoothhound inhabited relatively shallow waters (1 to 92 m) concentrated in the eastern Gulf in the late spring, but these results may be influenced by the timing of sampling across regions (Giresi et al. 2015).

Similar to other smoothhound species, Florida smoothhound have an annual reproductive cycle with parturition occurring in March and April (Jones et al. 2014). Brood sizes range from 6 to 14 pups (Jones et al. 2014). Little is known regarding early life stage habitat use and distribution. Smoothhound sharks generally prey on crustaceans, fish, molluscs, and other invertebrates (Cortes 1999).

The von Bertalanffy growth coefficient for both males and females is estimated as 0.26, and the intrinsic rate of increase is estimated as 0.21 (Jones et al. 2014; SEDAR 2015). Age at 50% maturity is estimated as 4.11 years for females and 3.28 years (Jones et al. 2014). Maximum observed age is 6 years and 9 years for females and males, though other similar species have maximum ages ranging to 14 years (Jones et al. 2014).

Florida smoothhound is managed as a complex with *M. canis* and *M. sinusmexicanus*. The last stock assessment of the HMS Gulf of America smoothhound shark complex was conducted in 2015 and found that the complex is not overfished and overfishing is not occurring ($N_{2012}/N_{MSY} = 1.68\text{--}1.83$; $F_{2012}/F_{MSY} = 0.07\text{--}0.35$) (SEDAR 2015).

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Galapagos shark - *Carcharhinus galapagensis*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = moderate ☐Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☐ (98% certainty)

Directional Effect = neutral (44% certainty)

Data Quality = 0% of scores ≥ 2

<i>Carcharhinus galapagensis</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.7	1.8		Low Moderate High Very High
	Habitat Specificity	2.1	1.7		
	Mobility and Dispersal of Early Life Stages	1.4	1.5		
	Other Stressors	1.7	0.7		
	Population Growth Rate	2.6	1.2		
	Prey Specificity	1.6	1.4		
	Reproductive Cycle	1.3	1.3		
	Reproductive Strategy Sensitivity	1.8	0.6		
	Sensitivity to Ocean Acidification	1.5	0.6		
	Sensitivity to Temperature	1.9	1.4		
	Site Fidelity	2.2	0.8		
	Specificity in Early Life History Requirements	1	1.4		
	Stock Size Status	2.5	1		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Galapagos shark (*Carcharhinus galapagensis*)

Contributors: J. Cudney, T. Curtis, T. Loughran, C. McCandless, B. Wetherbee

Overall Climate Vulnerability Rank: **Not applicable.**

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored at or above 2.5: Population Growth Rate (2.6), and Stock Size/Status (2.5). Galapagos sharks have a low intrinsic rate of population growth and delayed maturation (Kyne et al. 2019). Limited data on stock size and status are available.

Data Quality: 0% of the data quality scores were 2 or greater. Scoring reflects particularly low confidence in life history knowledge, particularly for Other Stressors, Reproductive Strategy Sensitivity, Sensitivity to Ocean Acidification, and Site Fidelity.

Climate Effects on Distribution and Abundance: Limited information is available regarding the effects of climate factors to the distribution and abundance of Galapagos sharks. A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) projected the majority of suitable habitat to be maintained, with minor poleward expansion (Diaz-Carballido et al. 2023). Habitat loss was concentrated in nearshore waters of the southeastern Atlantic and South Atlantic, near Ascension Island. Given this species' site fidelity to oceanic islands and seamounts, increasingly unsuitable habitat in these areas may increase the likelihood of localized extinction and may negatively impact abundance (Kyne et al. 2019). Additionally, Galapagos sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: Galapagos sharks are found most commonly in tropical waters surrounding oceanic islands (e.g., Bermuda) and seamounts in the western Atlantic, and are considered to be an occasional stray in the southeastern Atlantic (Castro 2011; Kohler and Turner 2019). They are generally absent from continental waters and rarely venture within coastal waters, instead inhabiting deeper depths near or below the mixed layer during the day and surface waters at night (Castro 2011). Galapagos sharks are usually found in temperatures greater than 20°C around oceanic islands or seamounts (Madigan et al. 2021). The average distance traveled by Galapagos sharks was 138 nm, though a subset of the population was found to travel from Bermuda to the East Coast of the United States and Caribbean regions (Kohler and Turner 2019). Dives below the thermocline are common (Madigan et al. 2021). There is incomplete knowledge about mobility and dispersal of early life stages, however tagging studies involving juvenile sharks (<200 cm total length (TL), Wetherbee et al. (1996)) have identified similar movement patterns as adults (e.g., Mitchell et al. 2021, Morales et al. 2021). Galapagos and dusky (*Carcharhinus obscurus*) sharks are closely related and morphologically cryptic, but are confirmed as separate species with no evidence of hybridization (Corrigan et al. 2017).

Little information is available regarding the reproductive habits of Galapagos sharks in the Atlantic. Galapagos sharks are viviparous, and are thought to have a biennial to triennial reproductive cycle (Castro 2011). Parturition is estimated to occur in late spring and early summer in nearshore waters

around Ascension Island after a 12-month gestation period (Castro 2011; Wetherbee et al. 1996). The average brood size estimate from the Pacific is 8.7 pups (75 to 89 cm TL) (Castro 2011; Wetherbee et al. 1996). Females reach maturity between 6.5 to 9 years, and males 6 to 8 years (Kyne et al. 2019). Diet reports from the Pacific indicate that Galapagos sharks are generalist predators, consuming teleosts, cephalopods, crustaceans, and marine mammals (Papastamatiou et al. 2006).

Little information is available regarding population growth of Galapagos sharks. Galapagos sharks are estimated to have low population growth, but there is no evidence of large-scale population declines (Kyne et al. 2019). Local depletion of Galapagos populations may occur throughout oceanic islands and seamounts, though there is evidence of returning populations to Saint Paul's Rocks (Hazin et al. 2018; Kyne et al. 2019). Landing of Galapagos sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Great hammerhead shark - *Sphyrna mokarran*

Overall Vulnerability Rank = high (92% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (97% certainty)

Directional Effect = neutral (67% certainty)

Data Quality = 77% of scores ≥ 2

<i>Sphyrna mokarran</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	3		Low Moderate High Very High
	Habitat Specificity	2.7	2.6		
	Mobility and Dispersal of Early Life Stages	1.7	2.6		
	Other Stressors	2.4	1.6		
	Population Growth Rate	3.4	2.7		
	Prey Specificity	2.1	2.3		
	Reproductive Cycle	1.5	2.6		
	Reproductive Strategy Sensitivity	2.2	1.3		
	Sensitivity to Ocean Acidification	1.7	1.2		
	Sensitivity to Temperature	2.5	2.4		
	Site Fidelity	2.2	2.6		
	Specificity in Early Life History Requirements	1.1	2.7		
	Stock Size Status	2.6	2.1		
	Sensitivity Score		moderate		
Exposure Factors	Bottom Salinity	3.1	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.3	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

Great hammerhead shark (*Sphyrna mokarran*)

Contributors: M. Drymon, B. Frazier, B. Keller, T. Loughran, H. Marshall, C. McCandless

Overall Climate Vulnerability Rank: **High** (92% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), Sea Surface Oxygen (3.9) and pH (4.0). Within the western North Atlantic, great hammerhead sharks are found in shallow lagoons, coral reefs, and deep pelagic tropical waters ranging from North Carolina to the Greater Caribbean (Guttridge et al. 2017).

Biological Sensitivity: **Moderate.** Four sensitivity attributes scored above 2.5: Habitat Specificity (2.7), Population Growth Rate (3.36), Sensitivity to Temperature (2.5), and Stock Size and Status (2.6). Great hammerhead sharks utilize a variety of sensitive habitats, including coral reefs, seagrass beds, and other shallow nearshore environments (Castro 2011; Guttridge et al. 2017). Adult stages vary in their vertical habitat use, spending time in waters generally between 5 and 140 m depth (Drymon and Wells 2017; NMFS SEFSC, unpublished data). The most recent estimate of the intrinsic rate of increase is 0.06 (SEDAR 2024).

Data Quality: 77% of the data quality scores were 2 or greater indicating moderate data quality.

Climate Effects on Distribution and Abundance: Little species-specific information is available regarding the effect of climate factors on abundance and distribution. There is limited evidence to suggest that great hammerhead sharks may be expanding their range north as primary prey species (tarpon (*M. atlanticus*) and blacktip sharks (*C. limbatus*) move northward (Bowers 2023; Griffin et al. 2023; Hammerschlag et al. 2011). Scalloped hammerhead shark, a congener of the great hammerhead shark, also has among the highest metabolic rate of ectothermic elasmobranchs (Lowe 2001). Rising temperatures may affect physiological performance given their high metabolic rate, exacerbated by reduced oceanic dissolved oxygen (Crear et al. 2019; Dell’Apa et al. 2023; Lowe 2001). Great hammerhead sharks may be affected by similar metabolic rates and subsequent stressors, but further research is needed. As a group, Sphyrnidae have some of the highest observed levels of at-vessel and post-release mortality of any shark (Binstock et al. 2023; Braccini and Waltrick 2019; Coelho et al. 2012; Eddy et al. 2016; Ellis et al. 2017; Gulak et al. 2015; Morgan and Burgess 2007; Whitney et al. 2021) suggesting that they are quite sensitive to stressors and physiological challenges. These dynamics may have direct impacts on the abundance, distribution, and ecology of this species.

Life History Synopsis: The great hammerhead shark is a large bodied shark with a circumtropical range. In waters of the United States, it is found from coastal to pelagic waters off the oceanic shelf from North Carolina south through the Florida Keys into the Gulf of America (Castro 2011). Large juvenile and adult great hammerhead sharks appear to use a variety of habitats, from shallow flats and coastal beaches, to oceanic shelf waters. Great hammerhead sharks appear to have a preferred temperature range (13 to 30°C), spending most of their time in shallow coastal waters (Guttridge et al. 2022; NMFS SEFSC, unpublished data). Great hammerhead sharks routinely cross international waters and are known to display site fidelity and residency in the Bahamas and the Florida Keys (Guttridge et al. 2017).

Great hammerhead sharks are a placental viviparous species. Although reproductive data are lacking, they likely have a biennial reproductive cycle with a <1 year gestation and an average brood size of 26 pups (Driggers et al. 2021). Few studies have documented parturition and pupping or nursery grounds in this species. For example, young of year were detected in Bulls Bay, South Carolina, as well as nearshore waters off of South Carolina, Georgia, and the Big Bend of Florida (Barker et al. 2017); however, low numbers were detected. MacDonald et al. (2021) reported captures of juvenile great hammerhead sharks (likely 1+ years old based on length) in Biscayne Bay, Florida and speculated that the area serves as a primary or secondary nursery for this species.

Great hammerhead sharks are thought to be a generalist predator, consuming a wide range of fishes; however preferred prey appear to be marine catfish, stingrays, tarpon and medium bodied sharks (Castro 2011). Age at 50% maturity is approximately 8 years old, and maximum longevity is 42 years (Moncrief-Cox et al. 2021; Passerotti et al. 2010; Piercy et al. 2010).

While tag and recapture data are limited, multiple satellite telemetry studies have shown great hammerhead sharks are a wide ranging species, and there is only one stock in United States waters confirmed by research into genetic stock structure (SEDAR 2024). The Southeast Data, Assessment, and Review is currently conducting a stock assessment for great hammerhead sharks.

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Gulf smoothhound - Gulf of America complex stock (*Mustelus sinusmexicanus*)Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (100% certainty)

Directional Effect = neutral (70% certainty)

Data Quality = 38% of scores ≥ 2

<i>Mustelus sinusmexicanus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	Low Moderate High Very High
Sensitivity Attributes	Adult Mobility	1.7	1.4		
	Habitat Specificity	1.8	2		
	Mobility and Dispersal of Early Life Stages	1.6	1.6		
	Other Stressors	1.6	0.6		
	Population Growth Rate	2.1	2.4		
	Prey Specificity	1.6	2.4		
	Reproductive Cycle	1.6	2		
	Reproductive Strategy Sensitivity	1.6	0.7		
	Sensitivity to Ocean Acidification	2	1.6		
	Sensitivity to Temperature	1.8	1.6		
	Site Fidelity	1.8	1.6		
	Specificity in Early Life History Requirements	1.1	1.6		
	Stock Size Status	1.3	2.4		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Gulf smoothhound - Gulf of America complex stock (*Mustelus sinuismexicanus*)

Contributors: D. Grubbs, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low.** No attribute means scored above 2.5.

Data Quality: 23% of the data quality scores were 2 or greater. Two attributes scored below 1.0: Reproductive Strategy Sensitivity (0.7) and Other Stressors (0.6). Gulf smoothhound reproduce annually (Jones et al. 2014). There is no other information available regarding the reproductive strategy of the species. There is no information available on other stressors that may affect the Gulf smoothhound including pollution effects, parasites, or disease.

Climate Effects on Distribution and Abundance: Little to no information regarding the effect of climate factors on Gulf smoothhound is available, though studies regarding similar species lend insight to possible stressors. A possibly advantageous trait for smoothhound species is the ability to buccal pump, rather than ram ventilate. Florida smoothhound (*Mustelus norrisi*), another congener of Gulf smoothhound, responded to hypoxic conditions by slowing their movement versus ram ventilating shark species increasing their swimming speed (Carlson and Parsons 2001). Buccal pumping allows smoothhounds to remain motionless while breathing, which may conserve energy and oxygen when encountering hypoxic conditions (Carlson and Parsons 2001). This may become advantageous if hypoxic areas increase in prevalence under future conditions.

As a shelf-associated species, *Mustelus* spp. in the Gulf of America are generally found in depths between 50 to 330 m and temperatures of 11 to 25°C (Drymon et al. 2020; Giresi et al. 2015; Jones et al. 2014; NMFS SEFSC, unpublished data). Though there have been no studies regarding distributional shifts of Gulf smoothhound, congener smooth dogfish (*Mustelus canis*) has been highlighted as an at-risk species due to projected effects of fishing and climate change (Cheung et al. 2018). Possible climate effects on populations of *M. canis*, and subsequently other smoothhound species, include habitat compression and diminished sensory capabilities. Populations of *M. canis* were found to respond negatively to warmer winter temperatures which may limit habitat availability and seasonal movements between shallow and deeper habitats (Morely et al. 2017; Pinsky et al. 2021). Rising water temperatures are of particular concern for species endemic to the Gulf of America where the upper 50 m of surface waters are warming at twice the global rate (Wang et al. 2023). Smoothhound sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). Additionally, *M. canis* exposed to projected ocean acidification conditions for the year 2100 (0.3 to 0.4 decline in pH) demonstrated impaired ability to track prey odors (Dixson et al. 2015). Despite having a generalist diet with prey items ranging from invertebrates to teleosts, smoothhound species may encounter difficulty predating on existing food sources under future ocean acidification conditions. Limited information is available regarding other stressors smoothhound species may encounter, particularly in the Gulf of America. Analysis of genetically verified *Mustelus* spp. indicated

that Gulf smoothhound inhabited depths from 51 to 233 m, primarily further west in the Gulf during the summer, but these results may be influenced by the timing of sampling across regions (Giresi et al. 2015).

Life History Synopsis: Gulf smoothhound shark is a small (max. length approx. 10 cm total length) coastal and shelf-affiliated shark species found in the Gulf of America from southern Florida to southern Texas, with highest concentrations found in the northwestern Gulf (Castro 2011; Drymon et al. 2020; Jones et al. 2014). Gulf smoothhound prefer temperate waters along the continental shelf in depths of 50 to 250 m, 75 to 85 km offshore (Giresi et al. 2015; Murawski et al. 2018; Drymon et al. 2020). Little is known about Gulf smoothhound shark life history or movement due to their similarity to other smoothhound species and resulting identification issues.

Similar to other smoothhound species, Gulf smoothhound reproduce annually with parturition occurring in late spring to summer months (Jones et al. 2014). Brood sizes range from 3 to 10 pups, 35 to 43 cm stretch total length at birth (Jones et al. 2014). Limited information is available regarding habitat use of early life stages and growth of Gulf smoothhound. Smoothhound sharks generally prey on crustaceans, fish, molluscs, and other invertebrates (Cortes 1999).

The von Bertalanffy growth coefficient for both males and females is estimated as 0.16, and the intrinsic rate of increase is estimated as 0.21 (Jones et al. 2014; SEDAR 2015). A combined analysis of Gulf smoothhound and smooth dogfish growth estimates age at 50% maturity of 4.11 years and maximum age of 14 years for females, and 3.28 years and 13 years for males (Jones et al. 2014).

Gulf smoothhound is managed as a complex with *M. canis* and *M. norrisi*. The last stock assessment of the HMS Gulf of America smoothhound shark complex was conducted in 2015 and found that the complex is not overfished and overfishing is not occurring ($N_{2012}/N_{MSY} = 1.68\text{--}1.83$; $F_{2012}/F_{MSY} = 0.07\text{--}0.35$) (SEDAR 2015).

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Lemon shark - *Negaprion brevirostris*

Overall Vulnerability Rank = high (37% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (75% certainty)

Directional Effect = negative (70% certainty)

Data Quality = 85% of scores ≥ 2

<i>Negaprion brevirostris</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.9	2.6		Low Moderate High Very High
	Habitat Specificity	3	2.5		
	Mobility and Dispersal of Early Life Stages	1.8	2.4		
	Other Stressors	2.8	1.9		
	Population Growth Rate	3.6	2.3		
	Prey Specificity	1.6	2.2		
	Reproductive Cycle	1.2	2.4		
	Reproductive Strategy Sensitivity	2.8	2.5		
	Sensitivity to Ocean Acidification	1.9	2.2		
	Sensitivity to Temperature	2.3	2.4		
	Site Fidelity	2.8	2.5		
	Specificity in Early Life History Requirements	1.6	2.7		
	Stock Size Status	1.9	1.7		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	3.1	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1.1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Lemon shark (*Negaprion brevirostris*)

Contributors: J. Carlson, D. Grubbs, E. Hoffmayer, B. Keller, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (37% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.9), pH (4.0), and Sea Surface Oxygen (3.9). Lemon sharks are generally found in temperatures of 17.8 to 33.9°C throughout subtropical to tropical coastal waters, often in nearshore habitats such as mangroves, reefs, and estuaries (Castro 2011; D. Grubbs, pers comm.; Kessel et al. 2014; NMFS SEFSC, unpublished data; Reyier et al. 2014; Williams et al. 2019). Given the propensity of juvenile lemon sharks to occupy shallow nursery grounds, it is possible that these life stages may experience periods of hypoxia (Hamilton 2014).

Biological Sensitivity: **Moderate.** Five sensitivity attribute means scored above 2.5: Habitat Specificity (3.0), Other Stressors (2.8), Population Growth Rate (3.6), Reproductive Strategy Sensitivity (2.8), and Site Fidelity (2.8). Lemon sharks are a long-lived species with a low intrinsic rate of increase (0.07) (Hansell et al. 2021). Both adult and early life stages show a moderate degree of site fidelity; adults, for example, undergo latitudinal seasonal migrations and return to overwintering grounds off Jupiter, Florida, and summer grounds in the Canaveral Array (Reiyer et al. 2014). Young of year to juvenile sharks have been observed to stay in or near natal nursery grounds despite habitat degradation or other stressors, and move throughout the nurseries depending on prey and predator abundance and the availability of sheltering habitat (Chapman et al, 2009; Stump 2013).

Data Quality: 85% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Prolonged dependence on vulnerable nearshore habitats, natal philopatry, and temperature-dependent movements may increase the vulnerability of lemon sharks to the effects of climate change (Hamilton 2014; Kessel et al. 2014; Stump 2013). Regarding distribution, a study using ecological niche modeling (e.g MaxEnt) and four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) evaluated changes in suitable habitat for several carcharhinid species, including lemon sharks (Diaz-Carballido et al. 2022). Widespread habitat loss of nearshore habitats throughout the Gulf of America and Caribbean Sea were projected under RCP8.5 (-4.2% global loss). Distributional shifts of populations throughout the Gulf of America may prove difficult given latitudinal limitations, ectothermic nature, site fidelity, and philopatry of lemon sharks, and the species' nearshore distributions.

These effects may be mitigated by resilience to habitat degradation, as evidenced by anthropogenic development influencing natural selection of lemon sharks in the Bahamas (DiBattista et al. 2010). In this instance, development and subsequent habitat degradation was found to not affect the number of reproductive adults, instead altering natural selection patterns to favor larger, faster-growing immature stages. However, habitat degradation may have differing effects in other regions and the observed resilience may therefore not occur in similar situations. In addition to habitat degradation, lemon sharks may be exposed to similar stressors that other shark species experience in nearshore estuaries. Immature bull sharks, for example, were found in abundance throughout freshwater creeks that had

been degraded through anthropogenic development, exposing them to elevated levels of contaminants and phycotoxins as a results of harmful algal blooms (Curtis et al. 2011; Curtis et al. 2013; Edwards et al. 2023; Johnson-Restrepo 2005; Johnson-Restrepo 2008). Nam et al. (2010) also found brevetoxins from harmful algal blooms to bioaccumulate within lemon sharks. It is unclear what effects these stressors will have on lemon shark abundance under climate change conditions.

Life History Synopsis: Lemon sharks are distributed through tropical to subtropical coastal waters of the western North Atlantic between New Jersey to Florida, and the Gulf of America and Caribbean Sea (Castro 2011). Lemon sharks occupy offshore and nearshore areas and also frequent insular habitats, such as mangroves and lagoons, to accommodate reproductive and foraging needs (Castro 2011). Adult lemon sharks throughout the southeastern United States and Caribbean exhibit seasonal climatic migrations, tied closely to water temperature (Kessel et al. 2014; Kohler and Turner 2019). In spring to summer months, lemon sharks are distributed throughout the mid to southeast United States Atlantic coast (Kohler and Turner 2019). By winter, lemon sharks have often migrated to the Florida coast and the Caribbean Sea (Kohler and Turner 2019). Site fidelity is evident throughout all life stages of lemon shark, characterized by returning to the same overwintering grounds annually (Reiher et al. 2014). Female lemon sharks have documented natal philopatry to nursery grounds in the Bahamas (Feldheim 2001).

Lemon sharks are yolk-sac placental viviparous and reproduce biennially (Feldheim et al. 2001). Mating occurs in spring to early summer with a gestation period of 10 to 12 months, with parturition occurring in spring (Barker et al. 2005; Castro 2011; Compagno 1984). Litters may be up to 18 pups (Feldheim et al. 2002). Age at maturity is estimated as 12 years for males and 13 years for females (Brown and Gruber 1988; Hansell et al. 2021). Neonates occupy shallow, nearshore nurseries throughout the southeastern Atlantic and Caribbean Sea (Barker et al. 2005; Castro 2011; Feldheim et al. 2001). The dependence of lemon sharks on nurseries decreases with age, as large juveniles begin expanding areas of core habitat use outside of these grounds (Chapman et al. 2009). Early life stages of lemon sharks were found to continually occupy nursery grounds despite habitat degradation, moving throughout nurseries to seek shelter and foraging opportunities (Stump 2013). However, such degradation in other situations may have significant negative effects on space-use patterns and these findings should not lead to assumptions that the species is resilient to any level of anthropogenic disturbance.

Lemon sharks are opportunistic, generalist feeders, primarily feeding on teleosts. Non-teleosts (largely crustaceans) make up approximately 10% of juvenile diets (Wetherbee et al. 1990). Similar to other shark species, lemon sharks have a conservative life history with slow growth rates and an estimated population growth rate of 0.07/yr and von Bertalanffy growth coefficient of 0.057/yr (Brown and Gruber 1988; Hansell et al. 2021). Maximum observed age is currently 20 years, though is projected as 37 years (Hansell et al. 2021). A formal SEDAR stock assessment has not been completed for lemon sharks. Hansell et al. (2021) conducted a stock assessment utilizing available information and found that lemon shark abundance was subject to depletion prior to 1990, but has since remained stable since that time and continues to be above the overfishing threshold.

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Longbill spearfish - *Tetrapturus pfluegeri*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (96% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 8% of scores ≥ 2

<i>Tetrapturus pfluegeri</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	☒ Low ☒ Moderate ☒ High ☒ Very High
Sensitivity Attributes	Adult Mobility	1.3	1.6		
	Habitat Specificity	1.8	1.6		
	Mobility and Dispersal of Early Life Stages	2.1	1.1		
	Other Stressors	2	1.3		
	Population Growth Rate	2.5	1.3		
	Prey Specificity	1.8	1.5		
	Reproductive Cycle	2.2	1.4		
	Reproductive Strategy Sensitivity	2	0.6		
	Sensitivity to Ocean Acidification	1.8	1.3		
	Sensitivity to Temperature	2	2		
	Site Fidelity	2	0.7		
	Specificity in Early Life History Requirements	2.4	0.5		
	Stock Size Status	2.4	0.5		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Longbill spearfish (*Tetrapturus pfluegeri*)

Contributors: J. Graves, D. Kerstetter, T. Loughran

Overall Climate Vulnerability Rank: **Not applicable.**

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low.** One sensitivity attribute mean scored in the moderate category: Population Growth Rate (2.5). There is limited information available regarding population growth or status of longbill spearfish.

Data Quality: 8% of the data quality scores were 2 or greater. The reproductive strategy sensitivity, site fidelity, specificity in early life history requirements, and stock size and status attributes received scores equal to or less than one. Little information is available for these attributes.

Climate Effects on Distribution and Abundance: No published studies have specifically examined the effect of climate factors on the population productivity or changes in distribution for longbill spearfish under climate change scenarios. Longbill spearfish utilize surface waters to 150 m depth in temperatures between 24 and 29°C, spending the majority of their life in areas above the thermocline (Hoolihan 2013; Kerstetter et al. 2009). Increases in water temperature and salinity brought on by rapid climate change could have significant impacts on all aspects of their life history; rate of growth and maturity, range, prey availability and utilization, spawning frequency and success as well as individual strength and longevity. Likewise, increasing acidity could have significant negative impacts on all aspects, especially feeding and larval supply. Regarding depth utilization, two longbill spearfish electronically tagged in the eastern Atlantic appeared to be restricted by dissolved oxygen levels greater than 3.5 ml/L, possibly affecting their ability to avoid warmer surface waters with decreased dissolved oxygen concentrations (Kerstetter et al. 2009).

Regarding the distribution of longbill spearfish, their available habitat might expand with warming water temperatures but their upper thermal and dissolved oxygen thresholds may also decrease their ability to use the regions historically and currently being utilized. Because their habitat utilization is so narrow, longbill and other spearfish species may be more severely impacted by oceanographic changes incurred from rapid climatic changes. Overall, the lack of knowledge of this species and the probable low population size make it especially susceptible to negative impacts brought on by a rapidly changing climate regime and its impacts on the ocean surface waters.

Life History Synopsis: The life history of longbill spearfish is largely informed by similar istiophorid species. It occurs in the Atlantic Ocean and Gulf of America between 40°N and 35°S latitudes (Collette et al. 2011), where it is occasionally caught as bycatch in the offshore pelagic longline fishery targeting Swordfish and large tunas. Longbill spearfish have been detected throughout the Caribbean Sea during a reproduction study, though this may be an effect of using fisheries dependent data (Arocha et al. 2007). Longbill spearfish is primarily an epipelagic species found above the thermocline, with daily maximum depths at 150 m and night depth distributions slightly shallower (Kerstetter et al. 2009), although depths to 240 m have been reported (Nóbrega et al. 2009).

Data are contradictory in the western North Atlantic regarding spawning, which has been described as during the winter by Robins (1975) and de Sylva & Breder (1997), but during the late summer and early fall by Arocha et al. (2007). They are assumed to be broadcast spawners like most other pelagic migratory fish species. Age at maturity or growth parameters have not been reported for this species, though females over 150 cm fork length were recorded to have higher gonad index values (> 1.0) (Arocha et al. 2007). Similar to other istiophorid species, longbill spearfish diet consists of pelagic fishes and squids (Nakamura 1985). There is no stock assessment within the western North Atlantic available for longbill spearfish.

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Longfin mako shark - *Isurus paucus*Overall Vulnerability Rank = moderate  (63% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = very high  (74% certainty)

Directional Effect = negative (59% certainty)

Data Quality = 0% of scores ≥ 2

<i>Isurus paucus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1	1.7		Low Moderate High Very High
	Habitat Specificity	1.2	1.8		
	Mobility and Dispersal of Early Life Stages	1.4	1.3		
	Other Stressors	1.4	0.9		
	Population Growth Rate	3.8	1		
	Prey Specificity	1.4	1.7		
	Reproductive Cycle	1.1	1.1		
	Reproductive Strategy Sensitivity	1.4	0.6		
	Sensitivity to Ocean Acidification	1.3	1.2		
	Sensitivity to Temperature	1.1	1.7		
	Site Fidelity	1.5	0.6		
	Specificity in Early Life History Requirements	1.2	1.2		
	Stock Size Status	2.4	1.1		
	Sensitivity Score		low		
Exposure Factors	Mixed Layer Depth	1.2	-		
	Magnitude of Sea Surface Temperature Gradient	1.5	-		
	Oxygen at 200m	2.2	-		
	pH	4	-		
	Primary Production	1.1	-		
	Sea Surface Oxygen	3.8	-		
	Sea Surface Salinity	3.1	-		
	Sea Surface Temperature	3.8	-		
	Exposure Score		very high		
Overall Vulnerability Rank		moderate			

Longfin mako shark (*Isurus paucus*)

Contributors: B. Curtis, T. Curtis, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate.** (63% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors were the drivers in determining this score: pH (4.0), Sea Surface Oxygen (3.8), and Sea Surface Temperature (3.8). Longfin mako sharks are found in tropical to temperate epipelagic waters, and are capable of diving to depths below 1000 m (Hueter et al. 2017). This species has been documented throughout the western North Atlantic, from Georges Bank to southern Brazil and the Gulf of America (Hueter et al. 2017; Kohler and Turner 2019). Longfin mako sharks exhibit diel vertical migrations (dives from 6 to 952 m, and temps from 5 to 29°C), occupying surface waters at night, and at depth during the day (Hueter et al. 2017).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Population Growth Rate (3.8). This species has an estimated intrinsic rate of increase (r) of 0.010 to 0.026 (Cortes et al. 2010). Length at maturity for females is 245 cm, while males mature at approximately 200 cm (Hueter et al. 2017; Varghese et al. 2017).

Data Quality: 0% of the data quality scores were 2 or greater. All sensitivity attributes scored below 2.0 for data quality, though three sensitivity attributes scored below 1.0: Other Stressors (0.9), Reproductive Strategy Sensitivity (0.6), and Site Fidelity (0.6). Limited species-specific information exists for all sensitivity attributes. Shortfin mako behavioral ecology supplies much of the reproductive and other knowledge for longfin mako shark.

Climate Effects on Distribution and Abundance: Longfin mako is a poorly known deepwater shark found globally throughout warm temperate to tropical waters. This species may be afforded some refuge from increasing water temperatures, as they are known to dive to deeper waters (>1000 m) (Hueter et al. 2017). Longfin mako sharks exhibit daily vertical migration, spending time at the surface at night and at depths of 200 m or more during the day, and experiencing temperature ranges from 5 to 29°C (Hueter et al. 2017). This species is also thought to be regionally endothermic, similar to congener *Isurus oxyrinchus* (Compagno 2001). The advantages of this physiological adaptation when encountering elevated ocean temperatures due to climate change remains uncertain (Dell'Apa et al. 2023; Harding et al. 2021). However, this species would likely be vulnerable to deep-drop fisheries in the high seas. Simpfendorfer et al. (2008) describe longfin mako shark as a species that exhibits one of the highest levels of risk of overexploitation by Atlantic pelagic longline fisheries.

Additionally, longfin mako sharks are estimated to have some of the lowest productivity of pelagic shark species (Cortes et al. 2015). Simpfendorfer et al. (2008) provided a low level of productivity of 0.010 to 0.014/yr for longfin mako sharks, and Cortes et al. (2010) estimate an intrinsic rate of increase of 0.010 to 0.026 for this species. The closest relative of this species, shortfin mako shark, is often used to fill life history information gaps. It is unclear how climate change will affect longfin mako shark abundance given uncertainty in life history parameters.

Life History Synopsis: Found in tropical and warm temperate waters, longfin mako sharks inhabit a range from Georges Bank to southern Brazil, and in the Gulf of America (Hueter et al. 2017; Killam and

Parsons 1986; Wakida-Kusunoki and de AndaFuente 2012; Kohler and Turner 2019). This species exhibits vertical migrations, spending time at the surface during the night, and diving deep (>1000 m) during the day (Hueter et al. 2017)

While the maximum reported size for longfin mako sharks is 427 cm (Castro 2011), mature females and males are approximately 245 and 200 cm in length, respectively (Hueter et al. 2017; Varghese et al. 2017). Females are viviparous and give birth to 2 to 8 pups (Garrick 1967, Guitart-Manday 1966, Gilmore 1983, Muñoz-Chápuli 1984, Casey 1986, Compagno 2001, Castro 2011). According to Castro (2011) the Caribbean Sea may be a pupping ground, considering that gravid females carrying full term embryos (>120 cm) have been caught between 18°N and 20°N of the western North Atlantic. Both juveniles and adults are presumed to prey on schooling fish and pelagic squid, as squid beaks have been found in stomachs (Hueter et al. 2017).

There is insufficient information on stock size status for this species. Landing of longfin mako sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Narrowtooth shark - *Carcharhinus brachyurus*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (94% certainty)

Directional Effect = neutral (99% certainty)

Data Quality = 0% of scores ≥ 2

<i>Carcharhinus brachyurus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.5	0.9		Low Moderate High Very High
	Habitat Specificity	1.7	1.1		
	Mobility and Dispersal of Early Life Stages	1.4	0.9		
	Other Stressors	1.7	0.7		
	Population Growth Rate	3.2	1.1		
	Prey Specificity	1.8	0.9		
	Reproductive Cycle	1.2	0.7		
	Reproductive Strategy Sensitivity	1.8	0.8		
	Sensitivity to Ocean Acidification	1.5	0.9		
	Sensitivity to Temperature	1.8	0.9		
	Site Fidelity	1.8	0.9		
	Specificity in Early Life History Requirements	1.1	0.6		
	Stock Size Status	2	0.7		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Narrowtooth shark (*Carcharhinus brachyurus*)

Contributors: L. Crawford, W. Driggers, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution. This species was reported in the Gulf of America by Applegate et al. (1979), which references two specimens from Baja California and Veracruz. Castro (2011) notes that the species is not present within the western North Atlantic, including the Gulf of America.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.2). The narrowtooth shark is a slow growing species. The species' affinity for temperate waters and nursery grounds at the highest latitude in its distribution is hypothesized to be the cause of the slow growth rate (Walter and Ebert 1991). The species is also long-lived (25 years males, 31 years females) and late to mature (16 years) (Drew et al. 2016); these parameters contribute to the slow growth rate of the population.

Data Quality: None of the data quality scores were 2 or greater. All sensitivity attributes except Population Growth Rate (1.1) and Habitat Specificity (1.1) had data quality scores below 1, indicating low data quality and high uncertainty in information available regarding the species.

Climate Effects on Distribution and Abundance: Little information regarding the effects of climate factors on narrowtooth sharks is available for the western North Atlantic Ocean. A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that loss of suitable habitat for narrowtooth shark may occur in the southeastern Atlantic (Diaz-Carballido et al. 2022). Remaining habitat is largely maintained under both projections. Additionally, narrowtooth sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021). It is unclear how climate change will affect narrowtooth abundance.

Life History Synopsis: Narrowtooth sharks are highly mobile (Cuevas et al. 2022). There is minimal information about the species in the Atlantic, as its distribution is patchy and largely present in the South Atlantic (NMFS 2006). Evidence from studies in South Africa indicates that the species displays high site fidelity (Rogers et al. 2022). Narrowtooth sharks in Australian waters display strong preference for habitats with temperate waters and sea grass beds (Drew et al. 2019). Notably, nursery grounds for the species are found at the highest latitudes of its range (Smale et al. 1991; Walter and Ebert 1991).

Growth information for narrowtooth shark is available for populations in Australia; the intrinsic rate of increase is 0.126 and 0.146 for males and females, respectively (Drew et al. 2016). Litter sizes average between 13 and 20 pups (total length 60 to 70 cm) (NMFS 2006). Reproductive cycle length is estimated as biennial, and gestation is thought to last a year (NMFS 2006).

Landing of narrowtooth sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Night shark - *Carcharhinus signatus*Overall Vulnerability Rank = moderate  (91% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = very high  (87% certainty)

Directional Effect = neutral (99% certainty)

Data Quality = 23% of scores ≥ 2

<i>Carcharhinus signatus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.3	1.6		Low Moderate High Very High
	Habitat Specificity	1.4	1.6		
	Mobility and Dispersal of Early Life Stages	1.2	1.8		
	Other Stressors	1.5	0.9		
	Population Growth Rate	3.4	2		
	Prey Specificity	1.6	1.9		
	Reproductive Cycle	1.2	2.2		
	Reproductive Strategy Sensitivity	1.6	1.1		
	Sensitivity to Ocean Acidification	1.3	1.4		
	Sensitivity to Temperature	1.4	2		
	Site Fidelity	1.7	0.8		
	Specificity in Early Life History Requirements	1	1.7		
	Stock Size Status	2.2	1		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3.2	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1.1	-		
	Sea Surface Oxygen	4	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Night shark (*Carcharhinus signatus*)

Contributors: J. Carlson, J. Cudney, W. Driggers, E. Hoffmayer, T. Loughran, C. McCandless, D. Ortiz

Overall Climate Vulnerability Rank: **Moderate** (91% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), pH (4.0) and Sea Surface Oxygen (3.9). This is a subtropical to tropical species that is typically found in subtropical waters near the outer continental shelf, often in deep waters (61-294 m, >275 m during day and >180 m at night) (Castro 1983; NMFS SEFSC, unpublished data) of the continental slope or canyons (Kohler and Turner 2019).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Population Growth Rate (3.4). Night sharks are a relatively long-lived fish, with maturation occurring between 8 years (males) and 10 years (females) of age, intrinsic rates of increase about 10% per year, moderate rebound potential, and an intermediate generation time compared to other sharks (Carlson et al. 2008). Most of the remaining sensitivity attributes evaluated had an average score of 2 or less. Scoring balances life history and offshore distribution of the species with uncertainty and limited information available.

Data Quality: 23% of the data quality scores were 2 or greater. Other Stressors (0.9), Site Fidelity (0.8), and Stock Size Status (1.0) had data quality scores of 1 or below.

Climate Effects on Distribution and Abundance: Changes in the distribution of night sharks have been linked to warming in the northwestern Atlantic. Diaz-Carballido et al. 2022 used ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) to project changes in suitable habitat. Night sharks were predicted to lose suitable habitat across the north and south Atlantic basins across all four scenarios, with notable loss of suitable habitat occurring in the Gulf of America, the Caribbean (including Cuban waters), and parts of the Florida Straits. However, areas off the southeast United States, notably between the Carolinas and east coast of Florida, were predicted to maintain high suitability for night sharks. Night sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). It is unclear how climate change will affect night shark abundance.

Life History Synopsis: Night sharks are distributed throughout deep tropical waters of the southwest Atlantic, the Caribbean Sea, and the Gulf of America (Baremore et al. 2019; Carlson et al. 2008; Castro 2011). Tagging studies suggest that night sharks can undergo large scale movements (Baremore et al. 2019; Kohler and Turner 2019). While potential seasonal aggregations have been noted in the literature (e.g., Baremore et al. 2019 describe a seasonal aggregation at Glover Reef Atoll in Belize), in general trends are more broadly described for the western North Atlantic (Carlson et al. 2008; Castro 2011; Schärer-Umpierre and Franqui-Rivera 2022). Night sharks are not known to be endemic to a particular location in United States waters. Areas of concentration include the Florida Straits and off the east coast of Florida and South Carolina (Carlson et al. 2008).

Although night sharks are not considered rare, there is limited information available concerning various life history characteristics. Night sharks are viviparous, giving birth to an average litter size of 11 pups

(50 to 60 cm total length) (Carlson et al. 2008; Hazin et al. 2000). Gravid females have been detected off the coast of Carolinas and Georgia, and copulation has been recorded during the month of February in the western South Atlantic (Carlson et al. 2008; Castro 2011). Age of maturity estimates are currently 8 and 10 years for males and females, respectively (Hazin et al. 2000). Night sharks are dietary generalists and typically feed on crustaceans, cephalopods, and both reef and benthopelagic fish (Castro 2011; Vaske Júnior et al. 2009).

The intrinsic rate of increase is estimated as 0.01/yr, and von Bertalanffy growth coefficient estimates are 0.11/yr and 0.12/yr for females and males, respectively (Carlson et al. 2008). Maximum age is currently recorded as 17 years, though theoretical longevity estimates are as great as 30.4 and 27.9 for females and males (Carlson et al. 2008). In the northwest Atlantic, trend analysis from data collected by at-sea observers suggests a steady decline in night shark abundance (9.4% from 1996 to 2017) (Carlson et al. 2021). Landing of night sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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North Atlantic albacore tuna - *Thunnus alalunga*Overall Vulnerability Rank = low ■ (95% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 54% of scores ≥ 2

<i>Thunnus alalunga</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.3	2.5		Low Moderate High Very High
	Habitat Specificity	1.5	2.3		
	Mobility and Dispersal of Early Life Stages	2.1	1.1		
	Other Stressors	1.5	1.5		
	Population Growth Rate	2.4	2.6		
	Prey Specificity	1.5	2.5		
	Reproductive Cycle	2	2.1		
	Reproductive Strategy Sensitivity	2	1.7		
	Sensitivity to Ocean Acidification	1.6	1.5		
	Sensitivity to Temperature	1.4	2.4		
	Site Fidelity	1.7	1		
	Specificity in Early Life History Requirements	2.5	0.9		
	Stock Size Status	1.4	2.3		
	Sensitivity Score	low			
Exposure Factors	Mixed Layer Depth	1.5	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.3	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.5	-		
	Sea Surface Salinity	2.7	-		
	Sea Surface Temperature	3.5	-		
	Exposure Score	high			
Overall Vulnerability Rank		low			

North Atlantic albacore tuna (*Thunnus alalunga*)

Contributors: B. Galuardi, T. Loughran, E. Orbesen, D. Snodgrass

Overall Climate Vulnerability Rank: **Low** (95% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.5), Sea Surface Oxygen (3.5), and pH (4.0). Albacore tuna are found in epi- and mesopelagic waters, favoring temperatures of 10 to 20°C and minimum oxygen concentrations of 3.7 ml/l (Graham et al. 1989).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 54% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: There has already been documentation of alterations in the timing of migrations, perhaps, by the climatic changes that have taken place over the last couple of decades (Dufour et al. 2010). Merino et al. (2019) described how the management of the albacore tuna fishery has already made adaptations in accordance with climatic changes. Using species distribution models to examine the potential impact of a warming ocean on 22 different tuna stocks, Erauskin-Extramiana et al. (2019) found that the north Atlantic stock of albacore tuna is predicted to shift northward. Most recently, Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal and temporal shifts of albacore tuna throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Albacore tuna catch was estimated to have shifted poleward 9 ± 1 km for every 1°C increase, and an average of 2 ± 1 km/year for 2019 relative to 2002.

As with other long-lived species that reach reproductive maturity late in (e.g., approximately halfway) their life cycle the albacore tuna population is possibly more susceptible to climatic impacts due to longer exposure to novel perturbations that the species has not been exposed to in multiple millennia. The long distance between feeding and spawning grounds also could expose the species to unsuitable conditions along the way hence blocking or disrupting their access to those regions. The portion of the water column that they primarily occupy exposes them to the area where the largest climatic impact will occur. The amount of temperature change could have a significant impact on prime prey items utilized by albacore. In addition to temperature changes the greater influence might be dealt from the variations in acidity (pH) given that they rely so heavily on invertebrate prey species. Additionally, lower oxygen availability in the Gulf of America may impact larval survival, further exacerbated by warmer sea surface temperatures (Dell’Apa et al. 2018; Dell’Apa et al. 2023; Wexler et al. 2011). Average body size of albacore tuna is projected to decrease by 21.7% by 2050 (Erauskin-Extramiana et al. 2023).

Life History Synopsis: Albacore tuna are distributed throughout the Indian, Pacific and Atlantic Oceans in tropical and temperate regions (Collette and Nauen 1983). The available evidence leads us to believe that albacore tuna are distributed in three distinct Atlantic populations (north, south and Mediterranean) (Arrizabalaga et al. 2004, Davies et al. 2011). They primarily inhabit the meso- to epipelagic zones of the water column between the tropics to roughly 45° N (Sagarminaga and

Arrizabalaga 2010). There appears to be multiple known spawning areas primarily in the western North Atlantic and Gulf of America whereas the prime feeding grounds are in the northeast Atlantic (Trenkel et al. 2014). Their migratory movements across the ocean appear to be linked with seasonal sea surface temperature fluctuations (Sagarminaga and Arrizabalaga 2010). This might imply that temperature is a significant factor influencing the habitat use choices of albacore tuna (Dufour et al. 2010).

Very little information is published on early life history and the various key aspects of it, and is largely assumed to mirror pelagic egg and larval stages of other tunas. There is some disagreement on maximum age but many studies have strong evidence to support it being at 10 years while others have shown that it is closer to 15 years. It appears that as the species is entering the reproductive maturity stage (approximately 100 cm) they are heavily reliant on crustaceans and cephalopods (Logan et al. 2013) with over 50% of their observed gut contents containing said items. Goni et al. (2011) demonstrated that crustaceans made up an even higher proportion of prey items in other areas of the North Atlantic. The sampled population seems to be dominated by males individuals with over a 4:1 ratio (de Zárate et al. 2004). Reproductive maturity appears to be reached at about 80 cm fork length which is about 5 years of age (de Zárate et al., 2004, Arocha et al. 2019).

The last stock assessment on North Atlantic albacore tuna was conducted in 2018 and found that the stock was not overfished or subject to overfishing ($SSB_{2021}/SSB_{MSY} = 2.19 (1.21-4.01)$; $F_{current}/F_{MSY} = 0.45 (0.29-0.71)$) (SCRS 2023).

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North Atlantic blue shark - *Prionace glauca*Overall Vulnerability Rank = low ■ (100% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = very high ■ (100% certainty)

Directional Effect = neutral (89% certainty)

Data Quality = 77% of scores ≥ 2

<i>Prionace glauca</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.7		Low Moderate High Very High
	Habitat Specificity	1.2	2.8		
	Mobility and Dispersal of Early Life Stages	1.1	2.5		
	Other Stressors	1.3	1.5		
	Population Growth Rate	2.3	2.5		
	Prey Specificity	1.2	2.8		
	Reproductive Cycle	1.1	2.4		
	Reproductive Strategy Sensitivity	1.1	1.1		
	Sensitivity to Ocean Acidification	1.2	1.6		
	Sensitivity to Temperature	1.1	2.6		
	Site Fidelity	1.4	2.3		
	Specificity in Early Life History Requirements	1	2		
	Stock Size Status	1.6	2.7		
	Sensitivity Score		low		
Exposure Factors	Mixed Layer Depth	1.6	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.3	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.4	-		
	Sea Surface Salinity	2.6	-		
	Sea Surface Temperature	3.4	-		
	Exposure Score		high		
Overall Vulnerability Rank		low			

North Atlantic blue shark (*Prionace glauca*)

Contributors: J. Carlson, B. Curtis, T. Loughran, H. Marshall, C. McCandless, M. Passerotti

Overall Climate Vulnerability Rank: **Low** (100% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors were the drivers in determining this score: pH (4.0), Sea Surface Oxygen (3.4), and Sea Surface Temperature (3.4). Adult blue sharks utilize offshore and nearshore habitats throughout the Atlantic, ranging from maritime Canada to the Caribbean Sea, and spending substantial amounts of time offshore (Kohler and Turner 2019; Vandeperre et al. 2014a). Juveniles utilize the warm Gulf Stream and Sargasso Sea for overwintering, moving offshore in winter months (Campana et al. 2011; Kohler and Turner 2019; Vandeperre et al. 2014a; Vandeperre et al. 2014b).

Biological Sensitivity: **Low.** Population Growth Rate (2.3) for North Atlantic blue sharks was the highest scoring Sensitivity Attribute, while all others remained relatively low. North Atlantic blue sharks are estimated to be one of the most productive shark species, though this depends on survivorship of neonates and small juveniles (Aires-da-Silva and Gallucci 2007). The age at maturity (4 to 5 years) is relatively young for pelagic sharks, as is the maximum age (16 to 21 years) (Carlson et al. 2023; Skomal and Natanson 2003). According to IUCN, spawning stock biomass in the North Atlantic is estimated to have declined between 53.9% to 79% over 30 years (Rigby et al. 2019).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Blue shark distribution is projected to shift poleward as a result of shifting oceanographic conditions. Braun et al. (2023) projected a 40 to 60% decline in suitable habitat within the western North Atlantic for blue shark and estimated a northern latitudinal shift, whereas other highly migratory species are projected to move northeast. Similarly, Crear et al. (2023) noted that the distribution of blue shark catch along the Northeast Shelf is largely influenced by sea surface temperature and is expected to shift poleward at a rate of 40 ± 1 km for every 1°C increase. Diaz-Carballido et al. (2022) used ecological niche modeling (MaxEnt) under four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) to project suitable habitat. While the majority of habitat for blue sharks within the north Atlantic is classified as maintained, there are losses along the northeastern Atlantic coast of the United States under RCP8.5. Additionally, blue sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). Rising temperatures may lead to shifts in vertical distribution, as blue sharks have been documented using deeper, cooler waters to behaviorally thermoregulate and forage (Campana et al. 2011).

Additionally, Vedor et al. (2021) found that blue sharks are sensitive to the vertical extent of the oxygen minimum zone, occupying shallower depths in response to decreased dissolved oxygen in deeper waters. These behaviors may increase the vulnerability of blue sharks to expansion of the oxygen minimum zone, limiting them to surface waters and furthering exposure to elevated temperatures and

pelagic longline fisheries (Dell’Apa et al. 2023; Vedor et al. 2021). It is unclear how climate change will affect blue shark abundance.

Life History Synopsis: Blue sharks are globally distributed throughout tropical to temperate epipelagic waters (Coelho et al. 2018; Druon et al. 2022). There are two stocks within the Atlantic Ocean, split between the north Atlantic and south Atlantic by 4°N latitude (Fitzmaurice et al. 2004; SCRS 2023). In the western Atlantic, the stock is distributed from Newfoundland to the Caribbean Sea and individuals commonly complete trans-Atlantic migrations (Campana et al. 2011; Kohler and Turner 2019). Blue sharks occupy a wide range of temperatures (3.9°C to 31.3°C, most commonly between 12°C and 20°C) and seasonally migrate between coastal surface waters in Canada and the northeastern United States in summer and fall, then dispersing to deeper offshore waters until spring (Campana et al. 2011; Howey et al. 2017; Kohler and Turner 2019; Nakano and Stevens 2008). These seasonal migrations are specifically prevalent in immature life stages, which utilize the Gulf Stream in the north-central Atlantic and Sargasso Sea as overwintering grounds (Campana et al. 2011; Vandeperre et al. 2014a). Similar seasonal movements are found throughout populations in the central (i.e., specifically neonates and smaller size classes) and eastern Atlantic. In addition, strong site fidelity is seen for all life stages except adult females in the central Atlantic (Vandeperre et al. 2014a).

Blue sharks are viviparous and have an annual reproductive cycle (Mas et al. 2023). Gestation lasts 9 to 11 months and culminates in an average minimum fecundity of 27 to 31 pups (45 to 50 cm total length at birth) (Mas et al. 2023). Nursery grounds have been identified in the North Atlantic, in the area encompassing the Azores, Mid-Atlantic Ridge, and Atlantis – Great Meteor seamount complex (Vandeperre et al. 2014a). Blue sharks reach 50% maturity at sizes of 190.9 and 192.5 cm fork length, and are one of the most productive shark species (mean finite rate of population increase = 1.23/year) though this is heavily dependent on survival of early life stages (Aires-da-Silva and Gallucci 2007; Viducic et al. 2022). Blue sharks are known to utilize anticyclonic eddies for foraging purposes, diving to mesopelagic depths during the day to feed on a variety of pelagic teleosts and cephalopods (Braun et al. 2019; Cortes 1999; McCord and Campana 2003). Campana et al. (2011) found a similar diel vertical migration pattern in migratory blue sharks within the Gulf Stream, attributed to feeding and behavioral thermoregulation purposes.

The last stock assessment for North Atlantic blue shark was conducted in 2023 and found that the stock is not overfished or subject to overfishing ($B_{2021}/B_{MSY}=1.00$ (0.75-1.30)/ $F_{2021}/F_{MSY} = 0.70$ (0.50-0.93)) (SCRS 2023). Spawning stock biomass is estimated to have declined between 53.9% (median) to 50 to 79% (highest probability) over 30 years (Rigby et al. 2019).

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North Atlantic shortfin mako shark - *Isurus oxyrinchus*Overall Vulnerability Rank = moderate  (60% certainty)Biological Sensitivity = moderate Climate Exposure = high Distributional Vulnerability Rank = very high  (94% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 69% of scores ≥ 2

<i>Isurus oxyrinchus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1	2.8		Low Moderate High Very High
	Habitat Specificity	1.3	2.6		
	Mobility and Dispersal of Early Life Stages	1	2.6		
	Other Stressors	1.3	1.7		
	Population Growth Rate	3.6	2.6		
	Prey Specificity	1.3	2.8		
	Reproductive Cycle	1.1	2		
	Reproductive Strategy Sensitivity	1.1	1.6		
	Sensitivity to Ocean Acidification	1.1	1.8		
	Sensitivity to Temperature	1.1	2.6		
	Site Fidelity	1.3	1.7		
	Specificity in Early Life History Requirements	1	2.2		
	Stock Size Status	3	2.4		
	Sensitivity Score		moderate		
Exposure Factors	Mixed Layer Depth	1.5	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.2	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.3	-		
	Sea Surface Salinity	2.9	-		
	Sea Surface Temperature	3.4	-		
	Exposure Score		high		
Overall Vulnerability Rank		moderate			

North Atlantic shortfin mako shark (*Isurus oxyrinchus*)

Contributors: J. Carlson, B. Curtis, B. Keller, J. Kneebone, T. Loughran, C. McCandless, M. Passerotti

Overall Climate Vulnerability Rank: **Moderate** (60% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors were the drivers in determining this score: pH (4), Sea Surface Oxygen (3.3), and Sea Surface Temperature (3.4). North Atlantic shortfin mako sharks are often found in temperate epipelagic and coastal waters from 17 to 27°C, though are resilient to temperature fluctuations from 8 to 30°C (Vaudo et al. 2017).

Biological Sensitivity: **High.** Two sensitivity attributes scored above 2.5: Population Growth Rate (3.6) and Stock Size and Status (3). The intrinsic rate of increase for North Atlantic shortfin mako sharks is estimated as 0.031-0.060/yr, and current relative biomass level is estimated as 0.57–0.95 B/B_{MSY} (Cortes 2017; SCRS 2023). The stock is currently overfished, and overfishing is occurring (SCRS 2023).

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Recreational catch records from 2002 to 2019 indicate a northward/northeastward shift in shortfin mako shark catch from Virginia to Maine has already occurred at an estimated rate of 4 km/year and 22 km for every 1°C increase in sea surface temperature (Crear et al. 2023). Species distribution modeling exercises also predict suitable habitat for shortfin mako shark will decrease by roughly 65% in the Gulf of America and 30% in the western North Atlantic by the end of the century (Braun et al. 2023). Under current climate warming projections, the species is also expected to be displaced northward by up to 120 km in the Gulf of America and 400 km in the northwest Atlantic (Braun et al. 2023). Additionally, juvenile stages may seek optimal temperatures by occupying deeper, cooler waters (Dell’Apa et al. 2023; Vaudo et al. 2016).

Shortfin mako sharks are regionally endothermic. The species can regulate its body temperature due to its high metabolic rate and a system of blood vessels called *rete mirabile* that function as a countercurrent heat exchange system to maintain heat within tissues and prevent cooling of the blood through the gills. Despite broad thermal tolerance capabilities due to the species’ endothermic capabilities, shortfin mako sharks show a strong preference for temperatures between 22 and 27°C (Vaudo et al. 2016). Additionally, juveniles were found to have limited thermoregulation capability compared to adult stages, indicating an ontogenetic shift in endothermy (Vaudo et al. 2016). While the resilience of endothermic species to elevated ocean temperatures due to climate change remains uncertain, the shortfin mako shark may be limited to changes in vertical distribution given its thermal preferences (Dell’Apa et al. 2023; Harding et al. 2021). It is unclear how climate change will affect shortfin mako shark abundance.

Life History Synopsis: The shortfin mako shark is a pelagic species found throughout all tropical and temperate ocean waters, including the North Atlantic Ocean (Compagno 2001). The species is highly migratory throughout all life stages, with juveniles and adults documented to make straight line horizontal movements in excess of 1000 km between North America, Europe (Spain and Portugal), northwest Africa, and South America (Vaudo et al. 2017; Kohler and Turner 2019). During migration, the species typically exhibits periods of both directional movement and extended residency, with residency

behavior being more common in shallow, continental shelf or slope waters (Byrne et al. 2019). Movements between the Gulf of America and broader western North Atlantic occur, but are uncommon (Vaudo et al. 2017; Kohler and Turner 2019; Gibson et al. 2021). The species has a broad thermal tolerance and occupies waters from 5.2 to 31.1°C (Vaudo et al. 2016), but typically occurs in temperatures between 22 and 27°C (Vaudo et al. 2016; Vaudo et al. 2017; Santos et al. 2021) and is less common in water temperatures over 27°C (Vaudo et al. 2016). Shortfin mako sharks routine and maximum metabolic rates are among the highest measured for any shark species (Sepulveda et al. 2007), and may cause the species to avoid areas of low dissolved oxygen (Abascal et al. 2011).

Their reproductive cycle is not well defined, but has been hypothesized to be as long as 3 years, inclusive of up to an 18 month resting period between pregnancies, which produce between 4 and 25 pups (Mollet et al. 2000). Shortfin mako sharks primarily feed upon teleost fishes, but also consume cephalopods, elasmobranchs, cetaceans, and crustaceans (Stillwell and Kohler 1982; Cortés 1999). In the western North Atlantic, bluefish (*Pomatomus saltatrix*) are a major inshore prey item and can make up 77.5% of diet by volume (Stillwell and Kohler 1982) and 92.6% of diet by weight (Wood et al. 2009). North Atlantic shortfin mako sharks have a very low rate of intrinsic population increase ($r_{\max} = 0.054$; Cortes 2016), largely due to females not reaching 50% maturity until 19 to 22 years (Natanson et al. 2006; Rosa et al. 2017; Natanson et al. 2020).

The last stock assessment on North Atlantic shortfin mako shark was conducted in 2019 and found that the stock was overfished and subject to overfishing ($B_{2015}/B_{MSY} = 0.57\text{--}0.95$; $F_{2015}/F_{MSY} = 1.93\text{--}4.38$) (SCRS 2023).

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North Atlantic swordfish - *Xiphias gladius*Overall Vulnerability Rank = low ■ (98% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = high ■ (100% certainty)

Directional Effect = neutral (94% certainty)

Data Quality = 92% of scores ≥ 2

<i>Xiphias gladius</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.3	2.7		Low Moderate High Very High
	Habitat Specificity	1.4	2.9		
	Mobility and Dispersal of Early Life Stages	2.1	2.3		
	Other Stressors	1.9	2.3		
	Population Growth Rate	2.4	2.9		
	Prey Specificity	1.4	2.7		
	Reproductive Cycle	2.2	2.7		
	Reproductive Strategy Sensitivity	2	2		
	Sensitivity to Ocean Acidification	2.1	1.9		
	Sensitivity to Temperature	2	2.5		
	Site Fidelity	1.9	2.3		
	Specificity in Early Life History Requirements	2.2	2.3		
	Stock Size Status	2.2	2.9		
	Sensitivity Score	low			
Exposure Factors	Mixed Layer Depth	1.6	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.4	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.2	-		
	Sea Surface Salinity	2.7	-		
	Sea Surface Temperature	3.2	-		
	Exposure Score	high			
Overall Vulnerability Rank		low			

North Atlantic swordfish (*Xiphias gladius*)

Contributors: J. Graves, D. Kerstetter, T. Loughran, G. Skomal

Overall Climate Vulnerability Rank: **Low** (98% certainty from bootstrap analysis).

Climate Exposure: **High**. Three exposure factors contributed to this score: Sea Surface Oxygen (3.2), Sea Surface Temperature (3.2) and pH (4.0). Swordfish occupy a broad thermal range (5 to 27°C), spending a considerable proportion of time at depth (400 to 800 m) during the day and returning to surface waters at night (Abid and Idrissi 2006; Lerner et al. 2013).

Biological Sensitivity: **Low**. No sensitivity attribute means scored above 2.5.

Data Quality: 92% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: The effects of climate change on swordfish has been the subject of several studies. Sundby et al. (2013) reported on the recent appearance of swordfish off Norway, suggesting a northward range expansion. Chang et al. (2013) and Schrippa et al. (2017) report that changes in swordfish distribution due to oceanographic conditions are likely from regime changes in large-scale condition shifts, such as the Atlantic Multidecadal Oscillation, which are in turn affected by climate change effects. Modeling of swordfish in Chile by Silva et al. (2015) suggested that climate change would shift the 18°C isotherm, thereby affecting feeding and reproduction movements and ultimately decreasing productivity. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. No significant shifts in distribution of swordfish catch were estimated to occur between 2002 and 2019, however, poleward movement was estimated to have shifted 4 ± 2 km for every 1°C increase.

More recently, other modeling by Erauskin-Extramiana et al. (2020) found overall global swordfish catch-per-unit-effort (CPUE) would decrease 22% by the end of the century under the RCP8.5 climate change scenario, the highest greenhouse gas concentration scenario. Swordfish CPUE was predicted to decline most in tropical waters and the distribution would expand poleward. Braun et al. (2023) also found a 20% decrease in swordfish suitable habitat in the Gulf of America, and 40% for the western North Atlantic. It is unclear how climate change will affect north Atlantic swordfish abundance, though a 7.3% decrease in average body size is projected by 2050 (Erauskin-Extramiana et al. 2023).

Life History Synopsis: The swordfish is a large, highly migratory teleost that reaches 455 cm fork length and 536 kg (IGFA 2023). The species has a broad circumglobal oceanic distribution from tropical to temperate pelagic waters from approximately 69°N to 50°S (FAO-FIGIS 2005). Swordfish will occasionally be seen over coastal shelf edges, especially smaller individuals. An apparent population with limited genetic exchange exists in the Mediterranean Sea (Kotoulas et al. 1995), which is also managed as a separate unit by the International Commission for the Conservation of Atlantic Tunas. Swordfish are known to undergo broad diel habitat shifts, ranging from the surface at night above the thermocline to over 500 m during the day (summarized by Collette et al. 2011, see also Collette 1995, Sedberry & Loefer 2001, Abascal et al. 2010, Abecassis et al. 2012, Lerner et al. 2013); the deepest reported depth reached 2878 m (Collette et al. 2011). Anecdotal evidence from pelagic fisheries and limited electronic

tagging data analysis suggests that larger individuals undergo broader diel depth changes and that minimum depths are shallower at night during times of the new moon with limited lunar illumination (Abascal et al. 2010, Abecassis et al. 2012, Fenton 2012). Individuals in the western Atlantic have demonstrated with electronic tagging data a generalized seasonal north-south migration, with some displaying a high degree of seasonal site fidelity (Neilson et al. 2014). Juveniles electronically tagged in the Azores made regional movements while adults tagged in the western North Atlantic moved between summer foraging grounds near the Grand Banks to winter habitats near the Antillean Arc (Braun et al. 2019). The United States pelagic longline fishery historically fishes the northern Caribbean during the winter, following the larger swordfish north through Florida, the Mid-Atlantic, and into the Grand Banks fishery during the summer. Similar broad-scale seasonal migrations are seen in other swordfish worldwide (e.g., Mediterranean: Teserpes et al. 2008, southeastern Pacific: Abascal et al. 2010, eastern Pacific: Sepulveda 2020).

Spawning appears to occur opportunistically in tropical waters, but seasonally at higher latitudes in the Pacific and Indian Oceans (Nishikawa and Ueyanagi 1974). Spawning is usually one female to one or occasionally two males (Palko et al. 1981). As with many scombroids, large females produce large numbers of small (1.6 to 1.8 mm) eggs; 29 million eggs were reported from a 272 kg female (Wilson 1984). Larvae grow quickly in epipelagic waters, with the Florida Straits being a nursery ground for swordfish in the western North Atlantic.

Swordfish traverse a broad vertical habitat range and have a generalist diet, including mesopelagic and pelagic fishes, crustaceans, and cephalopods (see reviews in Collette 1995, Nakamura 1997). Specific to the western North Atlantic, swordfish primarily consume cephalopod prey, and opportunistically feed on fishes and crustaceans (Scott and Tibbo 1974, Heemsoth et al. 2020). Swordfish in higher latitudes are known to exhibit so-called “basking” behavior that is putatively associated with feeding and thermoregulation, where individuals are found floating at the relatively warmer surface, particularly in the cooler waters of the western North Atlantic and the Mediterranean, where they were historically targeted by harpoon fisheries. However, this basking has been demonstrated worldwide through electronic tagging data (e.g., California: Sepulveda et al. 2010, north Pacific: Abecassis et al. 2012, Florida: Lerner et al. 2013).

Aging of swordfish is difficult due to small otoliths and a lack of all but miniscule scales as adults. Attempts with length-based methods (e.g., Ehrhardt 1992) found poor fits with standard growth models, and the use of anal fin spine cross sections (e.g., Berkeley and Houde 1983, Cerna 2009) resulted in slower growth estimates than those using otoliths (e.g., Wilson and Dean 1982). Nonetheless, swordfish grow rapidly as juveniles and reach an age of maturity at five years (Arocha & Lee 1996). Maximum longevity estimates range from 10 years in males and 12 years in females in western Pacific Waters (Sun et al. 2012), but may reach 15 years in the western North Atlantic (Arocha et al. 2003). Mean generation time estimates range from 5.6 years (Hinton and Maunder 2006) to 7 years (Collette et al. 2011). The last stock assessment on North Atlantic swordfish was conducted in 2022 and found that the stock was not overfished or subject to overfishing ($B_{2020}/B_{MSY} = 1.08 (0.71-1.33)$; $F_{2020}/F_{MSY} = 0.80 (0.64-1.24)$ (SCRS 2023)).

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Northwest Atlantic porbeagle shark - *Lamna nasus*Overall Vulnerability Rank = moderate  (88% certainty)Biological Sensitivity = moderate Climate Exposure = high Distributional Vulnerability Rank = very high  (73% certainty)

Directional Effect = negative (83% certainty)

Data Quality = 77% of scores ≥ 2

<i>Lamna nasus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.8		Low Moderate High Very High
	Habitat Specificity	1.4	2.6		
	Mobility and Dispersal of Early Life Stages	1.1	2.4		
	Other Stressors	1.3	1.4		
	Population Growth Rate	3.6	2.8		
	Prey Specificity	1.4	2.4		
	Reproductive Cycle	1.3	2		
	Reproductive Strategy Sensitivity	1.2	1.6		
	Sensitivity to Ocean Acidification	1.1	2		
	Sensitivity to Temperature	1.4	2.4		
	Site Fidelity	1.3	1.9		
	Specificity in Early Life History Requirements	1	2.2		
	Stock Size Status	2.9	2.4		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.5	-		
	Magnitude of Sea Surface Temperature Gradient	1.6	-		
	Oxygen at 200m	2.6	-		
	pH	4	-		
	Primary Production	1.2	-		
	Sea Surface Oxygen	3.4	-		
	Sea Surface Salinity	3.1	-		
	Sea Surface Temperature	3.2	-		
	Exposure Score	high			
Overall Vulnerability Rank		moderate			

Northwest Atlantic porbeagle shark (*Lamna nasus*)

Contributors: J. Carlson, L. Crawford, B. Curtis, E. Hoffmayer, T. Loughran, C. McCandless, M. Passerotti, J. Sulikowski

Overall Climate Vulnerability Rank: **Moderate** (88% certainty from bootstrap analysis).

Climate Exposure: **High.** Four exposure factors contributed to this score: pH (4), Sea Surface Oxygen (3.4), Sea Surface Salinity (3.1), and Sea Surface Temperature (3.2). Porbeagle sharks are found in continental shelves and the open ocean (Campana et al. 2010; Haugen et al. 2022).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (3.6) and Stock Size Status (2.9). Porbeagle sharks have a low intrinsic rate of increase, $r = 0.046\text{--}0.059/\text{yr}$ (Cortes 2020) and growth rate, $k = 0.07/\text{yr}$ (Natanson et al. 2002). Age at maturity is 8 for males and 13 for females, and a maximum age of 24 for males and 25 for females (Natanson et al. 2002).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: There is limited information available regarding the effect of climate change on porbeagle sharks within the western North Atlantic. Given the propensity of the species to migrate seasonally and occupy broad ranges of the water column to feed and find suitable temperatures, porbeagle sharks may be adaptable under climate change conditions. Further, porbeagle sharks are regionally endothermic, meaning the species can regulate its body temperature due to its high metabolic rate and a system of blood vessels called *rete mirabile*. The *rete mirabile* function as a countercurrent heat exchange system to maintain heat within tissues and prevent cooling of the blood through the gills. The advantages of this physiological adaptation when encountering elevated ocean temperatures due to climate change remains uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

The vulnerability of porbeagle sharks to climate is in part due to their stock status as overfished and low productivity. Overfishing is the greatest threat to porbeagle shark abundance; Northwest Atlantic stock populations declined rapidly in the 1960's, increased slightly until the 1990s, declined slowly until the early 2000's, and have remained relatively stable since (SCRS 2020). It is unclear how porbeagle shark abundance will be affected by climate change. The current biomass estimate is 201,000 sharks, and the B_{MSY} is 353,000 sharks (SCRS 2020).

Life History Synopsis: The porbeagle shark inhabits coastal waters, continental shelf waters, and open ocean. The species often inhabits the Gulf of Maine, Scotian Shelf, Georges Bank, deep continental shelf waters, and the Gulf Stream (Skomal et al. 2021). The species seasonally moves on and offshore using the Gulf of Maine and Georges Bank in the fall and moving offshore into the Gulf Stream in the winter (Haugen et al. 2022; Skomal et al. 2021). Young-of-the-year, juvenile, and adult porbeagle sharks are rarely caught at surface or below 200 m. Adults are capable of diving over 1000 m to forage. Tagging studies have revealed that porbeagle sharks encounter waters between 2 to 26°C; however, adult stages were generally caught between 5 to 10°C, and juveniles spend 97% of their time in waters between 6 and 22°C (Campana and Joyce 2003; Skomal et al. 2021).

Porbeagle sharks are viviparous and gestate for 8 to 9 months. Average litter size is 4 pups (Jensen et al. 2002). Porbeagle sharks are generalist feeders who consume 91% teleosts by weight. In spring months they consume cephalopods and pelagic fish, and in the fall they consume primarily groundfish (Joyce et al. 2002). Given the diversity of their diet and ability to forage at various depths, climate change will likely have minimal impact on porbeagle sharks via this life history characteristic.

Porbeagle sharks have a low intrinsic rate of increase, $r_{max} = 0.059$ (Cortes and Semba 2020) and growth rate, $k = 0.07/\text{yr}$ (Natanson et al. 2002). Age at maturity is 8 for males and 13 for females, and a maximum age of 24 for males and 25 for females (Natanson et al. 2002). The last stock assessment for Northwest Atlantic porbeagle shark was conducted in 2020 and found that the stock is overfished and overfishing is not likely occurring ($B_{2017}/B_{MSY} = 0.57$; $F_{2010-2018}/F_{MSY} = 0.413$) (SCRS 2023).

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Nurse shark - *Ginglymostoma cirratum*

Overall Vulnerability Rank = high (86% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (98% certainty)

Directional Effect = neutral (70% certainty)

Data Quality = 38% of scores ≥ 2

<i>Ginglymostoma cirratum</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	1.7		Low Moderate High Very High
	Habitat Specificity	2.1	2.8		
	Mobility and Dispersal of Early Life Stages	1.7	1.9		
	Other Stressors	2.8	1.8		
	Population Growth Rate	3	1.6		
	Prey Specificity	1.7	2.9		
	Reproductive Cycle	1.2	2.3		
	Reproductive Strategy Sensitivity	2.6	2.3		
	Sensitivity to Ocean Acidification	2.5	1.7		
	Sensitivity to Temperature	2.6	2.7		
	Site Fidelity	2.5	1		
	Specificity in Early Life History Requirements	1.2	1.8		
	Stock Size Status	1.8	1.2		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	3	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.3	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Nurse shark (*Ginglymostoma cirratum*)

Contributors: T. Daly-Engel, K. Hannan, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (86% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), Sea Surface Oxygen (3.9) and pH (4.0). Nurse sharks inhabit shallow and deep habitats (up to 75 m) ranging from coral reefs, seagrass beds, mangroves, to deeper reefs and rocky ledges or holes (Castro 2000; NMFS SEFSC, unpublished data). Water temperatures above 30°C appear to be intolerable for certain behaviors, such as aggregating and copulating, but these behaviors resume as the temperature drops (Pratt Jr. et al. 2022).

Biological Sensitivity: **Moderate.** Five sensitivity attribute means scored above 2.5: Other Stressors (2.8), Population Growth Rate (3.0), Reproductive Strategy Sensitivity (2.6), Sensitivity to Temperature (2.64), and Site Fidelity (2.52). Nurse sharks are particularly vulnerable to anthropogenic stressors due their use of nearshore, live bottom habitats (Carrier and Pratt 1998). This includes shallow sandflat habitat used for reproductive aggregations, which nurse sharks return to annually. Certain individuals have been documented to return to the same areas of the Dry Tortugas for 28 consecutive years for reproduction purposes (Pratt Jr. et al. 2022; Carrier et al. 1994).

Data Quality: 38% of the data quality scores were 2 or greater indicating low data quality.

Climate Effects on Distribution and Abundance: Prolonged dependence on vulnerable nearshore habitats and natal philopatry may increase the vulnerability of nurse sharks to the effects of climate change. The shallow waters that nurse sharks depend on are already showing climate effects that experts have predicted will expose this species to ill effects, including abnormally high bottom temperatures and low sea surface oxygen levels. Decreased oxygen and increased temperature have a dampening effect on pH, causing ocean acidification that can threaten populations of hard-shelled invertebrates like mollusks and crustaceans (Gazeau et al. 2013). Such creatures are considered a valuable food source for nurse sharks, in addition to a generally piscivorous diet (Castro 2011). Despite their high exposure to shallow-water warming, there have been relatively few studies conducted on nurse shark movement patterns, so scientists lack data to predict this species' capacity for making range shifts in the face of climate pressure (Kohler and Turner 2001; Pratt et al. 2018). Additionally, nurse sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Another factor shaping climate effects on nurse sharks is site fidelity, whereby species return to the same locations regularly, usually for feeding or reproduction. Site fidelity has been shown to play an important role in determining vagility and dispersal patterns in large sharks, especially with regards to reproductive site fidelity, or philopatry (Chapman et al. 2015). Long-term studies on the reproductive strategies of nurse sharks have shown that adults of both sexes rely heavily on long-term philopatry to shallow (<2m) habitat in the tropical Atlantic, returning regularly over many years (Pratt Jr. et al. 2022). It is not known whether individuals of other life stages exhibit such seasonal reliance on specific sites,

but anecdotal evidence has suggested some degree of site fidelity to artificial reefs among juveniles on the Atlantic coast of Florida (K Holloway-Adkins, pers. comm.).

Philopatry does not necessarily equate to inflexibility when it comes to founding new habitats, and nurse sharks have a long evolutionary history in shallow habitats that are regularly exposed to temperature extremes. Despite their sedentary reputations, nurse sharks are capable of moving long distances to suitable habitat (Hendon et al. 2013). Given their high ecological exposure, strong site fidelity, and known cessation of critical functions at high temperatures, nurse shark abundance may be affected by climate change.

Life History Synopsis: Nurse sharks are large, shallow, benthic sharks that are cosmopolitan in the western North Atlantic (Castro 2011). The biology and ecology of this species is relatively understudied among large charismatic sharks, though it has long been used as an ancestral biomedical model for human immunology (Bettcher et al. 2023; Criscitiello 2014; Pratt Jr. et al. 2022). A noted exception to this is nurse shark reproductive behavior, which has been documented in at least one study from the western North Atlantic that has lasted for nearly 30 years (Pratt Jr. et al. 2022).

In general, sharks' vagility is correlated with both size and habitat: larger species can move further than smaller ones, and pelagic sharks move further than benthic ones. Nurse sharks are both large and benthic, and their tendency to rest on the bottom while buccal pumping water over their gills has given them the reputation of leading a "sedentary" lifestyle (Bettcher et al. 2023). But a few direct tracking studies have shown that nurse sharks are capable of broad dispersal (300 to 540 km), including across the deep-water drop-off of the Gulf Stream that separates the Bahamas and United States Atlantic coast, and have average weekly movements of >7 km (Kohler and Turner 2001; Pratt et al. 2018). Nurse shark distribution is generally limited to eastern and central areas within the Gulf of America, due to associations with hard bottom habitats (Hannan et al. 2012).

Nurse sharks are viviparous and have a biennial reproductive cycle, giving birth to a median brood size of 34 pups (28 to 30.5 cm total length at birth) after a gestation period of 5 to 6 months (Castro 2000). Birth occurs over the course of several days in winter months (Castro 2000). Neonates are generally found in nearshore, shallow reef, rocky, macroalgae, sandy, and seagrass bed habitat (Bettcher et al. 2023). The environmental conditions of nurse shark neonate habitat were an average maximum depth of 20 m, average temperature of 26°C, and salinity of 36 ppt (Bettcher et al. 2023). There is recent uncertainty if these nearshore regions can be classified as nursery grounds as parturition occurs over several days and locations (Bettcher et al. 2023; Pratt Jr. et al. 2022). Age of maturity estimates vary; Pratt Jr. et al. 2000 uses an estimate of 15 years, Carlson et al. 2021 an estimate of 25 years, and Castro 2000 a total length of 223 to 231 cm for females and 214 to 214.6 cm for males. Diet of nurse sharks consists of teleosts and invertebrates (crabs, shrimp, squid, octopus, spiny lobster, sea urchin) (Castro 2000).

Little information is available regarding the population growth or stock status of nurse shark within the western North Atlantic. Maximum age estimates range from 35 to 43 years (Carlson et al. 2021; Pratt Jr. et al. 2022). Modest increases in abundance (1.9%/yr) were found in United States populations, however, declines were found throughout the Central Atlantic leading to a 20 to 29% population

reduction over the past 90 years (Carlson et al. 2021). Nurse sharks are federally managed within the United States and have no completed stock assessment available (NMFS 2022).

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Oceanic whitetip shark - *Carcharhinus longimanus*

Overall Vulnerability Rank = very high ■ (73% certainty)

Biological Sensitivity = high ■

Climate Exposure = very high ■

Distributional Vulnerability Rank = very high ■ (70% certainty)

Directional Effect = neutral (68% certainty)

Data Quality = 46% of scores ≥ 2

<i>Carcharhinus longimanus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.6		■ Low ■ Moderate ■ High ■ Very High
	Habitat Specificity	1.4	2.2		
	Mobility and Dispersal of Early Life Stages	1	1.9		
	Other Stressors	1.5	1.5		
	Population Growth Rate	3.2	2		
	Prey Specificity	1.2	2.2		
	Reproductive Cycle	1.3	1.8		
	Reproductive Strategy Sensitivity	1.2	1.2		
	Sensitivity to Ocean Acidification	1	1.6		
	Sensitivity to Temperature	2.3	2.2		
	Site Fidelity	2.3	2.2		
	Specificity in Early Life History Requirements	1	2		
	Stock Size Status	3.1	2		
	Sensitivity Score	high			
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.6	-		
	Sea Surface Salinity	2.2	-		
	Sea Surface Temperature	3.7	-		
	Exposure Score	very high			
Overall Vulnerability Rank		very high			

Oceanic whitetip shark (*Carcharhinus longimanus*)

Contributors: J. Carlson, E. Gutierrez, T. Loughran, C. McCandless, M. Passerotti

Overall Climate Vulnerability Rank: **Very High** (73% certainty from bootstrap analysis).

Climate Exposure: **Very High**. Three exposure factors contributed to this score: pH (4.0), Sea Surface Oxygen (3.7) and Sea Surface Temperature (3.7). Oceanic whitetip sharks are found throughout the world in tropical and subtropical waters, preferring temperatures greater than 20°C (Tolotti et al. 2017; Young and Carlson 2020). Temperatures above 28°C were found to impact vertical movement of oceanic whitetip sharks, suggesting behavioral thermoregulation to avoid warmer waters (Andrzejaczek et al. 2018; Tolotti et al. 2017).

Biological Sensitivity: **Moderate**. Two sensitivity attribute means scored above 2.5: Stock Size Status (3.1) and Population Growth Rate (3.2). The population of oceanic whitetip sharks has declined throughout the Atlantic since the 1990s, and by an estimated 88% in the Gulf of America (Young et al. 2017; Young and Carlson 2020). The species has a relatively low intrinsic rate of increase (0.33-0.081) and late age of maturity (6-9 years), leading to low population growth (Young and Carlson 2020; Young et al. 2017). In 2018, oceanic whitetip shark was listed as threatened throughout its range under the Endangered Species Act (NMFS 2018).

Data Quality: 46% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Limited information is available regarding the effects of climate factors to the distribution and abundance of oceanic whitetip sharks. As an ectothermic species, oceanic whitetips favor warm surface water temperatures (>20°C) (Young and Carlson 2020). A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) projected the majority of suitable habitat to be maintained, with possible poleward expansions as a result of this warm-water preference (Diaz-Carballido et al. 2023). Suitable habitat loss was concentrated in coastal waters throughout the Gulf of America, southeastern Atlantic, and Caribbean Sea.

Further resilience to climate change may be afforded to this species through behavioral thermoregulation. When encountering sea surface temperatures above 28°C in summer months, oceanic whitetips were found to thermoregulate by occupying deeper waters (>50 m) (Andrzejaczek et al., 2018; Dell'Apa et al. 2023). This behavioral thermoregulation tactic may indicate that oceanic whitetip sharks can seek refuge from warming sea surface temperatures by utilizing cooler, deeper depths (Andrzejaczek et al., 2018; Dell'Apa et al. 2023).

Overfishing is the greatest threat to the global oceanic whitetip abundance. Population declines within the northwestern Atlantic stock were recorded between the 1990 and 2000s, but likely occurred prior to 1990 as a result of pelagic longline fisheries (Rigby et al. 2019). Population levels of the northwestern Atlantic stock have since stabilized, though abundance remains low throughout the northwestern Atlantic and Gulf of America (Young et al. 2017). It is unclear how oceanic whitetip abundance will be affected by climate change.

Life History Synopsis: The oceanic whitetip shark is globally distributed throughout tropical to subtropical waters, often found in pelagic or shelf-associated waters (Young and Carlson 2020). Oceanic whitetip sharks spend nearly all their time in warm near-surface mixed layers (24 to 26°C, 25.5 to 50 m), periodically diving below 1,000 m (Carlson and Gulak 2012; Howey et al. 2016). These excursions largely occur at night, and last less than 30 minutes on average (Howey et al. 2016). Vertical distribution is influenced by seasonal changes in water temperature; as surface temperatures cool in winter, oceanic whitetip sharks are found closer to the surface (<50 m) and vice versa in summer (Andrzejczek et al. 2018). Oceanic whitetip sharks are highly mobile, generally following the extent of the Gulf Stream throughout the Atlantic, Gulf of America, and Caribbean Sea. Juvenile stages are less well-known but are similarly mobile to adult stages, but were found to return to the general area of origin (within 500 km) during tagging studies (Tambourgi et al. 2013; Tolotti et al. 2015). Site fidelity and philopatry has been documented within the Atlantic and Caribbean Sea (Howey-Jordan et al. 2013; Madigan et al. 2015; Tolotti et al. 2015).

Oceanic whitetips are viviparous, giving birth to an average litter size of 6 pups (maximum length at birth reported as 70 cm total length) after a gestation period of 10 to 12 months (Young and Carlson 2020; Lessa et al. 1999a). The reproductive cycle is assumed to be biennial, though further verification is required (Castro 2011; Tambourgi et al. 2013). Areas throughout the south Atlantic (Vitoria-Trindade seamount chain, Cuba, and Haiti) have been identified as possible nursery and pupping grounds (Aguilar et al. 2014; Domingo et al. 2007; Ruiz-Abierno et al. 2021a; Ruiz-Abierno et al. 2021b; Young and Carlson 2020). Age at maturity is estimated to be 6 to 9 years (Lessa et al. 1999b). Temporal differences in diet related to migratory patterns (i.e., inshore/shallow-offshore/deep movements) are evident (Madigan et al. 2015). This species is a top predator in pelagic environments, feeding on a variety of teleosts, cephalopods, crustaceans, marine mammal remains, marine reptiles, seabirds, and other elasmobranchs (Bonfil et al. 2008; Young et al. 2017). Most prey are taken from the upper 200 m of the water column but there is evidence of foraging during deep dives (Papastamatiou et al. 2020).

Oceanic whitetip sharks have a relatively high population growth rate compared to other pelagic shark species (population growth rate = 1.117) (Young and Carlson 2020). Von Bertalanffy growth coefficient is estimated as 0.099 within the south Atlantic (Lessa et al. 1999b). Maximum age is currently estimated at 25 years, though may extend to 36 years (Young and Carlson 2020).

Primary stressors to oceanic whitetip sharks globally include commercial fisheries and the international fin trade (Young et al. 2017). Oceanic whitetips are caught as bycatch throughout international pelagic commercial fisheries. Between the 1950's and 1990's, pelagic longline fisheries contributed to an estimated 50 to 88% population decline throughout the western North Atlantic and Gulf of America (Rigby et al. 2019; Young et al. 2017; Young and Carlson 2020). Since 2010, as a result of an International Commission for the Conservation of Atlantic Tunas (ICCAT) recommendation, it has been illegal for United States pelagic longline and recreational fishermen and pelagic longline fishermen of other ICCAT nations to harvest oceanic whitetip sharks (SCRS 2023). Since early 2024, the United States has prohibited the retention of oceanic whitetip sharks on all fishing gear (NMFS 2024). There is no completed stock assessment available for oceanic whitetip sharks. The oceanic whitetip was listed as threatened under the Endangered Species Act in 2018 (NMFS 2018).

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Roundscale spearfish - *Tetrapturus georgii*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = moderate ☐Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☐ (100% certainty)

Directional Effect = neutral (96% certainty)

Data Quality = 0% of scores ≥ 2

<i>Tetrapturus georgii</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	1.2		Low Moderate High Very High
	Habitat Specificity	1.6	1.5		
	Mobility and Dispersal of Early Life Stages	2.1	1.1		
	Other Stressors	2	1.2		
	Population Growth Rate	2.6	0.9		
	Prey Specificity	1.8	1.2		
	Reproductive Cycle	2.6	1.6		
	Reproductive Strategy Sensitivity	2.1	0.6		
	Sensitivity to Ocean Acidification	2	1.4		
	Sensitivity to Temperature	2.1	1.8		
	Site Fidelity	2	0.7		
	Specificity in Early Life History Requirements	2.3	0.7		
	Stock Size Status	2.6	0.8		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Roundscale spearfish (*Tetrapturus georgii*)

Contributors: J. Graves, D. Kerstetter, T. Loughran, D. Snodgrass

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Moderate.** Three sensitivity attribute means scored above 2.5: Population Growth Rate (2.6), Stock Size and Status (2.6), and Reproductive Cycle (2.6). There is limited to no knowledge of roundscale spearfish stock status, population growth, or reproductive behaviors.

Data Quality: 0% of the data quality scores were 2 or greater. This indicates low confidence in life history knowledge, particularly for Population Growth Rate, Reproductive Strategy Sensitivity, Site Fidelity, Specificity in Early Life History Requirements, and Stock Size and Status.

The validity of *Tetrapturus georgii* was somewhat questionable for many years. It was described morphologically in 1841 with a specimen from Madeira. However, reports differed between only an eastern Atlantic or an Atlantic-wide distribution. Spurred by reports of roundscale spearfish by commercial fisheries observers in the U.S. pelagic longline fishery, Shivji et al. (2006) described the species on a molecular level and confirmed its presence in the western Atlantic. The species appears to have been conflated with white marlin (*Kajikia albidus*) in both commercial and recreational catches for as long as records have been kept and beyond. More user-friendly identification characteristics were described by Beerkircher and Serafy (2011) so that the species could be more easily distinguished without having to wait on molecular work or having to remove the animal from the water for inspection of obscure meristic differences. Still, the fishery participants have not fully embraced the idea of field identification of the species on a large scale. Due to this, there is very little data available that would allow for exact definition of catch rates, life history traits and other general ecological aspects of this species.

Currently, any life-history information and understanding of roundscale spearfish can only be drawn from similar istiophorid species. White marlin is often presumed to be the most relevant proxy, especially given the historical conflation of the two species in catch records, notwithstanding the limited life-history information of that species as well. However, white marlin and roundscale spearfish have been separated into different genera (Collette et al. 2006) based on genetic analysis so it is possible that each species has distinct ecological niches. For example, Beerkircher et al. (2009) found significant differences in the proportion and seasonality of *T. georgii* vs. *K. albidus* in several regions of the Atlantic, suggesting that there is habitat partitioning between the species. Thus, using *K. albidus* as a representative for *T. georgii* may be problematic. The impacts of this confusion in regards to population levels and stock assessments is further described by Beerkircher et al. (2009). The descriptions below are primarily based on this comparison with *K. albidus* unless otherwise noted.

Climate Effects on Distribution and Abundance: No published studies have specifically examined the effect of climate factors on the population productivity or changes in distribution for roundscale spearfish under climate change scenarios. Roundscale spearfish utilize the uppermost regions of the water column all over the Atlantic Ocean from 29°S to 38°N (Bernard et al. 2013). This range is much

broader than previously suspected from historical catch reporting. Regardless of ocean depth they primarily occupy the realm above the thermocline, spending 80 to 95% of their time above 100 meters. The largest proportion of that use is no deeper than 50 meters (Horodysky et al. 2007, Hoolihan et al. 2015). Roundscale spearfish spend the majority of their time in water temperatures ranging from 24 to 28°C. The amount of time spent in these depths and temperatures demonstrates that they conduct the bulk of their life cycle here. Increases in water temperature and salinity brought on by rapid climate change could have significant impacts on all aspects of their life history; rate of growth and maturity, range, prey availability and utilization, spawning frequency and success as well as individual strength and longevity. Likewise, increasing acidity could have significant negative impacts on all aspects, especially feeding and larval supply. While dissolved oxygen requirements are poorly understood, a reduction in depth utilization has been observed for sailfish and blue marlin in areas of low dissolved oxygen (<3.5ml/l), and other billfish, such as roundscale spearfish may be similarly impacted (Dell'Apa et al. 2023; Kerstetter et al. 2009; Prince and Goodyear 2006; Prince et al. 2010).

Regarding the distribution of roundscale spearfish, it might be assumed that their available habitat might expand with warming water temperatures but their upper thermal and dissolved oxygen thresholds may also decrease their ability to use the regions historically and currently being utilized. Because their habitat utilization is so narrow and their population appears to be rather low roundscale spearfish may be more severely impacted by oceanographic changes incurred from rapid climatic changes. Overall, the lack of knowledge of this species and the probable low population size make it especially susceptible to negative impacts brought on by a rapidly changing climate regime and its impacts on the ocean surface waters.

Life History Synopsis: As mentioned above, very little is known about specific life history traits for roundscale spearfish and for all members of the istiophorid billfishes. Inferences on the roundscale spearfish species' life history will be drawn from similar istiophorid species.

Movements inferred from conventional tagging results show that billfish tend to utilize a large portion of the ocean and adjacent seas with seasonal movements from the United States mid- Atlantic to northern South America (Orbesen et al. 2008). Based on fisheries-dependent data, there is a possible connection between central and southern Caribbean Sea waters and the United States fisheries between the Bahamas and Cape Cod, Massachusetts. More accurate measures of horizontal movement patterns have been better described through the use of electronic tag technology, especially pop-up satellite tags. The lower sample size may not accurately represent the whole population, but it does demonstrate trans-oceanic movement as well as utilization of more restricted areas such as the Gulf of America.

Roundscale spearfish feeding habits tend to focus on a variety of prey species, mostly small sized epipelagic fishes and cephalopods (Pinheiro et al. 2010). Over half of the prey items are epipelagic fishes and cephalopods that have a heavy reliance on zooplankton within sargassum beds (Luckhurst 2015) which, depending on how that algae fares with sea surface temperatures changes may severely impact prey availability.

Reproductive maturity is attained at just under 150 cm lower jaw fork length (Oliveira et al. 2007). The majority of recreational and commercial catches are of mature adult animals (Goodyear and Prince

2003, Goodyear et al. 2003). Batch fecundity was between 770,000 and 880,000 oocytes for the most common sized females collected (Oliveira et al. 2007). Spawning takes place in the Spring in tropical waters from April to June (Arocha and Barrios 2009, Prince et al. 2005). The actual timing and method of spawning are unknown but they are assumed to be broadcast spawners like most other istiophorid species. The periodicity of spawning is unknown.

Roundscale spearfish appear to reach reproductive maturity between 3 to 6 years of age for both genders and reach a maximum age of 10 to 14 years (Die and Drew 2008, Krusic-Golub et al. 2022). The last stock assessment on roundscale spearfish and white marlin was conducted in 2019 (using data through 2017) and found that the stock was overfished but overfishing was not occurring (B_{2017}/B_{MSY} : 0.58 (0.27–0.87); F_{2017}/F_{MSY} : 0.65 (0.45–0.93)) (SCRS 2023).

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S3 File Supporting Information: Species Narratives

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Sand tiger shark - *Carcharias taurus*

Overall Vulnerability Rank = high (51% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (99% certainty)

Directional Effect = neutral (44% certainty)

Data Quality = 77% of scores ≥ 2

<i>Carcharias taurus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.8	2.5		Low Moderate High Very High
	Habitat Specificity	1.8	2.7		
	Mobility and Dispersal of Early Life Stages	1.3	2.6		
	Other Stressors	2.1	1.4		
	Population Growth Rate	3.6	2		
	Prey Specificity	1.6	2.5		
	Reproductive Cycle	1.3	2.5		
	Reproductive Strategy Sensitivity	2.2	2		
	Sensitivity to Ocean Acidification	1.6	1.9		
	Sensitivity to Temperature	1.8	2.4		
	Site Fidelity	3	2.2		
	Specificity in Early Life History Requirements	1.6	2.7		
	Stock Size Status	2.7	1.8		
	Sensitivity Score		moderate		
Exposure Factors	Bottom Salinity	2.7	-		
	Bottom Temperature	3.6	-		
	Mixed Layer Depth	1.4	-		
	pH	3.9	-		
	Primary Production	1.3	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

Sand tiger shark (*Carcharias taurus*)

Contributors: A. Carlisle, J. Kneebone, T. Loughran, C. McCandless, C. Peterson, G. Skomal

Overall Climate Vulnerability Rank: **High** (51% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.6), Sea Surface Oxygen (3.8) and pH (3.9). Sand tiger sharks are found in warm-temperate to tropical coastal and continental shelf waters (Branstetter, 2002). Adult stages are found to stay in temperatures of 17 to 23°C for 95% of the observed time, moving north into the Mid-Atlantic coastal ocean only when surface waters warm to approximately 18°C (Teter et al. 2015). Adults were also found to avoid cooler deep and shallow waters, occurring in deeper waters when sea surface temperature was 15 to 18°C (Haulsee et al. 2018).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 3.0: Population Growth Rate (3.6) and Stock Size and Status (2.7). There is currently no estimate available for relative biomass. Sand tiger sharks were listed as a Species of Concern in 1997, but have since shown greater genetic diversity indicating possible recovery of the stock (Gray et al. 2013).

Climate Effects on Distribution and Abundance: Given the species' reliance upon coastal embayments for foraging, particularly during the juvenile life stage (Kneebone et al. 2012), habitat alteration (e.g., increasing temperatures, decrease in dissolved oxygen, and phytoplankton blooms) may greatly impact their distribution and fitness. For example, water temperature is a predictor of sand tiger presence in the Delaware Bay region (Haulsee et al. 2018) and has been linked to sand tiger emigration from summer nursery habitat in Massachusetts (among other variables, like photoperiod and prey) (Kneebone et al. 2012). Thus, the coastal distribution of this species and its use of specific habitats, specifically shallow bays and estuaries, residency times within these habitats and timing of seasonal migration will likely be altered by increasing water temperatures. This may impact Habitat Areas of Particular Concern designated for all life stages of sand tiger sharks within Delaware Bay and young of the year and juveniles in coastal Massachusetts (NMFS 2017). Additionally, sand tiger sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021). Climate impacts on preferred prey species, particularly Atlantic menhaden (*Brevoortia tyrannus*), a species whose population dynamics are strongly influenced by climate effects (Buchheister et al. 2016; Kneebone et al. 2012), may also adversely impact sand tiger bioenergetics, distribution, and abundance.

It is also important to consider sand tiger preference for habitat with structure, whether this is natural areas like where deep channels meet shoals or artificial reefs and shipwrecks; sand tigers aggregate around these sites as evidenced by historical and recent data in Delaware Bay (C McCandless pers comm). Sand tigers have also shown preferences for artificial structures with increased complexity and vertical relief, potentially due to increased feeding opportunities (Paxton et al. 2020). With the expansion of offshore wind energy development planned along the United States Atlantic coast, one can infer the presence of wind turbines within sand tiger migratory routes or seasonal habitat may be attractive to the species. While these structures may provide increased feeding opportunities that can

promote animal fitness; alternately, they may reduce fitness by altering migratory routes and delaying arrival at more beneficial habitats such as nursery areas, increasing competition and/or predation, or increasing fishery interactions with the species or their prey (McCandless et al. 2023). Additionally, the operation of wind energy areas may also alter sand tiger behavior due to changes in hydrodynamics within the wind energy area and surrounding areas and the electromagnetic emissions from the high voltage cables between turbines and that export energy from the turbines back to land (McCandless et al. 2023). Both climate change and wind energy development have the potential to alter sand tiger habitat and influence sand tiger behavior and habitat use, and these stressors will likely operate concurrently with potential cumulative and synergistic effects on sand tiger bioenergetics, distribution and abundance.

Life History Synopsis: The sand tiger is a globally distributed, temperate species that occurs from the northern Gulf of America to the Gulf of Maine in the western North Atlantic (Compagno 1984). The species is highly migratory and occurs over a wide range of depths (1 to 188 m) and temperatures (9 to 31°C) and has been documented in salinity and dissolved oxygen ranging from 18.0 to 36.9 psu and 2.5 to 9.8, respectively (Kneebone et al. 2014; McCandless et al. 2007; Teter et al. 2015; Haulsee et al. 2018; C McCandless pers comm). Sand tigers are thought to be born at southern latitudes (off North Carolina and Florida) and initially migrate north to New England waters (Kneebone et al. 2014). Although small juveniles (83 to 94 cm fork length) within the size range for young of the year (<95 cm, Goldman et al. 2006) have been caught in Delaware Bay, the majority of juveniles encountered are considered age 1 or older (C McCandless pers comm). Juveniles make seasonal migrations along the continental shelf (waters <80 m) between summer habitat in Delaware Bay up into the Gulf of Maine and winter habitat off Cape Hatteras, North Carolina to central Florida, with evidence of an ontogenetic shift around age 1 to 2 years, in which individuals cease to migrate into Gulf of Maine waters (Kneebone et al. 2014; Kohler and Turner 2019). Compared to juveniles, adult migration is more latitudinally compressed, with individuals remaining resident in some locations year round and fewer individuals straying north of Cape Cod (Teter et al. 2015; Haulsee et al. 2018). Adult female movements are also linked to reproduction, with pregnant and non-pregnant females aggregating off North Carolina and Delaware, respectively, during the summer months (Wyffels et al. 2022). Ontogenetic shifts in depth and temperature are evident in the species, with young juveniles generally occupying shallower depths and cooler temperatures than adults.

Sand tigers are relatively slow growing, with males reaching sexual maturity around 6 to 7 years of age and females around 9 to 10 years (Gilmore et al. 1983; Goldman et al. 2006). Females produce only two young per pregnancy, which occurs either biennially or triennially; a 1 or 2 year resting period is evident after parturition (Bansemer and Bennett 2009; Branstetter and Musick 1994; Dicken et al. 2006; Castro 2009; Lucifora et al. 2002; Wyffels et al. 2022). Longevity is estimated at around 40 years (Passerotti et al. 2014); thus, a single female may only produce up to 14 to 20 young in her lifetime. Both juveniles and adults are generalist predators that consume lobsters, crabs, and squids, but prey primarily on fishes such as menhaden, summer flounder, skates, monkfish, sea robins, scup, spot, bluefish, butterfish, and tautog (Branstetter 2002; Kneebone et al. 2012). Larger adults also commonly feed on elasmobranchs, including skates and smaller sharks.

The population status of sand tigers in the western North Atlantic is uncertain (Carlson et al. 2009), but severe population decline has been postulated since the 1980s and 1990s (Musick et al. 2000). However, an updated trend analysis of CPUE for 1990 through 2019 revealed an annual rate of increase of 1.3% (Rigby et al 2021). Landing of sand tiger sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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S3 File Supporting Information: Species Narratives

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Sandbar shark - *Carcharhinus plumbeus*

Overall Vulnerability Rank = high (98% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (100% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 69% of scores ≥ 2

<i>Carcharhinus plumbeus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	2.7		Low Moderate High Very High
	Habitat Specificity	1.9	2.6		
	Mobility and Dispersal of Early Life Stages	1.5	2.7		
	Other Stressors	2.6	1.5		
	Population Growth Rate	3.5	2.6		
	Prey Specificity	1.6	2.6		
	Reproductive Cycle	1.2	2.4		
	Reproductive Strategy Sensitivity	2.1	1.8		
	Sensitivity to Ocean Acidification	1.4	1.5		
	Sensitivity to Temperature	1.8	2.5		
	Site Fidelity	2.6	1.9		
	Specificity in Early Life History Requirements	1.5	2.2		
	Stock Size Status	2.7	2.4		
	Sensitivity Score		moderate		
Exposure Factors	Bottom Salinity	3	-		
	Bottom Temperature	3.9	-		
	Mixed Layer Depth	1.1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

Sandbar shark (*Carcharhinus plumbeus*)

Contributors: A. Carlisle, J. Carlson, L. Crawford, W. Driggers, B. Frazier, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (98% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors contributed to this score: Bottom salinity (3.0), Bottom Temperature (3.9), pH (4), and Sea Surface Oxygen (3.8). The sandbar shark is distributed globally in warm-temperate and tropical oceans (Compagno 1984). The western North Atlantic population ranges from off Cape Cod, Massachusetts into the Gulf of America, found in temperatures between 8 and 31°C (Castro 2011; Kohler & Turner 2019; McCandless et al. 2007a; NMFS NEFSC & SEFSC, unpublished data). Neonates and juveniles use nursery habitat in nearshore and estuarine waters along the United States Atlantic coast with Delaware and Chesapeake Bays providing the principle nursery grounds for the species (Castro 1993; Merson & Pratt 2001; Grubbs & Musick 2002; Grubbs et al. 2007; Grubbs and Musick 2007; McCandless et al. 2007a; McCandless et al. 2007b; Merson & Pratt 2007). Juveniles have been documented in a variety of coastal habitats (tidal creek, tidal flat, channels, beach) over mud, sand, shell, and rock bottoms with temperatures between 15.0 and 31.0°C, salinities between 13 and 39 psu, dissolved oxygen between 2.1 and 8.3 mg/L, and depths between 0.8 and 29.3m (Rechisky & Wetherbee 2003; Grubbs and Musick 2007; McCandless et al. 2007a; McCandless et al. 2007b).

Biological Sensitivity: **Moderate.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.48). Sandbar sharks are large-bodied, slow growing sharks, and the species has a low population growth rate, estimated to be between 0.050 and 0.168/yr (Cortes 2016).

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: A study using ecological niche modeling (MaxEnt) under four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that sandbar sharks will gain suitable habitat across their global range. In United States waters they may gain habitat around New England, but they may lose habitat in the eastern Gulf of America along the coast of Florida (Diaz-Carballido et al. 2022). Additionally, sandbar sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Adults and juveniles display philopatry and seasonal site-fidelity to bays and estuaries (Merson & Pratt 2001; Grubbs et al. 2007; McCandless et al. 2007b; Portnoy et al. 2008; Merson and Pratt 2007). Some older juveniles and adults also exhibit seasonal site fidelity to non-natal habitat (Altobelli and Szedlmayer 2020; Collatos et al. 2020). Juveniles generally use nursery areas during warmer summer months and return to them seasonally for up to 12 years (Grubbs et al. 2007; McCandless et al. 2007b). High mobility has been demonstrated within the bounds of large nursery areas such as the Delaware and lower Chesapeake Bays (Rechisky and Wetherbee 2003; Grubbs et al. 2007; Grubbs and Musick 2007). Critical nursery areas for sandbar sharks are expected to warm and decrease in oxygen concentration due to climate change. Habitat modeling of the Chesapeake Bay revealed that the area of

suitable bottom habitat will decrease by 2100 but when accounting for the entire water column, the amount of suitable habitat will increase (Crear et al. 2020). Additionally, when exposed to very warm (32°C, <3.5 mg/L) water conditions, sandbar shark performance significantly decreased, elucidating the temperature and oxygen thresholds of the species and indicating that current nursery areas may become unsuitable (Crear et al. 2019). It is unclear how climate change will affect sandbar shark abundance.

Life History Synopsis: The sandbar shark is a large-bodied coastal species that occupies inshore and offshore habitats in temperate, subtropical, and tropical waters around the world, and in the western North Atlantic inhabits continental and insular shelf habitats from southern New England into the Gulf of America (Castro 2011). The species migrates great distances and displays seasonal residency and philopatry to bays and estuaries (Grubbs et al. 2007; Koher and Turner 2019; McCandless et al. 2007b).

Sandbar sharks are viviparous with a reproductive periodicity of 2 to 3 years and average brood size of 8 pups (average 46 cm fork length at birth) (Baremore and Hale 2012; Piercy et al. 2016; SEDAR 2017). Mating aggregations have been observed on the Florida shelf (Springer 1960). Parturition occurs from May to July after approximately 12 months gestation (Baremore and Hale 2012). Neonates occupy coastal bays and estuaries through the mid-Atlantic and south Atlantic United States coast (Castro 1993). Sandbar sharks reach 50% maturity at 151.6 and 154.9 cm fork length for males and females, respectively, translating to 12 (males) and 13 (females) years of age based on vertebral band-pair counts (Baremore and Hale 2012; Hale and Baremore 2013). Recent estimates using additional data have updated this to 142 cm fork length (males) and 153 cm fork length (females) (Driggers et al. 2017). Sandbar sharks are long-lived (over 30 years) and recent age-validation studies on this species indicate that vertebral band-pair deposition rates change with ontogeny and modeling tag-recapture data may be more accurate for age estimation (Andrews et al. 2011; Casey and Natanson 1992; Natanson and Deacy 2019). Determining age at 50% maturity using tag-recapture data and Baremore and Hale (2012) size at 50% maturity estimates resulted in ages of 17 and 18 years for males and females, respectively (Natanson and Deacy 2019). The intrinsic rate of increase for sandbar population is between 0.050 and 0.168 (Cortes 2016), and the von Bertalanffy growth coefficient for both sexes combined is 0.15 using vertebral band counts (Driggers et al. 2017) and ranges between 0.04 and 0.09 using tag-recapture growth models (Casey and Natanson 1992; Natanson and Deacy 2019). Population declines in the late 20th century led to reduced genetic diversity of the stock (Heist et al. 1995).

Sandbar sharks are generalist predators with a high trophic level, consuming invertebrates, teleosts, and elasmobranchs (Stillwell and Kohler 1993; Cortés 1999; Ellis and Musick 2007; Shiffman et al. 2014). Juvenile sandbar sharks primarily feed on teleosts with crustaceans contributing significantly to young of the year diets in Delaware Bay and South Carolina estuaries and small juveniles including young of the year in Virginia estuarine habitat (Stillwell and Kohler 1993; Ellis and Musick 2007; McElroy 2009; Shiffman et al. 2014). There is an ontogenetic shift toward teleost and elasmobranch prey with increasing size (Ellis and Musick 2007; McElroy 2009; Stillwell and Kohler 1993).

The last stock assessment of sandbar shark was conducted in 2017 and found that the stock was overfished but not subject to overfishing ($SSF_{2015}/SSF_{MSY} = 0.77$; $F_{2015}/F_{MSY} = 0.58$) (SEDAR 2018). Modeled catch and estimated fishing mortality of sandbar sharks peaked in late 1980s and mid 1990s and has

continued to decline. Estimated stock abundance and recruitment were stable across most of the 20th century and then declined rapidly between the 1980s and early 2000s. Abundance slowly increased in the 2010s up to the last year of data in 2015, while interannual variability makes the recruitment trend unclear in recent years.

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Scalloped hammerhead shark - *Sphyrna lewini*

Overall Vulnerability Rank = high (81% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (100% certainty)

Directional Effect = neutral (68% certainty)

Data Quality = 54% of scores ≥ 2

<i>Sphyrna lewini</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	2.7		Low Moderate High Very High
	Habitat Specificity	2	2.2		
	Mobility and Dispersal of Early Life Stages	1.3	2.4		
	Other Stressors	2.4	1.8		
	Population Growth Rate	3.2	2		
	Prey Specificity	1.6	2.3		
	Reproductive Cycle	1.2	2.3		
	Reproductive Strategy Sensitivity	2.1	1.8		
	Sensitivity to Ocean Acidification	1.4	1.4		
	Sensitivity to Temperature	1.8	2.4		
	Site Fidelity	2.2	1.9		
	Specificity in Early Life History Requirements	1.2	2.5		
	Stock Size Status	2.6	1.8		
	Sensitivity Score		moderate		
Exposure Factors	Bottom Salinity	3.1	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

Scalloped hammerhead shark (*Sphyrna lewini*)

Contributors: A. Carlisle, J. Carlson, D. Coffey, B. Curtis, M. Drymon, B. Frazier, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (81% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors were the drivers in determining this score: Bottom Temperature (3.8), pH (4), Sea Surface Oxygen (3.9). Shallow coastal and estuarine waters are used by young of year and juveniles, while mid- to shelf edge and oceanic habitats are utilized as the individuals mature (Castro 2011; Wells et al. 2018). Adults utilize tropical coastal and oceanic waters, and are generally found in surface waters between 13 and 32°C (Castro 2011; Drymon et al. 2020; J. Carlson pers comms; NMFS SEFSC, unpublished data). Scalloped hammerhead sharks utilize various water column depths with repeated diel vertical behaviors, diving at night to almost 1,000 m where temperatures reach 5.0°C (Hoffmayer et al. 2013; Gallagher and Klimley 2018; Hutchinson et al. 2023).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (3.2) and Stock Size and Status (2.6). Knowledge about temperature effects on reproduction is limited. Scalloped hammerhead sharks have been shown to have an intrinsic rate of increase (r) of 0.29 (Hayes et al. 2009), and ages at maturity for females at 16.1 years, and males at 11.3 years (Montcrief-Cox et al. 2021).

Data Quality: 54% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Scalloped hammerhead sharks have some of the highest metabolic rates observed in ectothermic elasmobranchs (Lowe 2001), which suggests that increasing temperatures may affect their physiological performance. This is exacerbated by the fact that warmer waters have less dissolved oxygen, so while metabolic rates (and oxygen and energetic requirements for aerobic respiration) increase with temperature, oxygen availability decreases. Together, rising temperatures and reduced oceanic dissolved oxygen may detrimentally affect scalloped hammerhead shark physiological performance given their high metabolic rate (Crear et al. 2019; Dell’Apa et al. 2023; Gallagher et al. 2014; Lowe 2001).

As a group, Sphyrnidae have some of the highest observed levels of at-vessel and post-release mortality of any shark (Binstock et al. 2023; Braccini and Waltrick 2019; Coelho et al. 2012; Eddy et al. 2016; Ellis et al. 2017; Gulak et al. 2015; Morgan and Burgess 2007; Whitney et al. 2021) suggesting that they are quite sensitive to stressors and physiological challenges. These dynamics may have direct impacts on the abundance, distribution, and ecology of this species.

This species is also known to exhibit behavioral thermoregulation by using different parts of the water column to select preferred temperatures (between 23 and 26°C), further suggesting a changing thermal environment may influence the geographic distribution of this species as well as the vertical distribution (Ketchum et al. 2014a). Climate change may also directly impact the ecology of this species, as warmer waters and oceanic regimes (e.g., El Nino events) are generally associated with reduced productivity and prey availability. During El Nino events, the trophic ecology of scalloped hammerhead sharks changes as they are forced to switch to a more generalist foraging strategy due to reduced ecosystem productivity

and prey abundance, with potentially significant impacts on their bioenergetics (Arnés-Urgellés et al. 2021; Rodriguez-Burgos et al. 2022). In addition, the fact that they utilize shallow coastal environments as nursery areas (Duncan and Holland 2006; Ebert et al. 2021) makes their early life history stages particularly vulnerable to changes in the environment and habitat.

Life History Synopsis: The scalloped hammerhead shark is a globally distributed species primarily found in warm temperate to tropical waters over continental shelves and adjacent oceanic waters (Ebert et al. 2021; Rigby et al. 2019; Wells et al. 2018). They are generally found in shallower waters <275 m though they have been observed to depths in excess of 1000 m (Moore and Gates 2015). Younger sharks inhabit shallower, nearshore habitats (including bays and estuaries), and larger sharks occur in more offshore habitats, including seamounts (Ebert et al. 2021; Rigby et al. 2019). Scalloped hammerhead sharks are migratory, moving inshore or poleward during the warmer months and offshore or towards lower latitudes during colder months, often migrating in large schools (Branstetter 1987; Clarke et al. 2015; Ebert et al. 2021; Rigby et al. 2019). Wells et al. 2018 identified bathymetric features as a key habitat preference for scalloped hammerhead sharks in the northern Gulf of America, noting that mid-shelf and shelf edge habitat is of particular importance. Multi-year electronic tagging studies have also revealed seasonal high site fidelity of adults to specific habitats and nurseries (Bessudeo et al. 2011, Ketchum et al. 2014b, Wells et al. 2018, Hutchinson et al. 2023).

Scalloped hammerhead sharks are a large shark, reaching a maximum size of 370 to 420 cm, that can live up to 39.5 years (Frazier et al. 2021). Like many elasmobranchs, male sharks mature at a lower size (148 cm fork length (FL)) and age (11.1 yrs) than females (184 cm FL, 16.1 yrs) (Montcrief-Cox et al. 2021). Scalloped hammerhead sharks are viviparous with a yolk sac placenta and gives birth to between 12 and 41 pups (31 to 57 cm at birth) in a biennial reproductive cycle (Branstetter 1987; Clarke et al. 2015; Ebert et al. 2021; Rigby et al. 2019). They have a diverse diet, though they feed primarily upon teleosts, other elasmobranchs, benthic invertebrates, and cephalopods (Bush and Holland 2002; Ebert et al. 2021; Flores-Martinez et al. 2017; Rigby et al. 2019; Torres Rojas et al. 2015).

Scalloped hammerhead sharks are caught around the world as target and bycatch in a variety of artisanal fisheries as well as coastal and pelagic commercial fisheries (Rigby et al. 2019), yet have some of the highest observed levels of at-vessel and post-release mortality (Braccini and Waltrick 2019; Coelho et al. 2012; Eddy et al. 2016; Gulak et al. 2015). The Central and Southwest Atlantic Distinct Population Segment is currently listed as threatened under the Endangered Species Act. The Carolina hammerhead shark (*S. gilberti*) is a recently described cryptic congener of the scalloped hammerhead shark, with overlapping distributions in the western North Atlantic (excluding the Gulf of America). Data on life history, movement, and feeding ecology of scalloped hammerhead sharks in the western North Atlantic likely includes data from both species (Barker et al. 2021). Southeast Data, Assessment, and Review is currently conducting a stock assessment that combines scalloped and Carolina hammerhead sharks into one stock.

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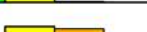
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Sevengill shark (*Heptanchias perlo*)Overall Vulnerability Rank = moderate  (70% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = very high  (86% certainty)

Directional Effect = neutral (95% certainty)

Data Quality = 0% of scores ≥ 2

<i>Heptanchias perlo</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.4	1.5		Low Moderate High Very High
	Habitat Specificity	1.4	2		
	Mobility and Dispersal of Early Life Stages	1.2	1.5		
	Other Stressors	1.6	0.7		
	Population Growth Rate	3.1	1.6		
	Prey Specificity	1.7	2		
	Reproductive Cycle	1.5	1.6		
	Reproductive Strategy Sensitivity	1.6	0.4		
	Sensitivity to Ocean Acidification	1.6	1.1		
	Sensitivity to Temperature	1.6	1.8		
	Site Fidelity	2.4	1		
	Specificity in Early Life History Requirements	1	1.7		
	Stock Size Status	2.2	0.5		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	2.9	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	Oxygen at 200m	1.7	-		
	pH	4	-		
	Primary Production	1.1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Sevengill shark (*Heptanchias perlo*)

Contributors: B. Curtis, T. Curtis, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (70% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), pH (4), and Sea Surface Oxygen (3.8). Sevengill sharks are found globally in subtropical waters between 27 and 720 m depth, but are most common in 100 to 450 m (Castro 2011; Barnett et al. 2012). Juvenile and adult sevengill sharks have been recorded as deep as 1000 m (Castro 2011). In 2018, a male sevengill shark was filmed within the Tongue of the Ocean, a deep ocean trench of the Bahamas, within the lower mesopelagic zone, at a depth of 718 m with a temperature recording of 9°C (Phillips et al. 2019).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 2.5: Population Growth Rate (3.1). Little information is available regarding the population growth rate of this species, outside of a low intrinsic rate of increase estimate (Barnett et al. 2012). Females produce 6 to 20 pups per reproductive cycle, the length of which is unknown (Barnett et al. 2012; Bařusta 2016).

Data Quality: 0% of the data quality scores were 2 or greater. Several attributes received data quality scores below 1.0: Other Stressors (0.7), Reproductive Strategy Sensitivity (0.4), and Stock Size and Status (0.5). Similarly, all other attributes received low data quality scores. Little to no information is available for this species depending on the attribute, and knowledge is largely limited to what is known of similar species.

Climate Effects on Distribution and Abundance: No information is available regarding the effects of climate factors on sevengill sharks. Sevengill sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). Additionally, sevengill sharks within the western North Atlantic are known to inhabit the mesopelagic depth zone, defined as 200 to 1000m (Gong et al. 2021). Climate modeling studies have shown that the mesopelagic is likely to be the first oceanic zone to experience severe deoxygenation (Gong et al. 2021), which may negatively affect the distribution of the sevengill and the food web that sustains this species.

Life History Synopsis: Sevengill sharks are a deep-water species found in subtropical waters between Cape Cod, Massachusetts to the Gulf of America in the western North Atlantic (Castro 2011). Similar to *Hexanchus nakamurai*, sevengill primarily predate on deep-water teleosts, with cephalopods and crustaceans also appearing in their diet (Barnett et al. 2012; Bracinni 2008). Little is known about the distribution patterns or reproductive cycle of sevengill in the western North Atlantic. Length at maturity is estimated as 75 to 85 cm and 90 to 105 cm total length (TL) for male and females, respectively; maximum length is reported as 107 cm and 137 cm for each sex (Barnett et al. 2012). Litter sizes are estimated between 6 to 20 pups, between 25 to 30 cm TL at birth (Barnett et al. 2012; Bařusta 2016; Castro 2011). Landing of sevengill sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Silky shark - *Carcharhinus falciformis*

Overall Vulnerability Rank = high (77% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (99% certainty)

Directional Effect = neutral (96% certainty)

Data Quality = 69% of scores ≥ 2

<i>Carcharhinus falciformis</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	2.4		Low Moderate High Very High
	Habitat Specificity	1.8	2.8		
	Mobility and Dispersal of Early Life Stages	1.2	2.5		
	Other Stressors	1.6	1.6		
	Population Growth Rate	3.2	2.4		
	Prey Specificity	1.6	2		
	Reproductive Cycle	1.2	2.4		
	Reproductive Strategy Sensitivity	1.8	2.1		
	Sensitivity to Ocean Acidification	1.4	1.5		
	Sensitivity to Temperature	2.1	2.2		
	Site Fidelity	2.2	0.8		
	Specificity in Early Life History Requirements	1.1	2.5		
	Stock Size Status	2.8	1.3		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.7	-		
	Sea Surface Salinity	2.2	-		
	Sea Surface Temperature	3.8	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Silky shark (*Carcharhinus falciformis*)

Contributors: J. Carlson, W. Driggers, B. Keller, T. Loughran, C. McCandless, B. Wetherbee

Overall Climate Vulnerability Rank: **High** (77% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.8), Sea Surface Salinity (3.7), and pH (4.0). Silky sharks are distributed throughout tropical to temperate oceanic and coastal waters that are typically warmer than 23°C, spending the majority of time in pelagic surface waters greater than 200 m depth (Castro 2011; Hutchinson et al. 2019).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (3.2) and Stock Size and Status (2.8). Based on intrinsic rate of increase estimates from the Equatorial Atlantic (0.043/yr), populations of silky sharks are slow growing (Santander-Neto et al. 2021). A 46 to 50% decrease in catch per unit effort was observed in the northwest Atlantic Ocean between 1950 and 1990, and the 2021 IUCN Red List report lists the population trend as decreasing (Rigby et al. 2021). A stock assessment has not been completed for silky sharks in the western North Atlantic.

Data Quality: 69% of the data quality scores were 2 or greater. One sensitivity attribute received a score less than 1: Site Fidelity (0.8). There is no known information regarding site fidelity of silky sharks.

Climate Effects on Distribution and Abundance: Possible distributional shifts in silky sharks are fairly well-represented in recent literature. A study using ecological niche modeling (e.g MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) evaluated distributional shifts of carcharhinids, including silky sharks (Diaz-Carballido et al. 2022). The majority of suitable habitat was maintained under these scenarios, with modest gains poleward and losses in nearshore waters of western Florida and the western North Atlantic coast (-5.4% global loss). Comparatively, Lezama-Ochoa et al. (2016) projected a 15.5% loss of suitable habitat in the western North Atlantic using MaxEnt and the IPCC A2 non-mitigation scenario. A similar study using CMIP5 evaluated suitable habitat of requiem and mackerel sharks in the Australian exclusive economic zone and found losses throughout the southwestern coast of Australia (Birkmanis et al. 2020). Silky sharks are predicted to move as a result of climatic oscillations (Lennert-Cody et al. 2016; Rigby et al. 2021). Additionally, silky sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Regarding abundance, silky sharks are a bycatch species of pelagic longline fisheries within the western North Atlantic. Mortality rates for silky sharks captured on pelagic longlines targeting swordfish and tunas have been shown to increase with soak time and further exacerbated by warmer temperatures and declines in dissolved oxygen (Dell’Apa et al. 2023; Gallagher et al. 2014). While the mortality of bycatch species such as silky shark may increase with rising sea surface temperatures and reductions in available dissolved oxygen, the effects of climatic factors on abundance outside of pelagic fisheries is unknown (Dell’Apa et al. 2023; Gallagher et al. 2014).

Life History Synopsis: Silky sharks are globally distributed throughout epipelagic tropical and temperate waters, found in the western North Atlantic from Cape Cod, Massachusetts to Florida, the Gulf of

Mexico, and the Caribbean Sea (Rigby et al. 2021). Information regarding the life history of silky sharks within the western Atlantic is limited, therefore the summary presented below also includes findings from the Pacific and Indian Oceans.

Silky shark habitat preferences include pelagic areas adjacent to shelf breaks, seasonally migrating between the mid to southeast Atlantic coast, throughout the Gulf of America, and the Florida Keys (Kohler and Turner 2019). Movements of silky sharks are influenced by water temperatures and productivity, with northern migrations occurring in spring and summer and southern migrations occurring in winter (Kohler and Turner 2019; Lopez et al. 2020). Movements between the equatorial western and eastern Atlantic were also detected (Kohler and Turner 2019). Other habitats may include reefs or coastal waters, though the species largely occupies surface waters to 500 m and rarely dives below the mixed layer (approximately 300 m recorded in the Indian Ocean) (Curnick et al. 2020; Hutchinson et al. 2019). Silky sharks have also been associated with seamounts and floating objects and schools of yellowfin and other tuna species (Filmlalter et al. 2021; Murray et al. 2023).

Silky sharks are viviparous and are suggested to have a biennial reproductive cycle in the southern Pacific (Galván-Tirado et al. 2014). Timing of reproductive activities varies regionally (Alejo-Plata et al. 2016; Galván-Tirado et al. 2014). In the Gulf of America, mating and parturition is thought to occur concurrently in spring and summer months, whereas pupping occurs throughout the year in the equatorial regions (Alejo-Plata et al. 2016; Branstetter 1987; Hazin et al. 2007). Average litter sizes are reported as 4 to 15 pups and approximately 75 cm total length at birth (Compagno 1984; Hazin et al. 2007). Size at maturity is estimated as 210 to 230 cm and 230 cm for males and females, respectively (Hazin et al. 2007). In the Pacific Ocean, juveniles were found to predate mainly on cephalopods, shifting to larger prey with ontogeny (teleosts including *Thunnus* spp.) (Duffy et al. 2015).

Recent population parameter information for silky sharks in the northwest Atlantic are sparse compared to southwest Atlantic populations. Within the Gulf of America, von Bertalanffy growth coefficient estimates from the 1980s to 1990s range between 0.101-0.15/ yr, and an intrinsic rate of increase of 0.043/yr in the equatorial Atlantic (Branstetter 1987; Bonfil et al. 1993; Bonfil 2008). Recent studies on the southwestern Atlantic suggest the von Bertalanffy growth coefficient estimate is 0.0987/yr (Santander-Neto et al. 2021). Silky sharks are caught as bycatch in pelagic longline fisheries, though the magnitude of population declines due to longlining has been disputed (Baum and Myers 2004; Burgess et al. 2005a; Baum and Myers 2005; Burgess et al. 2005b; Cortes et al. 2007; Rigby et al. 2021). No stock assessment is available for silky sharks.

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Sixgill shark (*Hexanchus griseus*)Overall Vulnerability Rank = moderate  (82% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = high  (76% certainty)

Directional Effect = neutral (89% certainty)

Data Quality = 31% of scores ≥ 2

<i>Hexanchus griseus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	2.1		Low Moderate High Very High
	Habitat Specificity	1.4	2.2		
	Mobility and Dispersal of Early Life Stages	1.2	1.6		
	Other Stressors	1.5	1.1		
	Population Growth Rate	3.6	1.6		
	Prey Specificity	1.8	2.1		
	Reproductive Cycle	1.2	1.8		
	Reproductive Strategy Sensitivity	1.8	0.3		
	Sensitivity to Ocean Acidification	1.6	1.4		
	Sensitivity to Temperature	1.6	2.4		
	Site Fidelity	1.9	1.8		
	Specificity in Early Life History Requirements	1	1.4		
	Stock Size Status	2.3	1		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	3.1	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	Oxygen at 200m	1.7	-		
	pH	4	-		
	Primary Production	1.1	-		
	Sea Surface Oxygen	4	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score		very high		
Overall Vulnerability Rank		moderate			

Sixgill shark (*Hexanchus griseus*)

Contributors: D. Coffey, T. Daly-Engel, D. Grubbs, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (82% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), Sea Surface Oxygen (4.0) and pH (4.0). Sixgill sharks are a demersal, deep water species, common in depths 300 to 1000 m and temperatures of 5 to 16°C (Castro 2011). They are found throughout the world in tropical and temperate waters, the greatest depth recorded being 2,500 m (Barnett et al. 2012; Castro 2011; Coffey et al. 2020).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.6). Little information is available regarding the population growth rate of sixgill sharks, with an intrinsic rate of increase of 1.4% per year (COSEWIC 2007).

Data Quality: 31% of the data quality scores were 2 or greater. The reproductive strategy sensitivity and stock size and status attributes received scores equal to or below 1, due to no available information regarding reproductive behaviors and limited information available regarding stock size and status. Genetic studies of sixgill sharks suggested low genetic diversity throughout the Atlantic stock, and a decline in the global population of the species (Finucci et al. 2020; Vella and Vella, 2017).

Climate Effects on Distribution and Abundance: Sixgill sharks depends heavily on the mesopelagic ocean zone between 200 and 1,000 m depth to feed on prey associated with the deep scattering layer, vertically migrating along continental and insular slopes between the 5 and 17°C thermocline in both the Atlantic and Pacific (Grubbs et al. 2008). Sixgill sharks are ectothermic and able to make large-scale movements, and may shift to deeper waters in response to local warming (Comfort 2012). This makes sixgill sharks vulnerable to climate effects including changes to bottom temperature, mixing, and especially deoxygenation. Recent climate studies have shown that the mesopelagic is likely to be the first oceanic zone to experience severe deoxygenation, which will negatively affect the food web that sustains sixgill sharks (Gong et al. 2021).

Life History Synopsis: The sixgill shark is among one of the larger species of deep-water sharks, reaching >480 cm total length (TL), though accurate measurements are few (Castro 2011). This species is circumglobal in the tropical, subtropical, and temperate deep ocean between 0 and 2,500 m, and most common between 100 and 1,000 m (Ebert et al. 2013). Sixgill sharks are known to undergo diel vertical migrations between approximately 5 and 17°C, staying near the bottom during the day and moving up at night to feed on dogfish sharks (genus *Squalus*) and other prey of the mesopelagic layer (Compagno et al. 2005). This behavior is consistent between ocean basins, though the depth ranges of these vertical migrations vary (Grubbs et al. 2008). Horizontal movement is high: satellite tags deployed on adult sixgill sharks for durations of 1 to 12 months in the central Pacific showed a maximum horizontal dispersal of 1,000 km and multiple recaptures were made in the same location after multiple years, but in the Gulf of America movements up to 700 km were observed in only 32 days (Comfort 2012; Grubbs, Daly-Engel, and Cotton, unpublished data). Indirect dependence on the deep scattering layer likely represents the greatest climate threat to sixgill sharks. Recent climate studies have shown that the mesopelagic is likely

to be the first oceanic zone to experience severe deoxygenation, which would have severe negative consequences on the food web that supports this species (Gong et al. 2021).

Intrinsic rates of increase and reproduction are low in sixgill sharks, similar to many deep-water elasmobranchs (Kyne and Simpfendorfer 2010). This is likely due to energy allocation that maximizes reproductive output in a light-poor environment where resources are scarce, and the low metabolic rates of elasmobranchs in general and deep-sea species in particular (Kyne and Simpfendorfer 2010; Martin and Palumbi 1993). This species is ovoviviparous with relatively large litters, giving birth to >50 young that average 60 to 80 cm TL, depending on the population (Castro 2011).

This species has not been aged due to its uncalcified vertebrae (Castro 2011). Greenland sharks (*Somniosus microcephalus*), which reach a similar size and belong to an ancestral genus with many physiological similarities to *Hexanchus*, have been estimated to live for two to four hundred years (Nielsen et al. 2016); however, caution should be used when assuming a positive size/age relationship when correlating the carbon-14 derived ages to the measured fish lengths as age is often decoupled from maximum length (Natanson et al. 2018). Sixgill shark likely does not have the same extreme lifespan estimated for the Greenland shark (Greenland shark primary habitat is colder, though their range overlaps with sixgill shark in much of the Atlantic including the Gulf of America), but its size and environment suggest this species is likely to live longer than most elasmobranchs (Cotton et al. 2020).

There are distinct genetic differences between populations of sixgill sharks in different biogeographic regions, especially between ocean basins, indicating low historical connectivity and gene flow (Vella and Vella 2017). This finding is consistent with observations of low horizontal dispersal, which could inhibit gene flow even on a geological scale. Genetic diversity is also low in this species, which could impact rebound rates following depletion (Martin et al. 1992; Vella and Vella 2017).

Sixgill shark is fished in commercial and artisanal/game fisheries in parts of its range, used in fish oil and meal production and consumed as a food source. Though exact population sizes of sixgill shark are unknown, it is found in low abundance and thought to be decreasing, a trend that is likely to continue as commercial fishing fleets expand into deeper and deeper water (Cotton and Grubbs 2015; Finucci et al. 2020). Landing of sixgill sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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Smalltail shark - *Carcharhinus porosus*Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = moderate ☐Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☐ (99% certainty)

Directional Effect = neutral (68% certainty)

Data Quality = 46% of scores ≥ 2

<i>Carcharhinus porosus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.3	2		Low Moderate High Very High
	Habitat Specificity	1.8	2.2		
	Mobility and Dispersal of Early Life Stages	1.4	2.1		
	Other Stressors	1.9	1.3		
	Population Growth Rate	2.8	1.9		
	Prey Specificity	1.4	2.3		
	Reproductive Cycle	1.4	2		
	Reproductive Strategy Sensitivity	1.6	1.3		
	Sensitivity to Ocean Acidification	1.3	1.1		
	Sensitivity to Temperature	1.8	2		
	Site Fidelity	2	1		
	Specificity in Early Life History Requirements	1	1.5		
	Stock Size Status	2.6	1.6		
	Sensitivity Score	moderate			
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score	NA			
Overall Vulnerability Rank		NA			

Smalltail shark (*Carcharhinus porosus*)

Contributors: W. Driggers, K. Hannan, E. Hoffmayer, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable.**

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (2.8) and Stock Size and Status (2.6). The intrinsic rate of increase is unknown for smalltail shark, however, the von Bertalanffy growth coefficient is estimated as 0.077/yr (Lessa et al. 2006). The stock size and status of smalltail shark is unknown, but there is evidence of a strong population decline and possible genetic bottleneck (Feitosa et al. 2020a; Tavares et al. 2013). The species was added to the Brazilian Red List of Threatened Species Ordinance (Feitosa et al. 2018).

Data Quality: 46% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Few studies have examined the effect of climate change on the abundance and distribution of smalltail sharks. A study using ecological niche modeling (e.g. MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that smalltail sharks showed the greatest loss of suitable habitat compared to 24 other carcharhinid species that inhabit Mexican waters (Diaz-Carballido et al. 2022). The largest change in distribution occurred in the northernmost range of the smalltail shark (i.e. northern Gulf of America) under RCP2.6 and RCP8.5, while the southernmost range changed minimally under the same climate projections (Diaz-Carballido et al. 2022). Additionally, smalltail sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). It is unclear how climate change will affect smalltail abundance.

Life History Synopsis: Smalltail sharks are found in shallow, tropical waters from the Gulf of America to southern Brazil, though rarely found north of Corpus Christi, Texas (Castro 2011). The majority of smalltail shark suitable habitat is shallow littoral waters, often associated with mud substrate in estuaries, bays, and river mouths (e.g. Castellanos 2010; Swift and Portnoy 2021). According to Feitosa et al. (2020)b, there is no change in habitat use as it relates to environmental parameters throughout life indicating all life stages co-occur, with the exception of seasonal habitat partitioning based on sex.

Smalltail sharks exhibit a biennial reproductive cycle with a 10 and 12 month gestation period (Menni and Lessa 1998, Lessa et al. 1999, de Campos Santos et al. 1999, Castro 2011). They have on average six young and there is a significant positive relationship between maternal size and brood size (Menni and Lessa 1998; Lessa et al. 1999, de Campos Santos et al. 1999; Gadig 2001).

Smalltail sharks are opportunistic predators that consume primarily teleost fishes (Lessa and Almeida 1997; Cortes 1999; Gadig 2001; Castro 2011). An examination of stomach contents in smalltail sharks from northern Brazilian coastal waters revealed that 80% of the diet is comprised of small teleosts including Sciaenidae (drums and croaker), Mugilidae (mullet), Engraulidae (anchovies) and Stromatidae (butterfish). Younger smalltail sharks had a more variable diet compared to adults. Adult male smalltail

sharks were found with small coastal sharks and rays in their stomachs (Lessa and Almeida 1997). The opportunistic feeding strategy and diversity of prey choices may afford flexibility in diet under changing climate conditions.

Smalltail sharks reach sexual maturity at 6 years (Lessa and Santana 1999) and have slow growth rates ($k = 0.077 - 0.101/\text{year}$, da Silva Batista and Silva 1995; Lessa and Santana 1998). Due to their slow growth and low fecundity, smalltail sharks are thus considered to be weakly resilient (Santana et al. 2020). Overexploitation of juveniles (<6 years old) off of Brazil's northern coast led to significant population declines in the 1990s (Lessa and Santana 1999) and led to local species extinction northeast and southeast of Brazil's northern coast (Santana et al. 2020).

Smalltail sharks have seen drastic declines in all parts of its range (Pollom et al. 2020). All life stages are found in nearshore, shallow, coastal areas and have been exploited by fisheries for the past several decades (Santana 2020). Landing of smalltail sharks through the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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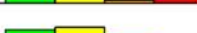
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Smooth dogfish - Atlantic stock (*Mustelus canis*)Overall Vulnerability Rank = moderate  (100% certainty)Biological Sensitivity = low Climate Exposure = very high Distributional Vulnerability Rank = high  (99% certainty)

Directional Effect = neutral (86% certainty)

Data Quality = 85% of scores ≥ 2

<i>Mustelus canis</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	2.8		Low Moderate High Very High
	Habitat Specificity	1.8	2.8		
	Mobility and Dispersal of Early Life Stages	1.4	2.4		
	Other Stressors	1.8	2.4		
	Population Growth Rate	2	2.8		
	Prey Specificity	1.4	2.9		
	Reproductive Cycle	2.1	2.9		
	Reproductive Strategy Sensitivity	1.9	2.2		
	Sensitivity to Ocean Acidification	2	1.8		
	Sensitivity to Temperature	1.6	2.9		
	Site Fidelity	2	1.2		
	Specificity in Early Life History Requirements	1.1	2.3		
	Stock Size Status	1.3	2.7		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	2.8	-		
	Bottom Temperature	3.7	-		
	Mixed Layer Depth	1.3	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Smooth dogfish - Atlantic stock (*Mustelus canis*)

Contributors: C. Bangley, T. Loughran, C. McCandless, G. Skomal

Overall Climate Vulnerability Rank: **Moderate** (100% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.7), pH (4.0) and Sea Surface Oxygen (3.8). Smooth dogfish are a bottom-dwelling species found in coastal and inshore waters throughout the northeastern United States during spring to fall, moving south to offshore waters in the southeastern United States during winter (Kohler and Turner 2019; SEDAR 2015). Smooth dogfish in nearshore and estuarine waters off the United States Atlantic Coast have been documented over seagrass, mud, sand, shell, and rocky bottoms, with temperatures between 9.8 and 32.4°C, salinities between 12.6 and 38.0 psu, dissolved oxygen between 2.0 and 15.6 mg/l, and depths from 0.3 to 39.0m (Bangley et al. 2018, C. McCandless pers comm; McCandless et al. 2007). Additionally, smooth dogfish have been documented offshore of North Carolina through New Jersey in March at depths down to 259 m and temperatures down to 5.3°C during the NEFSC Spring Bottom Trawl Survey (NMFS NEFSC unpublished data).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 85% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Smooth dogfish, a species managed as an Atlantic and Gulf of America stock, has been highlighted throughout several studies as an at-risk species under future climate conditions (Cheung et al. 2018; Dixon et al. 2015; Morely et al. 2017; Pinsky et al. 2021). Climate effects on the Atlantic stock of smooth dogfish include distributional shifts and diminished sensory capabilities from ocean acidification (Hare et al. 2016). As an ectothermic temperate species, smooth dogfish responded negatively to warmer winter temperatures (Morely et al. 2017; Pinsky et al. 2021). Warmer temperatures coinciding with seasonal shallower habitat use could cause a poleward shift in distribution and alter the timing of migrations. Examination of NEFSC trawl survey data suggest a shift north in the smooth dogfish center of distribution over the last decade (OceanAdapt 2023).

Additionally, early life stages of smooth dogfish may be impacted by loss of seagrass habitats, at least in North Carolina waters where juvenile smooth dogfish abundance was greatest in these habitats (Bangley and Rulifson 2017; Bangley et al. 2018; Waycott et al. 2009). Young of the year and small juveniles surveyed in estuarine nursery habitats further north during the spring and summer were primarily caught over mud or sand bottoms (C. McCandless pers comm). Efforts are underway to restore eelgrass (*Zostera marina*) habitat in these historically depleted areas up through the northern extent of the species range, with one notable success story that should help safeguard Virginia's coastal waters to climate change (Orth et al. 2020; Williamson 2023).

Regarding sensory capabilities, smooth dogfish exposed to projected ocean acidification conditions for the year 2100 (0.3 to 0.4 decline in pH) demonstrated impaired ability to track prey odors (Dixon et al. 2015). Despite having a generalist diet with prey items ranging from invertebrates to teleosts, smooth dogfish may encounter difficulty predating on existing food sources under future ocean acidification conditions.

A study on a similar species, Florida smoothhound (*M. norrisi*), demonstrated that buccal pumping species responded to hypoxic conditions by slowing their movement versus ram ventilating shark species increasing their swimming speed (Carlson and Parsons 2001). Buccal pumping allows smoothhounds and similar species to remain motionless while breathing, which may conserve energy and oxygen when encountering hypoxic conditions (Carlson and Parsons 2001).

Life History Synopsis: Smooth dogfish is a small (max. length approx. 150 cm total length) coastal and shelf-affiliated shark found along the western Atlantic Coast, occupying specific depth and temperatures throughout their range (up to 260 m, 5 to 23°C for smooth dogfish - Atlantic stock) (Bockus et al. 2020; Castro 2011; NMFS NEFSC unpublished data). Adult stages are driven by water temperatures to migrate annually to the northern extent of the species' range in Cape Cod, Massachusetts, during spring months followed by a southern migration to the eastern coast of North Carolina to Florida in fall months (Castro, 2011; Kohler et al. 2014; Kohler and Turner, 2019; SEDAR 2015). Adult and juvenile smooth dogfish are seasonally abundant in nearshore and estuarine habitats of the Mid-Atlantic coast during migratory periods (Kohler et al. 2014; Kohler and Turner, 2019). Adult males were found further north during fall migrations when compared to females, indicating sexual segregation (Dell'Apa et al. 2018).

The spring migration aligns with parturition in May to mid-July, indicating that coastal estuaries from North Carolina to Massachusetts, including Chesapeake and Delaware bays, serve as important nursery grounds for early life stages (Skomal 2007; Bangley et al. 2018; Bangley and Rulifson 2017; McCandless et al. 2007; SEDAR 2015). Gestation periods last approximately 11 months before giving birth to an average of 9 pups (SEDAR 2015). Smooth dogfish are generalist predators throughout all life stages, with prey species including a variety of crustaceans, mollusks, annelids, and teleosts (McElroy 2009; Montemarano et al. 2016; Perry 2017). In Delaware Bay, their diet is dominated by invertebrates with a significant ontogenetic shift in prey consumption from small less mobile invertebrates (shrimp, annelids, isopods, and bivalves) as young of the year to benthic macroinvertebrates (primarily large crabs), but also a few teleost species, as juveniles age 1 through adults (McElroy 2009).

The Atlantic stock of smooth dogfish is managed separately from the Gulf of America stock of smooth dogfish, which is part of a complex including *M. sinuamexicanus* and *M. norrisi*. The von Bertalanffy growth coefficient for males and females of the Atlantic stock of smooth dogfish is estimated as 0.44 and 0.292 respectively, and natural mortality is estimated as 0.78 (SEDAR 2015). Females reach maturity at approximately 4.4 years, and 2.5 years for males (SEDAR 2015). The last stock assessment on the Atlantic stock of smooth dogfish was conducted in 2015 and found that the stock is not overfished and overfishing is not occurring ($SSF_{2012}/SSF_{MSY} = 1.96-2.81$; $F_{2012}/F_{MSY} = 0.61-0.99$) (SEDAR 2015).

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Smooth dogfish - Gulf of America complex stock (*Mustelus canus*)Overall Vulnerability Rank = NA ☐ (NA% certainty)Biological Sensitivity = low ☒Climate Exposure = NA ☐Distributional Vulnerability Rank = high ☒ (99% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 69% of scores ≥ 2

<i>Mustelus canis</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	Low Moderate High Very High
Sensitivity Attributes	Adult Mobility	1.2	2.7		
	Habitat Specificity	1.8	2.3		
	Mobility and Dispersal of Early Life Stages	1.5	2		
	Other Stressors	1.7	2.2		
	Population Growth Rate	2.2	2.6		
	Prey Specificity	1.6	2.5		
	Reproductive Cycle	1.6	2.7		
	Reproductive Strategy Sensitivity	1.8	1.1		
	Sensitivity to Ocean Acidification	1.9	1.8		
	Sensitivity to Temperature	1.8	2.4		
	Site Fidelity	1.5	1		
	Specificity in Early Life History Requirements	1.1	1.6		
	Stock Size Status	1.3	2.5		
	Sensitivity Score		low		
Exposure Factors	Bottom Salinity	-	-		
	Bottom Temperature	-	-		
	Mixed Layer Depth	-	-		
	pH	-	-		
	Primary Production	-	-		
	Precipitation	-	-		
	Sea Surface Oxygen	-	-		
	Surface Wind Stress Magnitude	-	-		
	Exposure Score		NA		
Overall Vulnerability Rank		NA			

Smooth dogfish - Gulf of America complex stock (*Mustelus canis*)

Contributors: D. Grubbs, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Not applicable** due to uncertain species distribution.

Climate Exposure: **Not applicable** due to uncertain species distribution.

Biological Sensitivity: **Low**. No sensitivity attribute means scored above 2.5.

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Smooth dogfish, a species managed as an Atlantic and Gulf of America stock, has been highlighted throughout several studies as an at-risk species under future climate conditions (Cheung et al. 2018; Dixon et al. 2015; Morely et al. 2017; Pinsky et al. 2021). Climate effects on the Gulf of America stock of smooth dogfish include diminished sensory capabilities from ocean acidification and distribution shifts, specifically habitat compression due to limited available shelf habitat (Hare et al. 2016). As an ectothermic temperate species, smooth dogfish responded negatively to warmer winter temperatures (Morely et al. 2017; Pinsky et al. 2021). Warmer temperatures coinciding with seasonal shallower habitat use could limit habitat availability and movements of smooth dogfish in the Gulf of America, where the upper 50 m of surface waters are warming at twice the global rate (Pinsky et al. 2021; Wang et al. 2023). Additionally, smooth dogfish exposed to projected ocean acidification conditions for the year 2100 (0.3 to 0.4 decline in pH) demonstrated impaired ability to track prey odors (Dixon et al. 2015). Despite having a generalist diet with prey items ranging from invertebrates to teleosts, smooth dogfish may encounter difficulty predating on existing food sources under future ocean acidification conditions.

A study on a similar species, Florida smoothhound (*M. norrisi*), demonstrated that buccal pumping species responded to hypoxic conditions by slowing their movement versus ram ventilating shark species increasing their swimming speed (Carlson and Parsons 2001). Buccal pumping allows smoothhounds and similar species to remain motionless while breathing, which may conserve energy and oxygen when encountering hypoxic conditions (Carlson and Parsons 2001).

Life History Synopsis: Smooth dogfish is a small (max. length approx. 150 cm total length) shark species found throughout the Florida Keys to the United States-Mexico border, most common in temperate waters of the northern Gulf of America shelf (Castro 2011; Jones et al. 2014). As a shelf-associated species, *Mustelus* spp. in the Gulf of America are generally found in depths between 50 to 330 m and temperatures of 11 to 25°C (Drymon et al. 2020; Giresi et al. 2015; Jones et al. 2014; NMFS SEFSC, unpublished data). Analysis of genetically verified *Mustelus* spp. indicated that smooth dogfish prefer deeper waters (64 to 408m) compared to *M. norrisi* and *M. sinusmexicanus*, with 408 m as the greatest depth ever recorded for the species (Giresi et al. 2015). This may indicate a preference for or tolerance of cooler temperatures similar to smooth dogfish in the Atlantic (Giresi et al. 2015).

Little is known about smooth dogfish movement due to their similarity to other smoothhound species and resulting identification issues. Smooth dogfish largely feed on invertebrates, but opportunistically feed on teleosts and molluscs as well (Churchill et al. 2015; McElroy 2009; Montemarano et al. 2016).

Smooth dogfish reproduce annually with parturition occurring in July (Jones et al. 2014). Brood sizes range from 3 to 20 pups (Conrath and Musick 2002; Jones et al. 2014). Limited information is available regarding habitat use of early life stages and growth of smooth dogfish in the Gulf of America. The von Bertalanffy growth coefficient for both males and females is estimated as 0.13, and the intrinsic rate of increase is estimated as 0.281 (Jones et al. 2014; SEDAR 2015). A combined analysis of Gulf smoothhound and smooth dogfish growth estimates age at 50% maturity of 4.11 years and maximum age of 13 years for females, and 3.28 years and 11 years for males (Jones et al. 2014).

Smooth dogfish is managed as a complex with *M. sinusmexicanus* and *M. norrisi*. The last stock assessment of the HMS Gulf of America smoothhound shark complex was conducted in 2015 and found that the complex is not overfished and overfishing is not occurring ($N_{2012}/N_{MSY} = 1.68\text{--}1.83$; $F_{2012}/F_{MSY} = 0.07\text{--}0.35$) (SEDAR 2015).

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Smooth hammerhead shark - *Sphyrna zygaena*

Overall Vulnerability Rank = high (63% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (95% certainty)

Directional Effect = neutral (79% certainty)

Data Quality = 62% of scores ≥ 2

<i>Sphyrna zygaena</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.5	2.4		Low Moderate High Very High
	Habitat Specificity	1.7	2.2		
	Mobility and Dispersal of Early Life Stages	1.2	2.1		
	Other Stressors	2	1.3		
	Population Growth Rate	3.3	2.6		
	Prey Specificity	1.6	2.5		
	Reproductive Cycle	1.2	2.1		
	Reproductive Strategy Sensitivity	1.8	1.6		
	Sensitivity to Ocean Acidification	1.3	1.4		
	Sensitivity to Temperature	1.7	2.2		
	Site Fidelity	2	1		
	Specificity in Early Life History Requirements	1.1	2.5		
	Stock Size Status	2.9	1.5		
	Sensitivity Score		moderate		
Exposure Factors	Bottom Salinity	2.4	-		
	Bottom Temperature	3.6	-		
	Mixed Layer Depth	1.1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.6	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

Smooth hammerhead shark (*Sphyrna zygaena*)

Contributors: J. Carlson, R. Logan, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **High** (63% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.6), pH (4.0), and Sea Surface Oxygen (3.6). Smooth hammerhead sharks are found throughout temperate to tropical waters from the Bay of Fundy to the Florida Keys and the Caribbean, though are rarely found in the southeastern United States and Gulf of America (Castro 2011; Deacy et al. 2020; Miller 2016). This species spends the majority of time in the upper 20 m of surface waters or in shallow, nearshore environments (Miller 2016) but may sporadically dive to depths of 100 to 260 m (Santos and Coelho 2018).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Population Growth Rate (3.3) and Stock Size and Status (2.9). Smooth hammerhead sharks are a slow-growing species, reaching maturity at 9 years of age and a maximum age of 21 years (Coelho et al. 2011). The intrinsic rate of increase for this species is estimated as 0.225/yr, and the von Bertalanffy growth coefficient as 0.09 (Cortes et al. 2015; Rosa et al. 2017). Limited information regarding stock status is available for smooth hammerhead sharks in the Atlantic Ocean.

Data Quality: 62% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Little species-specific information is available regarding the effect of climate factors to abundance and distribution. Scalloped hammerhead sharks, a congener of smooth hammerhead sharks, have high metabolic rates as ectothermic elasmobranchs (Lowe 2001). Rising temperatures may affect physiological performance given their presumed high metabolic rate, exacerbated by less available dissolved oxygen (Crear et al. 2019; Dell’Apa et al. 2023; Lowe 2001). Smooth hammerhead sharks may be affected by similar metabolic rates and subsequent stressors, but further research is needed. As a group, Sphyrnidae have some of the highest observed levels of at-vessel and post-release mortality of any shark (Binstock et al. 2023; Braccini and Waltrick 2019; Coelho et al. 2012; Eddy et al. 2016; Ellis et al. 2017; Gulak et al. 2015; Morgan and Burgess 2007; Whitney et al. 2021) suggesting that they are quite sensitive to stressors and physiological challenges. These dynamics may have direct impacts on the abundance, distribution, and ecology of this species.

Life History Synopsis: Smooth hammerhead sharks are distributed globally throughout tropical to temperate coastal and pelagic environments, and are considered to be the most oceanic of Sphyrnidae species (Santos and Coelho 2019). Adults are generally solitary and prefer warmer sea surface temperatures above 23°C, often occupying offshore areas and occasionally diving to depths over 200 m (Miller 2016; Santos and Coelho 2018). Juveniles tend to occupy inshore, continental shelf habitat and may occasionally form aggregations while seasonally migrating between the New York Bight in summer and Cape Hatteras, NC in winter (Logan et al. 2020; Miller 2016).

Smooth hammerhead sharks are viviparous and are assumed to have a biennial reproductive cycle (J. Carlson pers comm). Females give birth in summer months after a gestation period of 10 to 11 months (Miller 2016; White et al. 2006). Litter sizes range between 20 to 50 pups, with a total average length

between 49 to 64 cm (Miller 2016). Several nursery grounds throughout the Pacific Ocean, South and East Atlantic, and the Mediterranean Sea though nursery grounds in the western North Atlantic have not been identified (Miller 2016). Adult and juvenile stages maintain a similar generalist diet of teleosts, elasmobranchs, crustaceans, marine mammals, and particularly cephalopods (Estupiñán-Montaña et al. 2018; Miller 2016). Smooth hammerhead sharks are a long-lived species, growing at a faster rate in the first four years and living to an estimated maximum age of over 20 years (Coelho et al. 2011; Miller 2016). Southeast Data, Assessment, and Review is currently conducting a stock assessment for smooth hammerhead sharks.

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Spinner shark - *Carcharhinus brevipinna*Overall Vulnerability Rank = moderate ■ (69% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = high ■ (92% certainty)

Directional Effect = neutral (83% certainty)

Data Quality = 54% of scores ≥ 2

<i>Carcharhinus brevipinna</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	1.9		Low Moderate High Very High
	Habitat Specificity	1.5	2.8		
	Mobility and Dispersal of Early Life Stages	1.6	2.6		
	Other Stressors	2.5	1.8		
	Population Growth Rate	2.4	2.5		
	Prey Specificity	2.4	2.8		
	Reproductive Cycle	1.3	2.6		
	Reproductive Strategy Sensitivity	1.8	1.9		
	Sensitivity to Ocean Acidification	1.6	1.3		
	Sensitivity to Temperature	1.7	2.5		
	Site Fidelity	2.2	0.8		
	Specificity in Early Life History Requirements	1.2	2.5		
	Stock Size Status	2	1.5		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	2.7	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1.1	-		
	pH	4	-		
	Primary Production	1.2	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.8	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Spinner shark (*Carcharhinus brevipinna*)

Contributors: C. Bangley, A. Carlisle, J. Carlson, K. Hannan, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (69% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), pH (4.0), and Sea Surface Oxygen (3.8). Spinner sharks are found throughout surface waters to depths of 200 m, ranging from temperate to tropical coastal and continental shelf habitats (J. Carlson pers comm; NMFS SEFSC, unpublished data; Rigby et al. 2020). Populations of spinner sharks in the Gulf of America and Atlantic appear to make latitudinal seasonal migrations (Castro 2011; Kohler and Turner, 2019).

Biological Sensitivity: **Low.** One sensitivity attribute mean received a score of 2.5: Other Stressors. Early life stages of spinner sharks are found in coastal habitats which may be degraded through anthropogenic activities (Knip et al. 2010; Plumlee et al. 2018; Swift and Portnoy, 2021). This includes possible exposure to mercury (Reistad 2018) and inflammatory responses from pollution (Karsten and Rice 2004). Additionally, spinner sharks are exposed to hypoxic areas by entering estuaries to feed on aggregating prey along the edges of low dissolved oxygen areas (Pickens et al. 2022).

Data Quality: 54% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: There is little direct information on how the changing climate will impact spinner sharks. A study using ecological niche modeling (MaxEnt) using four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5 projected habitat loss throughout the south Atlantic, while north Atlantic habitat was largely maintained (Diaz-Carballido et al. 2023). Additionally, spinner sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Spinner sharks have been observed to have relatively high at-vessel mortality rates (58%) in commercial trawl fisheries as well as longline fisheries (4% to as high as 97%), potentially related to their being obligate ram ventilators (Ellis et al. 2017, Lotti et al. 2011, White et al. 2020). At-vessel mortality significantly increases with decreased dissolved oxygen and increased soak-time (for longlines), which suggests that this species may be vulnerable to any increase in physiological stress, such as those associated with increases in temperature and concurrent declines in dissolved oxygen. In addition, the fact that they utilize very shallow coastal environments (Castro 1993; Bethea et al. 2015; Swift and Portnoy 2021) as nursery areas may make their early life history stages particularly vulnerable to changes in water temperature, dissolved oxygen, habitat, water quality, and harmful algal blooms.

Life History Synopsis: Spinner sharks are globally distributed in warm (Gulf of America: 24.1 to 29.5°C) temperate to tropical continental shelf waters less than 200 m in depth (J. Carlson pers comm; NMFS SEFSC, unpublished data; Weigmann et al. 2016). They are benthopelagic and commonly found in shallow (<30 m), inshore waters, including off beaches and in bays and estuaries, but are rarely observed in pelagic, offshore waters (Rigby et al. 2020). They are highly migratory and known to form large

schools, seasonally moving inshore and offshore or latitudinally with changing water temperatures (Ebert et al. 2021).

Spinner sharks are viviparous with a biennial reproductive cycle and give birth to larger litter sizes (4-8 pups per litter, 47 to 60 cm fork length (FL)) as females increase in size (Joung et al. 2005; Palmrose 2021; Rigby et al. 2019). Size at maturity is reported as 140 cm FL for females and 130 cm FL for males between 7 to 10 years of age (Carlson and Baremore 2005; Geraghty et al. 2013; Geraghty et al. 2016; Palmrose 2021). While their age and growth dynamics vary across their broad distribution, they are relatively large, attaining a maximum length of 176 to 220 cm FL (maximum observed age of 17.5 years) (Carlson and Baremore 2005). Spinner sharks consume a variety of prey, including other elasmobranchs and cephalopods, but they are primarily piscivores, feeding on small schooling fishes (Bethea et al. 2004). This species exhibits an unusual behavior from which it derives its name, where they will leap from the water, spinning along their long axis, when attacking prey.

There is no stock assessment available for spinner sharks. However, analysis of multiple relative indices of abundance within a dynamic factor model indicated an increase in relative abundance from the late 1990s to 2014 (Peterson et al. 2017).

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Tiger shark - *Galeocerdo cuvier*Overall Vulnerability Rank = moderate ■ (82% certainty)Biological Sensitivity = low ■Climate Exposure = very high ■Distributional Vulnerability Rank = very high ■ (82% certainty)

Directional Effect = neutral (98% certainty)

Data Quality = 69% of scores ≥ 2

<i>Galeocerdo cuvier</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	2.3		Low Moderate High Very High
	Habitat Specificity	1.4	2.7		
	Mobility and Dispersal of Early Life Stages	1.2	2.8		
	Other Stressors	1.8	1.8		
	Population Growth Rate	2.4	2.3		
	Prey Specificity	1.2	3		
	Reproductive Cycle	1.2	2.6		
	Reproductive Strategy Sensitivity	2.1	1.9		
	Sensitivity to Ocean Acidification	1.4	1.6		
	Sensitivity to Temperature	2	2.6		
	Site Fidelity	2.5	2.1		
	Specificity in Early Life History Requirements	1.2	2.4		
	Stock Size Status	2.2	1.3		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3	-		
	Bottom Temperature	3.8	-		
	Mixed Layer Depth	1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Precipitation	1.1	-		
	Sea Surface Oxygen	3.9	-		
	Surface Wind Stress Magnitude	1.2	-		
	Exposure Score	very high			
Overall Vulnerability Rank		moderate			

Tiger shark (*Galeocerdo cuvier*)

Contributors: D. Coffey, L. Crawford, T. Daly-Engel, J. Kneebone, T. Loughran, C. McCandless

Overall Climate Vulnerability Rank: **Moderate** (82% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Bottom Temperature (3.8), Sea Surface Oxygen (3.9) and pH (4.0). Tiger sharks are found throughout tropical and temperate waters worldwide. In the Atlantic, tiger sharks range from Cape Cod, Massachusetts to the Gulf of America and Caribbean Sea, though many stray beyond this range (Castro 2011; Hammerschlag et al. 2022). Tiger sharks become oceanic as they reach maturity, preferring shallow depths above 60 m and waters above 22°C (Vaudo et al. 2014; Afonso and Hazin 2015; Ajemian et al. 2020; Hammerschlag et al. 2022). Tiger sharks may spend considerable time within 5 m of the surface, but generally exhibit ‘yo-yo’ diving behavior (Nakamura et al. 2011) throughout the water column with frequent dives to mesopelagic depths (maximum reported depth of 1,112 m; Vaudo et al. 2014; Afonso and Hazin 2015).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 69% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Distribution and migration timing of tiger sharks in the western North Atlantic has already been altered by ocean warming, causing increased interactions with fisheries (Hammerschlag et al. 2022; Lea et al. 2018). Tiger sharks tracked with satellite tags between 2010 and 2019 revealed that the species has highly variable migration timing and extent, and they are migrating more poleward and earlier in the year. Catch data for tiger sharks in the western North Atlantic between 1980 and 2018 showed that catch densities have shifted northward as sea surface temperatures have increased (Hammerschlag et al. 2022). Additionally, tiger sharks are ectothermic, relying on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021).

Ecological niche modeling (MaxEnt) under four future climate change scenarios (RCP2.6, RCP4.5, RCP6.0, and RCP8.5) found that tiger sharks will maintain the majority of their suitable habitat in the western North Atlantic but may lose habitat in the eastern Gulf of America along the coast of Florida (Diaz-Carballido et al. 2022). The spatial distribution of young-of-the-year and juvenile tiger sharks is most abundant off of the Atlantic coast of Florida, Georgia, and South Carolina in the southeast United States, and off the Florida Panhandle, Mississippi, and Alabama in the Gulf of America (Driggers et al. 2008). Moreover, satellite tracking of immature and mature tiger sharks in the Gulf of America revealed core use areas along the shelf-edge region between Texas and Louisiana and the edge of the Mississippi-Alabama shelf (Ajemian et al. 2020). Therefore, a loss of suitable habitat in the eastern Gulf of America may have limited impacts on tiger shark abundance.

The occurrence of tiger sharks off Cocos Island in the eastern Tropical Pacific increased with rising sea surface temperatures, albeit non-significantly. Tiger sharks were observed more frequently during La Nina events and were less abundant during El Nino events. The authors attributed the variation in abundance to shifting prey availability in response to changing sea surface temperatures and

hypothesized that the wide thermal tolerance of tiger sharks may allow them to acclimate or adapt to rising water temperatures (Osgood et al. 2021).

Life History Synopsis: Tiger sharks are found globally throughout tropical to temperate coastal and pelagic waters. In the western North Atlantic, their distribution ranges from Cape Cod, Massachusetts to the Caribbean Sea (Castro 2011; Kohler and Turner 2019; Hammerschlag et al. 2022). The species is most commonly encountered in summer months near North and South Carolina, Florida, and in the Gulf of America, though individuals have been known to stray beyond the identified distribution (Castro 2011). Certain populations of tiger sharks are seasonally migratory, moving offshore in summer months and returning to coastal areas for the remainder of the year (Kohler and Turner 2019; Lea et al. 2015). Additional populations of tiger sharks remain in coastal waters throughout southeastern United States waters year-round (Castro 2011). Movements between the Gulf of America and Atlantic, in addition to trans-Atlantic movements, have been recorded (Kohler and Turner 2019; Vaudo et al. 2014). Two studies have shown that an ontogenetic shift in habitat use occurs between juvenile and adult sharks in the western North Atlantic at approximately 260 to 270 cm total length (TL) (Lea et al. 2015; McClain et al. 2022). Juvenile tiger sharks in the western North Atlantic and Gulf of America have been found to utilize coastal shelf habitat, increasing movement to deeper, offshore areas as they reach adult stages (Ajemian et al. 2020; Natanson et al. 2023a).

Reproductive cycles follow a biennial to triennial pattern (Hawaii: birth once every three years, Atlantic: suggested biennial cycle) (Castro 2009; Whitney and Crow 2006). Mating activities are predicted to occur in late fall and winter, followed by a 15 to 18 month gestational period with parturition in late spring to summer (Driggers et al. 2008; Sulikowski et al. 2016). In the Atlantic and Gulf of America, both mating and parturition are thought to occur near pupping areas (Driggers et al. 2008). Site fidelity thought to be related to foraging or reproduction has been recorded in the Hawaiian Islands, Bahamas, and Gulf of America (McClain et al. 2022; Papastamatiou et al. 2013; Sulikowski et al. 2016). In the Bahamas, site fidelity to Tiger Beach is thought to provide refuge to immature females and gravid adults with parturition occurring elsewhere (Sulikowski et al. 2016). Pupping areas for young-of-year tiger sharks were found between 27°N to 35°N along the western North Atlantic coast of the United States, 31°N to 33°N encapsulating the most important of the identified pupping grounds (Driggers et al. 2008). In the Atlantic and Gulf of America, a recent study found signatures of genetic partitioning between pupping areas, which is thought to be the result of biparental philopatry for the purposes of mating (McClain et al. 2022). Litter sizes in Hawaiian populations range from 3 to 57 pups (mean embryos per litter: 32.6) and sizes range from 80 to 90 cm TL at birth (Whitney and Crow 2006).

Tiger sharks display an ontogenetic shift in diet, prey size, and prey diversity (Lowe et al. 1996; Aines et al. 2018). Young-of-the-year tiger sharks consume primarily teleosts and molluscs and shift to consuming primarily teleosts as juveniles. Juvenile tiger sharks demonstrate selective feeding of gastropod feet (Aines et al. 2018). Adults tiger sharks are opportunistic foragers that have been observed to consume teleosts, molluscs, birds, cephalopods, reptiles, marine mammals, crustaceans, anthropogenic solid waste, plants, worms, cnidarians, other elasmobranchs (Lowe et al. 1996; Aines et al. 2018).

The intrinsic rate of increase for tiger sharks is estimated to be high, with $r_{\max}$ values ranging from 0.054 to 0.366 (Cortes 2016; Smukall et al. 2022). Von Bertalanffy growth coefficients are 0.131 and 0.124 for males and females, respectively (Kneebone et al. 2008). Age at 50% maturity is 9.5 years for males and 11.6 years for females (Natanson et al. 2023b). Maximum age is projected as 27 and 29 years for males and females, respectively (Kneebone et al. 2008). Relative abundance trends from a fishery-independent survey near Bimini, the Bahamas, from 1984 to 2019 suggest abundance of tiger sharks has been stable since the 1980s. In comparison, the abundance near the southeast United States has slowly increased in the past decade (Smukall et al. 2022). No stock assessment is available for tiger sharks.

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West Atlantic bluefin tuna - *Thunnus thynnus*Overall Vulnerability Rank = moderate  (77% certainty)Biological Sensitivity = moderate Climate Exposure = high Distributional Vulnerability Rank = high  (99% certainty)

Directional Effect = neutral (83% certainty)

Data Quality = 92% of scores ≥ 2

<i>Thunnus thynnus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	 Low  Moderate  High  Very High
Sensitivity Attributes	Adult Mobility	1.3	3		
	Habitat Specificity	1.7	2.8		
	Mobility and Dispersal of Early Life Stages	2	2.6		
	Other Stressors	1.8	2.1		
	Population Growth Rate	3.2	2.3		
	Prey Specificity	1.6	2.5		
	Reproductive Cycle	2.8	2.6		
	Reproductive Strategy Sensitivity	2.6	2.6		
	Sensitivity to Ocean Acidification	1.6	1.9		
	Sensitivity to Temperature	1.7	2.5		
	Site Fidelity	2.1	2.2		
	Specificity in Early Life History Requirements	2.7	2.2		
	Stock Size Status	2.6	2.3		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.6	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.4	-		
	pH	4	-		
	Primary Production	1.4	-		
	Sea Surface Oxygen	3.3	-		
	Sea Surface Salinity	2.6	-		
	Sea Surface Temperature	3.3	-		
	Exposure Score	high			
Overall Vulnerability Rank		moderate			

West Atlantic bluefin tuna (*Thunnus thynnus*)

Contributors: B. Galuardi, J. Kneebone, T. Loughran

Overall Climate Vulnerability Rank: **Moderate** (77% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.3), Sea Surface Oxygen (3.3), and pH (4). Bluefin tuna inhabit temperate to subtropical waters throughout the western and eastern Atlantic and Mediterranean Sea, are generally found in epipelagic waters, and spend time at depth for foraging and thermoregulation purposes (Fromentin and Fonteneau 2001).

Biological Sensitivity: **Moderate.** Five sensitivity attribute means scored above 2.5: Population Growth Rate (3.2), Reproductive Cycle (2.8), Reproductive Strategy Sensitivity (2.6), Specificity in Early Life History Requirements (2.7), and Stock Size and Status (2.6). Bluefin tuna are a relatively long-lived fish with a varying population growth rate (Restrepo et al. 2010). The West Atlantic stock of bluefin tuna is historically smaller than the East Atlantic stock, with an uncertain level of mixing between stocks (Andrews et al. 2021; MacKenzie et al. 2009). West Atlantic bluefin tuna seasonally migrate to spawning grounds in the northern Gulf of America, with evidence of additional spawning grounds located in the Slope Sea and southern Gulf (Cruz-Castán et al. 2019; Hernández et al. 2022; Richardson et al. 2016; Wilson et al. 2015).

Data Quality: 92% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Several studies have been done pertaining to Atlantic bluefin tuna life history and climate factor effects, including migrations, foraging behavior, and thermal tolerances. Hansell et al. (2022) reported that between 1993 and 2020, distribution of larger size classes of Atlantic bluefin tuna has expanded by 96 km²/year with a 1 km/year shift north and 2 km/year shift east. These results are in agreement with Erauskin-Extramiana et al. (2019), which projects Atlantic bluefin tuna distribution moving poleward to the northeast Atlantic. Braun et al. (2023) found that habitat loss for Atlantic bluefin tuna is projected to be greatest in the Gulf of America (~60%), with an additional ~20% loss in the northwestern Atlantic. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal and temporal shifts of small and large bluefin tuna throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Since 2002, small and large bluefin tuna catch was estimated to have shifted poleward 8 ± 1 km and 3 ± 1 km for every 1°C increase, and an average of 10 ± 1 and 4 ± 1 km/year, respectively. Timing of the first 25% of large bluefin tuna catch was estimated to have shifted 80 days earlier off Massachusetts from 2019 relative to 2002.

Abundance of Atlantic bluefin tuna may be affected by changing oceanographic conditions throughout the Gulf of America. Bluefin tuna are regionally endothermic, regulating their body temperature to a higher temperature than their surrounding environment. Temperatures throughout Atlantic bluefin tuna spawning grounds in the Gulf of America tend to be at the upper limit of their thermal tolerance, approaching 30°C (Teo et al. 2007). Recent studies indicate that the effects of climate change, namely

warming sea surface temperatures, may make early spring more suitable for spawning bluefin, ultimately impacting the timing of spawning behavior and migrations (Muhling et al. 2011). Larval stages benefit from the warmer water temperatures found throughout spawning grounds in the Gulf of America but may be adversely affected by rising temperatures and subsequent low dissolved oxygen under future conditions (Dell’Apa et al. 2023; Hazen et al. 2016; Reglero et al. 2014; Teo et al. 2007). These findings are further supported by Trueman et al. (2023), which demonstrated that juvenile stages prefer temperatures of 25°C to 26°C and temperatures above 28°C may contribute to declines in recruitment, particularly in the Gulf of America. Erauskin-Extramiana et al. (2023) also projected a decrease of 17.2% in average body size of Atlantic bluefin tunas by 2050. Rising sea surface temperatures may also limit the ability of adult bluefin to regulate their temperatures effectively, leading to overheating and diminished capability to withstand low oxygen environments (Braun et al. 2023; Hazen et al. 2016; Teo et al. 2007).

Life History Synopsis: Atlantic bluefin tuna are a highly migratory species, managed as two stocks: West Atlantic and East Atlantic. Atlantic bluefin tuna are habitat generalists, distributed from Labrador and Newfoundland to the Gulf of America and Caribbean in the western Atlantic (NMFS 2006). Atlantic bluefin tuna occupy temperate to subtropical waters throughout the western and eastern Atlantic and Mediterranean Sea and are generally found in epipelagic waters, with seasonal nearshore migrations for foraging purposes. This species can tolerate a wide range of temperatures (3°C to 30°C) and frequently dive to depths of 1000 m for foraging and thermoregulation purposes (Block et al. 2001).

Early life history of West Atlantic bluefin tuna consists of a 48-hour egg period and three to four week pelagic larval stage. Larval stages are dependent on eddies and current edges for warm, nutrient-rich waters that supply zooplankton prey (Shropshire et al. 2021). Juveniles reach maturity after 5 years, gradually expanding their thermal tolerance and distribution as they grow to a maximum age of 40 years (Arregui et al. 2018; Corriero et al. 2019; Druon et al. 2016; Marcek et al. 2016; SCRS 2020). Adults and juveniles aggregate along ocean fronts, primarily upwelling areas and meso-scale oceanographic structures associated with the general circulation of the North Atlantic. Adults migrate to spawning grounds, spawning for 6 to 8 weeks between April and June in the northern Gulf of America and March to April in the southern Gulf of America (Cruz-Castán et al. 2019; Druon et al. 2016; Shropshire et al. 2021). Skipped spawning is also evident throughout bluefin stocks (Aarestrup et al. 2022). Recent studies in the Slope Sea have provided evidence of West Atlantic bluefin spawning grounds in the Slope Sea during summer months, influenced by Gulf Stream extent (Hernández et al. 2022; Richardson et al. 2016; Rypina et al. 2021).

Adult and juvenile stages are opportunistic, generalist predators. Stomach analysis revealed 20 species of fish and about 10 invertebrates, ranging from jellyfish and salps to demersal and sessile species such as octopus, crabs and sponges (Butler et al. 2015; Logan et al. 2015). Adults primarily feed on herring and sand lance in the West Atlantic. Prey species may also include mackerel, menhaden, silver hake, and cephalopods.

Mixing between West and East Atlantic stocks has been documented, but the frequency of stock mixing is uncertain as is the stock status (Andrews et al. 2021; MacKenzie et al. 2009; SCRS 2023). The last stock assessment on the West Atlantic bluefin tuna was conducted in 2021 and found that the stock was not subject to overfishing ($F_{\text{Current (2018-2020)}}/F_{0.1} = 0.53 (0.49-0.58)$ and the estimated probability of overfishing,

is <1%) (SCRS 2023). The International Commission for the Conservation of Atlantic Tunas recently adopted a management procedure for Atlantic bluefin tuna that will provide a framework for management decisions and actions to reach conservation targets while accounting for uncertainty in stock status.

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West Atlantic sailfish - *Istiophorus platypterus*

Overall Vulnerability Rank = high (90% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (99% certainty)

Directional Effect = neutral (67% certainty)

Data Quality = 77% of scores ≥ 2

<i>Istiophorus platypterus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.6	2.3		Low Moderate High Very High
	Habitat Specificity	1.8	2.8		
	Mobility and Dispersal of Early Life Stages	2	2.1		
	Other Stressors	2.2	2.2		
	Population Growth Rate	2	2.1		
	Prey Specificity	1.6	2.7		
	Reproductive Cycle	2.7	2.4		
	Reproductive Strategy Sensitivity	2.4	2		
	Sensitivity to Ocean Acidification	2	1.5		
	Sensitivity to Temperature	2	2.2		
	Site Fidelity	2.1	1.9		
	Specificity in Early Life History Requirements	2.5	1.8		
	Stock Size Status	2.6	2.1		
	Sensitivity Score		moderate		
Exposure Factors	Mixed Layer Depth	1.2	-		
	Magnitude of Sea Surface Temperature Gradient	1.5	-		
	Oxygen at 200m	2.2	-		
	pH	4	-		
	Primary Production	1.1	-		
	Sea Surface Oxygen	3.8	-		
	Sea Surface Salinity	3.1	-		
	Sea Surface Temperature	3.9	-		
	Exposure Score		very high		
Overall Vulnerability Rank		high			

West Atlantic sailfish (*Istiophorus platypterus*)

Contributors: B. Galuardi, R. Logan, T. Loughran, E. Orbesen, D. Richardson

Overall Climate Vulnerability Rank: **High** (90% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Four exposure factors contributed to this score: Sea Surface Temperature (3.9), Sea Surface Salinity (3.1), Sea Surface Oxygen (3.8), and pH (4.0). West Atlantic sailfish spend nearly all time in surface waters, occupying temperatures between 12 and 33°C (frequently 22 to 33°C) epipelagic waters but will dive to over 100 m during the day, likely in search of prey (Lam et al. 2016).

Biological Sensitivity: **Moderate.** Three sensitivity attribute means scored equal to or above 2.5: Reproductive Cycle (2.7), Specificity in Early Life History Requirements (2.5), and Stock Size and Status (2.6). Sailfish spawning grounds are found through the slope and abyssal regions of the northern Gulf of America, which experiences dramatic shifts in salinity, temperature, nutrient input, dissolved oxygen, and anthropogenic activity (Simms et al. 2010). Warmer water temperatures were found to skew sex ratio towards female abundance (Mourato et al. 2018).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Recent literature has evaluated the distribution of sailfish and other billfish species in response to climate change. Braun et al. 2023 projected an overall increase in suitable habitat in the western North Atlantic by approximately 20%, but a possible decrease of suitable habitat in the Gulf of America exceeding 10%. Poleward shifts in distribution of other highly migratory species caught recreationally have been documented in recent years (Crear et al. 2023). Oxygen levels are hypothesized to be one factor limiting the dive depth of sailfish, and thus any shoaling of the oxycline may result in a vertical compression of sailfish habitat particularly in the eastern Atlantic (Stramma et al. 2012). Lower oxygen availability in the Gulf of America may also impact larval survival, further exacerbated by warmer sea surface temperatures (Dell’Apa et al. 2018; Dell’Apa et al. 2023; Simms et al. 2010).

Life History Synopsis: Sailfish are primarily a tropical to subtropical species that can occasionally be caught in more temperate latitudes. This species is more coastal than other billfish species, but migrations into the open ocean do occur. Genetic differences do exist between Atlantic and Indo-Pacific sailfish and there is some question on a global scale about whether sailfish are two species (Atlantic: *I. albicans*; Indo-Pacific: *I. platypterus*) or a single pan-global species (*I. platypterus*). Within the Atlantic, tagged sailfish have yet to be documented migrating between the western and eastern Atlantic, supporting the designation of two separate Atlantic sailfish stocks (Orbesen et al. 2008). However, to date, cross-Atlantic genetic differences in sailfish have not been demonstrated (McDowell and Graves 2002, Ferrete et al. 2023), perhaps because of the evolutionary time-scales of genetic homogeneity (Graves and McDowell 2015).

Sailfish exhibit a relatively shallow distribution, typically found above the thermocline, with feeding targeting more surface oriented small pelagic species, especially fishes (Jolley 1977). While the majority of their time is spent in near surface waters, sailfish often undertake short duration movements to 150 m depth (Kerstetter et al. 2011) have been reported to dive as deep as 340 m (Hoolihan et al. 2011).

While the full spatial extent of spawning and the relative importance of different spawning grounds is unknown, the available data indicates that sailfish have a protracted and widespread spawning distribution. In the western Atlantic, larval sailfish have been sampled off Barbados and the Atlantic coast of the United States, as well as in the Gulf of America and Straits of Florida (Richardson et al. 2009). Reproductive studies indicate that spawning south of the equator off Brazil occurs from December-April and in the eastern Caribbean from May to October (Mourato et al. 2018).

Individual sailfish spawn many batches of approximately 1.1 mm diameter eggs over a protracted period. Eggs hatch within a few days and larvae grow rapidly, reaching 10 mm standard length (SL) in about 12 days (Richardson et al. 2009). The early juvenile stage (> 4 cm) is rarely sampled, but by about 1 year of age sailfish are estimated to reach 74 cm lower jaw fork length (LJFL) (ICCAT 2023). Female sailfish reach 50% maturity at about 146 cm LJFL (Mourato 2018). Larval stages of sailfish are planktivores, shifting to copepods and small teleosts as they reach 7 mm SL (Luthy et al. 2005; Tidwell et al. 2008). Adults have a generalist diet of various pelagic teleost species (Jolley 1977, Rosas-Alayola et al. 2002).

The maximum lifespan of Atlantic sailfish is estimated as between 10 and 15 years of age. The intrinsic rate of increase is estimated as 0.132/yr, and von Bertalanffy growth coefficient as 0.607 and 0.583 for females and males (Mourato and Carvalho 2017; Erhardt and Deleveau 2006). Reported catches for the West Atlantic sailfish stock have been below maximum sustainable yield, however, it is believed that there is a significant number of unreported landings occurring. The last stock assessment on West Atlantic sailfish was conducted in 2023 found that the stock remains overfished, but overfishing is not occurring ($B_{2021}/B_{MSY} = 0.96$ (0.59-1.49); $F_{2021}/F_{MSY} = 0.585$ (0.364-0.952)) (SCRS 2023).

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West Atlantic skipjack tuna - *Katsuwonus pelamis*Overall Vulnerability Rank = low ■ (100% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = high ■ (99% certainty)

Directional Effect = neutral (83% certainty)

Data Quality = 100% of scores: 2

<i>Katsuwonus pelamis</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.1	3		Low Moderate High Very High
	Habitat Specificity	1.4	3		
	Mobility and Dispersal of Early Life Stages	1.8	2.6		
	Other Stressors	1.6	2.3		
	Population Growth Rate	1.4	2.5		
	Prey Specificity	1.4	3		
	Reproductive Cycle	1.2	2.8		
	Reproductive Strategy Sensitivity	1.8	2.7		
	Sensitivity to Ocean Acidification	1.8	2.2		
	Sensitivity to Temperature	2	2.8		
	Site Fidelity	1.1	2.8		
	Specificity in Early Life History Requirements	2.2	2.6		
	Stock Size Status	1.2	2.8		
	Sensitivity Score	low			
Exposure Factors	Mixed Layer Depth	1.2	-		
	Magnitude of Sea Surface Temperature Gradient	1.5	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.1	-		
	Sea Surface Oxygen	3.4	-		
	Sea Surface Salinity	2.4	-		
	Sea Surface Temperature	3.5	-		
	Exposure Score	high			
Overall Vulnerability Rank		low			

West Atlantic skipjack tuna (*Katsuwonus pelamis*)

Contributors: B. Galuardi, W. Goldsmith, T. Loughran, D. Snodgrass

Overall Climate Vulnerability Rank: **Low** (100% certainty from bootstrap analysis).

Climate Exposure: **High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.5), Sea Surface Oxygen (3.4), and pH (4.0). Skipjack tuna are found throughout epipelagic, oxygen-rich surface waters with preferred temperatures of 20 to 30°C. Dissolved oxygen content generally restricts skipjack tuna to waters above the thermocline, with minimum dissolved oxygen thresholds of 3.0-3.5 mL/L (Barkley et al. 1978).

Biological Sensitivity: **Low.** No sensitivity attribute means scored above 2.5.

Data Quality: 100% of the data quality scores were 2 or greater indicating very high data quality.

Climate Effects on Distribution and Abundance: Several studies have described the impact of climate change to skipjack tuna, most recently ongoing latitudinal and temporal shifts along the Atlantic coast of the United States. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. Results provided evidence of latitudinal and temporal shifts of skipjack tuna throughout the United States Atlantic coast since the early 2000s, driven in part by rising sea surface temperatures. Since 2002, skipjack tuna catch was estimated to have shifted poleward 4 ± 1 km for every 1°C increase, and an average of 3 ± 1 km/year. Timing of the first 25% of skipjack tuna catch was estimated to have shifted earlier by 3 ± 1 days/year for 2019 relative to 2002.

Further impacts of climate change on skipjack tuna distribution and abundance populations are uncertain, but the changes in habitat suitability will likely alter the habitat choices of skipjack tuna (Muhling et al. 2015). It is likely that due to the species' young age of maturity and ability to spawn across numerous areas and times throughout the year that they will likely be less impacted than some other tuna species. Dueri et al. (2014) noted that skipjack tuna could expand their habitat suitability through approximately 2050 but would suffer significant losses over the following 50-year time period. Erauskin-Extramiana et al. (2019) found that most tuna stocks were predicted to shift poleward; however, yellowfin tuna and skipjack tuna were the only species predicted to increase in abundance in their core tropical areas. Skipjack tuna's high dependence on invertebrates as a food source, however, may have a negative impact on the species' success due to ocean acidification. Average body size of skipjack tuna is projected to decrease by 15.5% by 2050 (Erauskin-Extramiana et al. 2023).

Life History Synopsis: Skipjack tuna are a tropical to temperate species, generally found in epipelagic waters between 16 and 28°C with a preference for temperatures between 18 and 6°C, forming mixed schools with other species of tuna (Boyce et al. 2008). Skipjack tuna are managed as two stocks, one in the West Atlantic and East Atlantic.

Reproductive activity has been documented throughout the year, across the Atlantic (Benevenuti Soares et al. 2018; Cayré and Farrugio 1986). Spawning occurs throughout the Gulf of America and Caribbean Sea in the western Atlantic, and the Gulf of Guinea in the eastern Atlantic (Cayré and Farrugio 1986).

Similar to other tuna species, skipjack tuna have a pelagic egg and larval stage. Skipjack tuna have been referred to as the fastest-growing of the tuna species, often achieving a growth rate as high as 0.95 m per year (Murua et al. 2017). The West Atlantic stock reaches maturity at about 50 cm (Batts 1972) which is roughly 5 to 7 months of age. A significant proportion of the prey items sought out by skipjack tuna are invertebrates with various mesopelagic shrimp species having the highest Index of Relative Importance (Ankenbrandt 1989). Menard et al. (2000) also showed a high level of dependence on crustaceans and cephalopods.

The last stock assessment on West Atlantic skipjack tuna was conducted in 2022 and found that the stock was not overfished or subject to overfishing ($B_{2020}/B_{MSY} = 1.60$ (0.90 – 2.87); $F_{2020}/F_{MSY} = 0.41$ (0.19 – 0.89)) (SCRS 2023).

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Whale shark - *Rhincodon typus*

Overall Vulnerability Rank = high (89% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (91% certainty)

Directional Effect = negative (58% certainty)

Data Quality = 77% of scores ≥ 2

<i>Rhincodon typus</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1	2.8		Low Moderate High Very High
	Habitat Specificity	1.7	2.6		
	Mobility and Dispersal of Early Life Stages	1.3	2.2		
	Other Stressors	2.2	2.1		
	Population Growth Rate	3.6	2.3		
	Prey Specificity	1.7	2.6		
	Reproductive Cycle	1.5	2		
	Reproductive Strategy Sensitivity	1.8	1.3		
	Sensitivity to Ocean Acidification	2.4	2.1		
	Sensitivity to Temperature	1.6	2.4		
	Site Fidelity	2.6	2.2		
	Specificity in Early Life History Requirements	1	1.4		
	Stock Size Status	1.9	1.7		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.4	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.1	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.7	-		
	Sea Surface Salinity	2.3	-		
	Sea Surface Temperature	3.8	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

Whale shark (*Rhincodon typus*)

Contributors: L. Crawford, T. Curtis, E. Hoffmayer, T. Loughran, C. McCandless, B. Wetherbee

Overall Climate Vulnerability Rank: **High** (89% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Sea Surface Temperature (3.8), pH (4.0), and Sea Surface Oxygen (3.7). Whale sharks are found throughout tropical to subtropical waters from the Bay of Fundy to the Gulf of America (Castro 2011). This species spends the majority of time in the upper 100 m of surface waters within upwelling areas (Castro 2011; Rowat and Gore 2007; Tyminski et al. 2015; Wilson et al. 2006).

Biological Sensitivity: **Moderate.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.6). Whale sharks are relatively long-lived elasmobranchs (approximately 50 years; Ong et al. 2020), with moderate population growth rates (0.22; Holmberg et al. 2008). Adults are extremely mobile and typically occupy surface waters to feed but can dive up to 1000 m to feed on zooplankton (Dove and Pierce 2021). Whale sharks have narrow temperature and salinity preferences and tend to occupy upwelling areas (Castro 2011).

Data Quality: 77% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: Whale sharks have shown site fidelity to feeding grounds that coincide with fish spawning, phytoplankton blooms, and coral spawning events (Dove and Pierce 2021; Heyman et al. 2001). As filter feeders, whale sharks are vulnerable to shifts in primary production, ocean acidification, and anthropogenic impacts of pollution. Ocean acidification and warming in particular may affect the availability and quality of sensitive prey such as fish and invertebrate eggs and larvae. Increased scale and frequency of red tides may also pose additional threats to whale sharks in the Gulf of America.

Whale sharks are ectothermic and rely on their environment to regulate body temperature. The degree to which rising ocean temperatures impact the distribution of ectothermic species versus endothermic species is uncertain (Dell’Apa et al. 2023; Harding et al. 2021). A recent study predicting current and future global distribution of whale sharks using general linear mixed-effects models and an ensemble of five global circulation models estimated a slight poleward shift in distribution and an overall range contraction (2.5 to 7.4%) in Atlantic by 2070 (Sequeira et al. 2014). In addition, Afonso et al. (2014) found that sea surface temperature helped predict whale shark occurrence in the eastern Atlantic Ocean and suggested a slight expansion in distribution as warmer waters penetrate into higher latitudes in response to climate change. However, Rowat et al. (2021) mentioned that their broad thermal tolerance and wide distribution might buffer whale sharks from direct mortality resulting from changing water temperature overtime.

Life History Synopsis: The whale shark is a large-bodied, filter feeding species with a circumglobal distribution in warm temperate and tropical marine waters (Compagno 2001; Jaffa and Taher 2007; Sequeira et al. 2014). Although whale sharks are known to use the entire water column from the surface down to over 2000 m (Hoffmayer et al. 2021; Graham et al. 2006; Tyminski et al. 2015), they are

thought to spend most of their time in the top 100 m (Wilson et al. 2006; Rowat and Gore 2007; Tyminski et al. 2015).

Knowledge on movement and habitat use patterns for whale sharks in the western Atlantic Ocean remains limited despite the growing number of tagging studies in recent years. Whale sharks are the largest fish in the ocean and have the ability to make large-scale movements; however, the majority of the documented movements in this region have been relatively limited. For example, Hoffmayer et al. (2021) tagged 48 whale sharks in the northern Gulf of America with satellite tags and none of the sharks left the Gulf of America. Hueter et al. (2013) tagged 35 whale sharks in the southern Gulf of America near the Yucatan Straits and reported that sharks moved in multiple directions throughout the Gulf of America, northwestern Caribbean Sea, and the Straits of Florida, but only a few exhibited long-distance movements. Some more recent tagging work has revealed some broader movements north into offshore waters of the New England region where whale sharks were rarely encountered in the past (C. Braun unpublished data, R. de la Parra Venegas unpublished data).

Unlike other locations throughout the world, the western Atlantic is one of the few areas where whale sharks rarely occur in coastal waters, but rather form aggregations along the continental shelf edge (Hoffmayer et al. 2021). There are two known aggregation sites in the region, including Ewing Bank and the Yucatan Straits (de la Parra Venegas et al. 2011; Hueter et al. 2013; McKinney et al. 2017; Hoffmayer et al. 2021) where primarily juvenile male sharks (4 to 8 m length) congregate during the summer months to feed on increased densities of prey, including little tunny spawn (Hoffmayer et al. 2007; de la Parra Venegas et al. 2011; Hoffmayer et al. 2021).

The whale shark is considered a yolk-dependent species, where pregnant females retain egg cases *in utero* until they hatch and then give live birth (Parsons et al. 2008). Gestation period has been estimated to be 9 to 12 months (Pierce et al. 2021), with a litter size as large as 300 pups (Joung et al. 1996). The reproductive cycle appears to be biennial (Pierce et al. 2021). Size at maturity has been estimated to be 9 to 10 m in females (Colman 1997) and 7 to 8 m in males (Norman and Stevens 2007; Ramírez-Macías et al. 2012; Robinson et al. 2016). Whale sharks are reported to have a relatively low growth coefficient ($k = 0.021\text{--}0.088/\text{year}$), late age at maturity (17 to 29 years), and a large maximum size (15.8 to 19.6 m) and longevity (80 to 130 years) (Hsu et al. 2014; Perry et al. 2018; Meeken et al. 2020; Ong et al. 2020). Due to these characteristics, the species is considered vulnerable to recovery from population depletions.

Whale sharks are considered opportunistic filter feeders and feed on a wide variety of planktonic and nektonic prey including small crustaceans such krill, crab larvae and copepods, and small schooling fishes such as sardines, anchovies, and mackerels (Last and Stevens 1994; Colman 1997; Gunn 1999; Hoffmayer et al 2007; Rohner 2015).

An overall population decline of 50% over the last 75 years was reported by the IUCN Red List (Pierce and Norman 2016). The Atlantic subpopulation showed a lower decline of 30% resulting in a subpopulation assessment of Vulnerable (Pierce and Norman 2016). In addition, the IUCN recovery category for whale sharks is largely depleted, due to the species being threatened across its range along with the aforementioned declines in abundance (Pierce et al. 2021). Landing of whale sharks through

the commercial or recreational fishery has been prohibited since the year 2000, and no completed stock assessment is available (NMFS 1999).

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White marlin - *Kajikia albida*

Overall Vulnerability Rank = high (91% certainty)

Biological Sensitivity = moderate

Climate Exposure = very high

Distributional Vulnerability Rank = high (99% certainty)

Directional Effect = neutral (90% certainty)

Data Quality = 46% of scores ≥ 2

<i>Kajikia albida</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	2.4		Low Moderate High Very High
	Habitat Specificity	1.6	2.5		
	Mobility and Dispersal of Early Life Stages	2.2	1.6		
	Other Stressors	1.9	1.6		
	Population Growth Rate	2.8	1.7		
	Prey Specificity	1.6	2		
	Reproductive Cycle	2.3	2		
	Reproductive Strategy Sensitivity	1.9	1.6		
	Sensitivity to Ocean Acidification	1.9	1.6		
	Sensitivity to Temperature	2.4	2.2		
	Site Fidelity	1.9	1.9		
	Specificity in Early Life History Requirements	2.2	1.4		
	Stock Size Status	3.1	2.1		
	Sensitivity Score	moderate			
Exposure Factors	Mixed Layer Depth	1.5	-		
	Magnitude of Sea Surface Temperature Gradient	1.4	-		
	Oxygen at 200m	2.2	-		
	pH	4	-		
	Primary Production	1.3	-		
	Sea Surface Oxygen	3.6	-		
	Sea Surface Salinity	2.7	-		
	Sea Surface Temperature	3.7	-		
	Exposure Score	very high			
Overall Vulnerability Rank		high			

White marlin (*Kajikia albida*)

Contributors: W. Goldsmith, J. Graves, D. Kerstetter, T. Loughran, E. Orbesen, D. Snodgrass

Overall Climate Vulnerability Rank: **High** (91% certainty from bootstrap analysis).

Climate Exposure: **Very High.** Three exposure factors contributed to this score: Sea Surface Oxygen (3.6), Sea Surface Temperature (3.7), and pH (4.0). White marlin spend the majority of their life in epipelagic waters in temperatures between 24 and 29°C, though recent studies showed white marlin occupying cooler temperatures down to 7.8°C (Hoolihan 2012; Hoolihan et al. 2013).

Biological Sensitivity: **Moderate.** Two sensitivity attribute means scored above 2.5: Stock Size and Status (3.1) and Population Growth Rate (2.8). White marlin are considered overfished, but overfishing is not occurring and little is known about their population growth rate (SCRS 2023).

Data Quality: 46% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: White marlin spend over 80% of their time in near surface waters with a preference for sea surface temperatures between 24 and 29°C (Hoolihan et al. 2015, Horodysky et al. 2007, Prince et al. 2005). While the majority of their time is spent in the upper 25 m of the water column, short-duration dives have been measured to depths exceeding 300 m (Hoolihan et al. 2015). White marlin are found at higher latitudes in the summer months, and in the western North Atlantic they are associated with steep drop-offs and canyons along the edge of the continental shelf (Nakamura 1985). As sea surface temperatures continue to rise, white marlin could experience both a range expansion to higher latitudes and a loss of overall suitable habitat in lower latitudes. Braun et al. 2023 reported a 20% decrease in white marlin suitable habitat throughout the Gulf of America, and slightly lower levels of habitat loss in the western North Atlantic. Crear et al. (2023) used recreational fishery dockside intercept data gathered between 2002 and 2019 throughout the Northeast Shelf to determine changes in the timing and location of highly migratory species catch. No shifts in white marlin catch as a result of increasing sea surface temperatures were detected, though poleward movement at an estimated rate of 2 ± 1 km/year was found for 2019 relative to 2002. Additionally, while dissolved oxygen requirements are poorly understood, a reduction in diving behavior has been observed for sailfish and blue marlin in areas of low dissolved oxygen (< 3.5 ml/l), and white marlin would likely be similarly impacted (Dell'Apa et al. 2023; Prince and Goodyear 2006; Prince et al. 2010).

Life History Synopsis: White marlin are an epipelagic predator found throughout the Atlantic Ocean from 45°N to 45°S (Nakamura 1985). White marlin are targeted by recreational anglers often along the shelf edge habitat or other nearshore oceanographic features (de Sylva and Davis 1963), and are captured as bycatch in pelagic longline fisheries throughout their range. Conventional tagging studies have observed both trans-Atlantic and trans-Equatorial movements, which is consistent with the current hypothesis of one Atlantic-wide stock (Orbesen et al. 2008). Several studies have examined the genetic population structure of white marlin, which also support the hypothesis of one Atlantic-wide stock (Mamoozadeh et al. 2017, Graves and McDowell 2006).

There is a high level of uncertainty associated with the biology and early life history of white marlin, including age, growth, and mortality estimates (Drew et al. 2006). Life history information is further

confounded by the fact that historically roundscale spearfish (*Tetrapturus georgii*) were reported as white marlin. While roundscale spearfish were first reported as a separate species in 1963 by Robins and de Sylva, the species was not generally accepted until genetic confirmation and morphological differences were documented over four decades later (Shivji et al. 2006), meaning studies prior to 2006 could have represented a mix of the two species.

50% of white marlin are estimated to be mature at 139 cm lower jaw fork length (LJFL) for males and 149 cm LJFL for females (Oliveira et al. 2007), and individuals may spawn up to four times during their reproductive season. Spawning has been observed in the Caribbean (Dominican Republic, and Puerto Rico) and the Sargasso Sea from April to June, and off Brazil from November to March (Arocha and Bárríos 2009; de Sylva and Breder 1997; Prince et al. 2005; Luckhurst and Arocha 2016; Amorim et al. 1998).

White marlin exhibit sexual dimorphism beginning around 160 cm with all fish over 200 cm LJFL being females (Arocha and Bárríos 2009). The primary method of estimating age of white marlin is through reading a cross section of fin spines; however, vascularization, especially in older fish, results in loss of the innermost rings, complicating age estimates (Kopf et al. 2008). Maximum age estimates derived from spine readings are approximately 13 years (Drew et al. 2006); however, estimates resulting from conventional tag recaptures have exceeded 15 years (Orbesen et al. 2008). Ontogenetic diet shifts to teleosts occur as blue marlin grow, eventually encompassing a variety of pelagic species (flying fish, *Thunnus* spp., dolphinfish, and squids) (Hoolihan 2013).

The last stock assessment on white marlin (and roundscale spearfish, *Tetrapturus georgii*) was conducted in 2019 (using data through 2017) and found that the stock was overfished but overfishing was not occurring (B_{2017}/B_{MSY} : 0.58 (0.27–0.87); F_{2017}/F_{MSY} : 0.65 (0.45–0.93)) (SCRS 2023). The International Commission for the Conservation of Atlantic Tunas (ICCAT) implemented an annual landings limit of 355 t for white marlin and roundscale spearfish combined in 2022 and requires the live release of white marlin/roundscale spearfish to the extent possible (SCRS 2023). In addition, ICCAT's Standing Committee on Research and Statistics cautions that if catches continue to exceed the 355 t landings limit, the rebuilding of the stock could proceed at a slower pace or lead to further stock declines (SCRS 2023). The next stock assessment for white marlin is not scheduled, but could occur as early as 2025.

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White shark - *Carcharodon carcharias*Overall Vulnerability Rank = low ■ (46% certainty)Biological Sensitivity = low ■Climate Exposure = high ■Distributional Vulnerability Rank = very high ■ (55% certainty)

Directional Effect = neutral (92% certainty)

Data Quality = 85% of scores ≥ 2

<i>Carcharodon carcharias</i>		Attribute Mean	Data Quality	Expert Scores Plots (tallies by bin)	
Sensitivity Attributes	Adult Mobility	1.2	3		Low Moderate High Very High
	Habitat Specificity	1.5	2.6		
	Mobility and Dispersal of Early Life Stages	1.3	2.8		
	Other Stressors	1.8	2		
	Population Growth Rate	3.6	2.6		
	Prey Specificity	1.6	2.7		
	Reproductive Cycle	1.6	2.5		
	Reproductive Strategy Sensitivity	1.4	2		
	Sensitivity to Ocean Acidification	1.3	1.9		
	Sensitivity to Temperature	1.6	2.8		
	Site Fidelity	2.5	2.2		
	Specificity in Early Life History Requirements	1	2.2		
	Stock Size Status	2.4	1.9		
	Sensitivity Score	low			
Exposure Factors	Bottom Salinity	3.2	-		
	Bottom Temperature	3.5	-		
	Mixed Layer Depth	1.4	-		
	pH	4	-		
	Primary Production	1.3	-		
	Precipitation	1	-		
	Sea Surface Oxygen	3.3	-		
	Surface Wind Stress Magnitude	1.1	-		
	Exposure Score	high			
Overall Vulnerability Rank		low			

White shark (*Carcharodon carcharias*)

Contributors: L. Crawford, D. Crear, T. Curtis, T. Loughran, C. McCandless, G. Skomal

Overall Climate Vulnerability Rank: **Low** (46% certainty from bootstrap analysis).

Climate Exposure: **High.** Four exposure factors contributed to this score: Bottom salinity (3.2), Bottom Temperature (3.5), pH (4), and Sea Surface Oxygen (3.3). White sharks migrate and feed in coastal and nearshore habitats during all life stages, using the majority of the water column (Bowlby et al. 2022, Curtis et al. 2018, Franks et al. 2021, James et al. 2022; Skomal et al. 2017).

Biological Sensitivity: **Low.** One sensitivity attribute mean scored above 3.0: Population Growth Rate (3.6). White sharks are long-lived elasmobranchs with relatively low population growth rates, estimated between 0.025 and 0.05/yr (Bowlby & Gibson 2020).

Data Quality: 85% of the data quality scores were 2 or greater.

Climate Effects on Distribution and Abundance: White sharks are highly mobile and have dispersive early life stages (Curtis et al. 2018, James et al. 2022). The species occupies coastal and nearshore habitats (typically <100m) (Bowlby et al. 2022, Franks et al. 2021, Skomal et al. 2017) with only large, subadult and adult animals venturing offshore and into deep waters presumably to feed (Jorgensen et al. 2012, Franks et al. 2021, Skomal et al. 2017) .

Few studies have examined abundance and distribution trends of Atlantic white sharks. However, it is well-documented that water temperature plays an important role in seasonal distributions and migration patterns across life stages (Bastien et al. 2020; Curtis et al. 2014; Curtis et al. 2018; Franks et al. 2021; Shaw et al. 2021; Skomal et al. 2017). The western North Atlantic white shark population declined significantly in the 1970s and 1980s but has steadily increased since the early 2000s (Curtis et al. 2014). Possible explanations for increasing white shark abundance include population recovery due to fishery management measures (1993 NMFS Fishery Management Plan, NMFS 1993; listed as prohibited species in 1997, NMFS 1997), northward shift due to climate change (Bastien et al. 2020), and increased populations of pinniped prey (Curtis et al. 2014; Skomal et al. 2012).

Increased efforts to tag and track the species with acoustic and satellite tags has elucidated the distributions, ranges, and site fidelity of white sharks in the western North Atlantic Ocean, ranging from the Gulf of America to Atlantic Canada. White sharks display site fidelity to waters off of Massachusetts (Franks et al. 2021; Skomal et al. 2017) and off of Atlantic Canada in the summer months (Franks et al. 2021, Bowlby et al. 2022). Overwintering residency occurs in the southeastern United States and Gulf of America (Franks et al. 2021; James et al. 2022). Juvenile white sharks tend to migrate more northward in the Canadian Maritimes region and are 5 times more likely to be found in Canadian waters compared to large adults. This indicates that juveniles make use of a larger range of suitable coastal habitat (Bowlby et al. 2022) and may be flexible to climate-induced range shifts.

Young of the year (YOY) white sharks use the New York Bight as a nursery area in the summer months and display site fidelity to waters off of North and South Carolina in the winter months (Curtis et al. 2018; James et al. 2017; Shaw et al. 2021). Changing ocean conditions in the New York Bight due to

climate change may result in a northward shift in suitable habitat for YOY white sharks, which may cause them to interact with larger white sharks off of Massachusetts (Tobey Curtis, unpublished data). Predation by larger animals in these expanded areas may negatively affect YOY white sharks (Shaw et al. 2021). Though climate change may disrupt the suitability of residency locations, the high variability in migration path, timing, and residency duration could afford white sharks flexibility to climate change (Franks et al. 2021).

Lewis et al. (2023) examined the time of emergence (year when temperatures will exceed a species' tolerance) for fish in the Scotian Shelf and Bay of Fundy conservation areas under climate change scenarios RCP 2.6 and RCP 8.5. White sharks were predicted to experience the least thermal stress across the conservation areas and had a predicted habitat loss of 0% by 2100 under both emission scenarios (Lewis et al. 2023). However, white sharks may be exposed to such thermal stress further south in their range.

White sharks are regionally endothermic. The species can regulate its body temperature due to its high metabolic rate and a system of blood vessels called *rete mirabile* that function as a countercurrent heat exchange system to maintain heat within tissues and prevent cooling of the blood through the gills. White sharks cannot thermoregulate like mammals, however, and thus slowly respond internally to external changes in water temperature. Increases in muscle temperature when white sharks swim into warmer waters is delayed by several hours due to the countercurrent heat exchange system (Carey et al. 1982). The advantages of this physiological adaptation when encountering elevated ocean temperatures due to climate change remains uncertain (Dell'Apa et al. 2023; Harding et al. 2021).

Life History Synopsis: White sharks migrate and feed in coastal and nearshore habitats during all life stages, using the majority of the water column (Casey and Pratt 1985; Bowlby et al. 2022, Curtis et al. 2018, Franks et al. 2021, James et al. 2022; Shaw et al. 2021; Skomal et al. 2017). Juveniles occupy nearshore environments and use nursery grounds located in the New York Bight (Curtis et al. 2018). Juveniles overwinter in 15 to 21°C waters less than 100m off the coasts of the Carolinas (James et al. 2022). Adults typically annually migrate between the southeastern United States and Gulf of America in the winter and Atlantic Canada and New England during summer. Older white sharks make long migrations into deeper waters and the open ocean as far East as the Mid-Atlantic ridge (Bowlby et al. 2022; Franks et al. 2021; Skomal et al. 2017).

Juvenile white sharks consume bony fish and small elasmobranchs (Casey and Pratt 1985; Curtis et al. 2014), and some juveniles have been observed to scavenge North Atlantic right whale carcasses (Taylor et al. 2013). Large white sharks consume bony fish, sea turtles, squid, smaller elasmobranchs, and marine mammals including cetaceans and pinnipeds (Curtis et al. 2014, Skomal et al. 2012). Scavenging on whale carcasses is thought to be a major component of the adult white shark diet (Carey et al. 1982; Casey and Pratt 1985).

White sharks have a low intrinsic rate of increase, between 0.025 and 0.05/yr (Bowlby & Gibson 2020) and a relatively slow growth rate ($k = 0.159$, Tanaka et al. 2011). The species lives up to 73 years old (Hamady et al. 2014) and matures at 26 years and 33 years old for males and females, respectively (Natanson and Skomal 2015). White sharks gestate for 1 year to 20 months and birth between 6 and 10

pups in each litter in a 1 male to 1.1 females sex ratio (Christiansen et al. 2014). White sharks suffered a steep decline in population in the 1970s and 1980s that compromised the genetic variability of the Northwest Atlantic stock and led to inbreeding (O’Leary et al. 2015).

The status of Atlantic white sharks varies across countries and governing bodies. In the United States, white sharks are a prohibited species (NMFS 1997). The stock status of the white shark population in United States waters is unknown and there have been no stock assessments. However, relative abundance analyses indicate increasing abundance in the western North Atlantic Ocean since 2000 (Curtis et al. 2014). In 2013, a status review of the white shark in United States Pacific waters was conducted, and it was concluded that listing was not warranted (Dewar et al. 2013). In Canada, the Northwest Atlantic population of white sharks are listed on Schedule I of the Canadian Species at Risk Act and have been designated as endangered since 2006 (designation reconfirmed in 2021; COSEWIC 2006, COSEWIC 2021).

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